BUSINESS STATISTICS USING EXCEL

J. ALEJANDRO GELVES



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Introduction

"Whatever you would make habitual, practice it; and if you would not make a thing habitual, do not practice it, but accustom yourself to something else." – *Epictetus*

This book is a companion to a course in business statistics. It teaches the core ideas of statistics and how to apply them in Excel. Each chapter introduces a few concepts, works through examples, and ends with practice problems. The material takes effort, but steady work will get you through it.

I owe thanks to my business statistics students, especially Haipei Liu, Addison Alvine, and Sara Blair, for their feedback and careful reviews. Any remaining errors are mine.

Learning from Data

Statistics is about learning from the world. We want to know how customers behave, whether a new process works, or where a market is headed, but we can never observe everything. Instead, we collect **data**, a small and imperfect view of what we care about, and use it to draw conclusions about the bigger picture.

That last step comes with a catch. A sample never perfectly mirrors the population it came from, so two analysts with different samples will get different answers. **Probability** lets us measure how far our data might mislead us. With it, we can test a **hypothesis**, an idea we hold about the world, and decide whether the evidence supports it or whether we are just seeing noise.

You already reason this way in everyday life. You form impressions from limited experience, update them when new evidence arrives, and try not to be fooled by coincidence. Statistics teaches you to do the same thing carefully.

Why Excel?

Excel is the most widely used data tool in business. Finance, accounting, marketing, and operations teams use it every day to organize data, find patterns, and report results. It is available almost everywhere, and the skills you learn here will be useful in your first internship.

Excel can handle every technique in this book, from descriptive statistics and pivot tables to probability, regression, and hypothesis testing. Each chapter shows you how, with step-by-step instructions, screenshots, and downloadable datasets. You do not need any programming experience.

Excel in the Era of AI?

AI tools can now run an analysis in seconds, but you still need to understand the data and the logic to know whether the results make sense. Working in Excel builds that understanding. You learn to organize information, check your assumptions, and follow a calculation from start to finish. These habits make you a better user of AI, because you will be able to question its output instead of simply accepting it.

Roadmap

The book follows the same path as learning from data. **Descriptive Statistics** shows you how to summarize, visualize, and explore a data set. **Regression** introduces tools for measuring relationships between variables and making predictions. **Probability** gives you a way to model the uncertainty that comes from working with samples. **Inference** brings these pieces together so you can reason from a sample to the population and test hypotheses. The final chapter, **Regression and Inference**, uses those tools to judge how reliable a regression model is.

If you are new to spreadsheets, or just want a quick reference, start with the **Excel Basics** chapter at the end of the book. It covers how Excel organizes data (cells, ranges, worksheets, and the Data Analysis ToolPak) and lists every function used in the book.

How to Use This Book

Every chapter has the same structure. It starts with key concepts, formulas, and worked examples. Next, it shows how to carry out each technique in Excel, followed by an **Excel Function Summary**. Many chapters include a dataset you can download to follow along, and the online version adds **Try it** boxes: interactive graphs you can adjust to see a concept in action. Each chapter ends with **Exercises**. Try each problem on your own before clicking to reveal the answer.

How do I get Excel?

William & Mary students can download Office 365, which includes Excel, at https://software.wm.edu/office_365_d1/.

Part I

Describe

1 Descriptive Stats I

Understanding the nature and classification of data is crucial for effective analysis and decision-making. Data are the building blocks of insights, providing a foundation for businesses, researchers, and policymakers to make informed choices. How we collect and categorize data significantly impacts how it is analyzed and interpreted. Below I highlight key types of data and their unique characteristics to help you get started with data analysis. In the arc of this book, this is where learning from data begins: before we can summarize what we have observed or infer what it means for the world, we first have to understand the raw material itself — what kind of data we have collected and how it is organized.

1.1 Data and Types of Data

Let's start by defining data and the types of data structures that you might encounter when conducting analysis. **Data** are facts and figures collected, analyzed and summarized for presentation and interpretation. Data can be classified as:

- **Cross Sectional Data** refers to data collected at the same (or approximately the same) point in time. *Ex: NFL standings in 1980 or Country GDP in 2015.*
- **Time Series Data** refers to data collected over several time periods. *Ex: U.S. inflation rate from 2000-2010 or Tesla deliveries from 2016-2022.*
- **Structured Data** resides in a predefined row-column format (tidy). *Ex: spreadsheet data.*
- **Unstructured Data** does not conform to a pre-defined row-column format. *Ex: Text, video, and other multimedia.*

EXAMPLE

Consider a retail store analyzing its sales performance. If the store collects data on the total revenue generated *by each location* on Black Friday, it is cross-sectional data. On the other hand, if the store tracks *weekly sales for the past year* to observe trends, it is time series data. Structured data, like sales figures stored in *spreadsheets*, allows for easy comparison and analysis. Meanwhile, customer feedback gathered from *social media posts and video reviews* represents unstructured data, requiring advanced tools to extract meaningful insights.

1.2 Data Sets

A **data set** contains all data collected for a particular study. Understanding the structure the data set aids in communicating findings to stakeholders clearly and effectively. Data sets are composed of:

- **Elements** are the entities on which data are collected. *Ex: Football teams, countries, and individuals.*
- **Variables** are a set of characteristics collected for each element. *Ex: Goals scored, GDP, weight.*
- **Observations** are the set of measurements obtained for a particular element. *Ex: Salah, 20 (goals), 15 (assists). US, 2.3 (inflation), 4.5% (federal interest rate).*

Elements	Variable 1	Variable 2
Element 1	#	#
Element 2	#	#
Element 3	#	#
...	...	...

EXAMPLE

Consider the dataset on electric vehicles (EVs) displayed below:

Make	Model	Range_km	TopSpeed_kmh	Price_pounds	Charge_kmh
Tesla	Model 3	415	201	39,990	690
Tesla	Model S	652	250	79,990	750
Nissan	Leaf	270	144	26,995	100
BMW	i3	260	150	37,955	100
Audi	e-tron	400	200	60,000	470
Ford	Mustang	483	180	42,995	750
...	...	...	...	...	...

In this data set, each row represents an electric vehicle model, making the elements the specific EV models rather than the manufacturers. The variables collected for each model include:

- Make: The manufacturer of the EV.
- Model: The specific name of the EV model.

- Range_km: Driving range in kilometers on a full charge.
- TopSpeed_kmh: Maximum speed in km/h.
- Price_pounds: Price in pounds (£).
- Charge_kmh: Charging speed in kilometers per hour.

An example observation is "Tesla Model 3," with the following data: Make: Tesla, Model: Model 3, Range_km: 415, TopSpeed_kmh: 201, Price_pounds: 39,990, Charge_kmh: 690.

1.3 Scales of Measurement

Understanding scales of measurement is crucial for analyzing and interpreting data effectively in business. By distinguishing between categorical (e.g., marital status, satisfaction ratings) and numerical data (e.g., profits, prices), you'll know what methods to use for analysis. Knowing whether data is nominal, ordinal, interval, or ratio ensures your analysis and conclusions are accurate and relevant.

The **scales of measurements** determine the amount and type of information contained in each variable. In general, variables can be classified as **categorical** or **numerical**.

- **Categorical** (qualitative) data includes labels or names to identify an attribute of each element. Categorical data can be **nominal** or **ordinal**.
  - With **nominal** data, the order of the categories is arbitrary. Numbers (if used) serve only as identifiers. Common statistics used are based on counting frequencies and the mode (most common category) *Ex: Marital Status, Race/Ethnicity, or NFL division.*
  - With **ordinal** data, the order or rank of the categories is meaningful, but intervals between ranks are not necessarily equal or quantifiable. Common statistics are based on median, percentiles, or rank-order correlations. *Ex: Rating, Difficulty Level, or Spice Level.*
- **Numerical** (quantitative) include numerical values that indicate how many (discrete) or how much (continuous). The data can be either **interval** or **ratio**.
  - With **interval** data, the distance between values is expressed in terms of a fixed unit of measure. The zero value is arbitrary and does not represent the absence of the characteristic. Ratios are not meaningful. Common statistics are the arithmetic mean, standard deviation, or the Pearson correlation, *Ex: Temperature or Dates.*
  - With **ratio** data, the ratio between values is meaningful. The zero value is not arbitrary and represents the absence of the characteristic. All statistics are permissable including geometric mean, and the coefficient of variation. *Ex: Prices, Profits, Wins.*

EXAMPLE

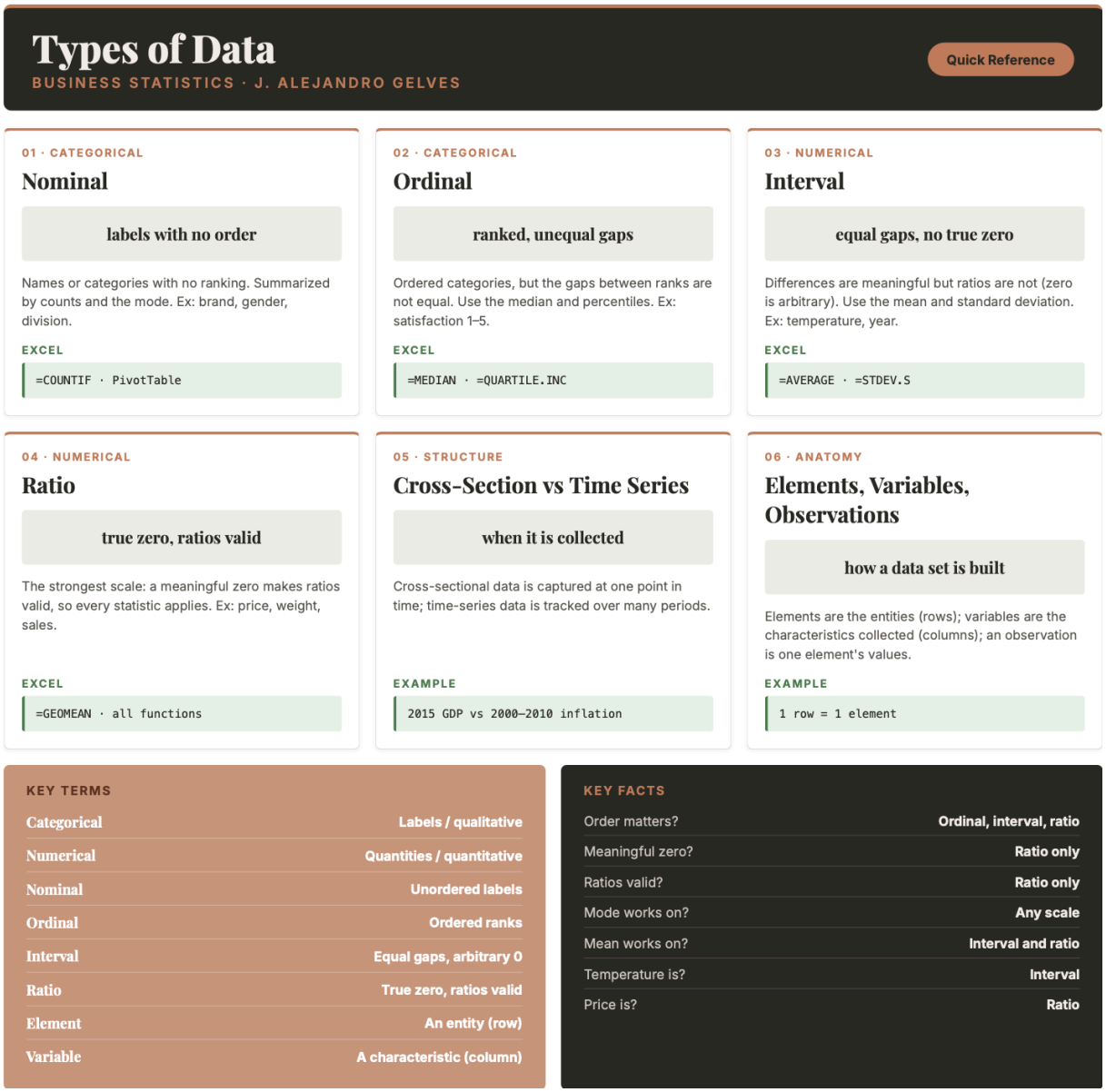
Let's keep using the EV example. Consider the new data set below:

Car	Brand	Range	Rating	Year
Model 3	Tesla	415	4	2021
Model S	Tesla	652	5	2021
Leaf	Nissan	270	3	2021
i3	BMW	260	3	2021
e-tron	Audi	400	4	2021
Mustang	Ford	483	4	2021
...	...	...	...	...

The variables can be classified as follows:

- Car (Categorical - Nominal), consists of names of cars, which are labels used to identify each row. The order of these names does not matter, making it nominal data.
- Brand (Categorical - Nominal) represents the manufacturer of the car (e.g., Ford, Audi). These are labels with no inherent order, making it nominal data.
- Range (Numerical - Ratio), refers to the car's driving range in miles. It is numerical and ratio because it has a meaningful zero (a car with zero range cannot move), and ratios are meaningful (e.g., a car with 250 miles range has double the range of one with 125 miles).
- Rating (Categorical - Ordinal) represents a rank or score (e.g., 4, 3, 2). The order matters, as higher ratings indicate better performance. However, the intervals between ratings are not consistent, so it is ordinal data.
- Year (Numerical - Interval) represents a point in time. While numerical, it is interval data because the zero point is arbitrary (e.g., year 0 does not indicate the "absence" of time), and ratios are not meaningful (e.g., 2020 is not "twice as late" as 1010).

1.4 Chapter Summary Cheat Sheet



1.5 Exercises

The following exercises will help you test your knowledge on the Scales of Measurement. In these exercises you will:

- Identify numerical and categorical data.
- Classify data according to their scale of measurement.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you formulate your own answer and double checked your work for any mistakes.

Exercise 1

A bookstore has compiled data set on their current inventory. A portion of the data is shown below:

Title	Price	Year Published	Rating
Frankenstein	5.49	1818	4.2
Dracula	7.60	1897	4.0
...	...	...	...
Sleepy Hollow	6.95	1820	3.8

1. Which of the above variables are categorical and which are numerical?

Answer on page 166 →

2. What is the measurement scale of each of the above variable?

Answer on page 166 →

Exercise 2

A car company tracks the number of deliveries every quarter. A portion of the data is shown below:

Year	Quarter	Deliveries
2016	1	14800
2016	2	14400
...	...	...
2022	3	343840

1. What is the measurement scale of the Year variable? What are the strengths and weaknesses of this type of measurement scale?

Answer on page 166 →

2. What is the measurement scale for the Quarter variable? What is the weakness of this type of measurement scale?

Answer on page 166 →

3. What is the measurement scale for the Deliveries variable? What are the strengths of this type of measurement scale?

Answer on page 167 →

Exercise 3

A coffee shop chain collects customer feedback data for analysis. For each visit/transaction recorded in their database, the following variables are stored:

- Customer ID: A unique identifier assigned to each registered customer (e.g., C001234, C005678).
- Satisfaction Level: Customers rate their overall experience as 1 (Very Dissatisfied), 2 (Dissatisfied), 3 (Neutral), 4 (Satisfied), or 5 (Very Satisfied).
- Visit Duration: The time spent in the store in minutes (e.g., 15, 42, 8).
- Beverage Type: The main drink ordered (e.g., Latte, Cappuccino, Americano, Tea, Other).

Answer on page 167 →

2 Descriptive Stats II

Understanding and visualizing data distributions is a fundamental step in data analysis. It provides critical insights into the underlying characteristics of the data, which directly impact decision-making, model performance, and interpretation. Below, we introduce tabular and visual techniques to describe your data. This is the *describe* stage of learning from data: we organize the raw observations into tables and pictures so their patterns become visible, before attempting to draw any conclusions from them.

2.1 Frequency Distributions (Categorical)

A **frequency distribution** is perhaps the most valuable tool for summarizing categorical data. It illustrates with a table the number of items within distinct, non-overlapping categories. An alternative known as the **relative frequency** quantifies the proportion of items in each category relative to the total number of observations. You can calculate it by taking the frequency of a particular class (f_i), and dividing it by the number of observations n. Relative frequency helps contextualize the data by highlighting the significance of each category compared to the whole.

The frequency distribution can be illustrated by using a bar plot. A **bar plot** illustrates the frequency distribution (or relative frequency) of categorical data. It includes the classes in the horizontal axis and frequencies or relative frequencies in the vertical axis and has gaps between each bar.

EXAMPLE

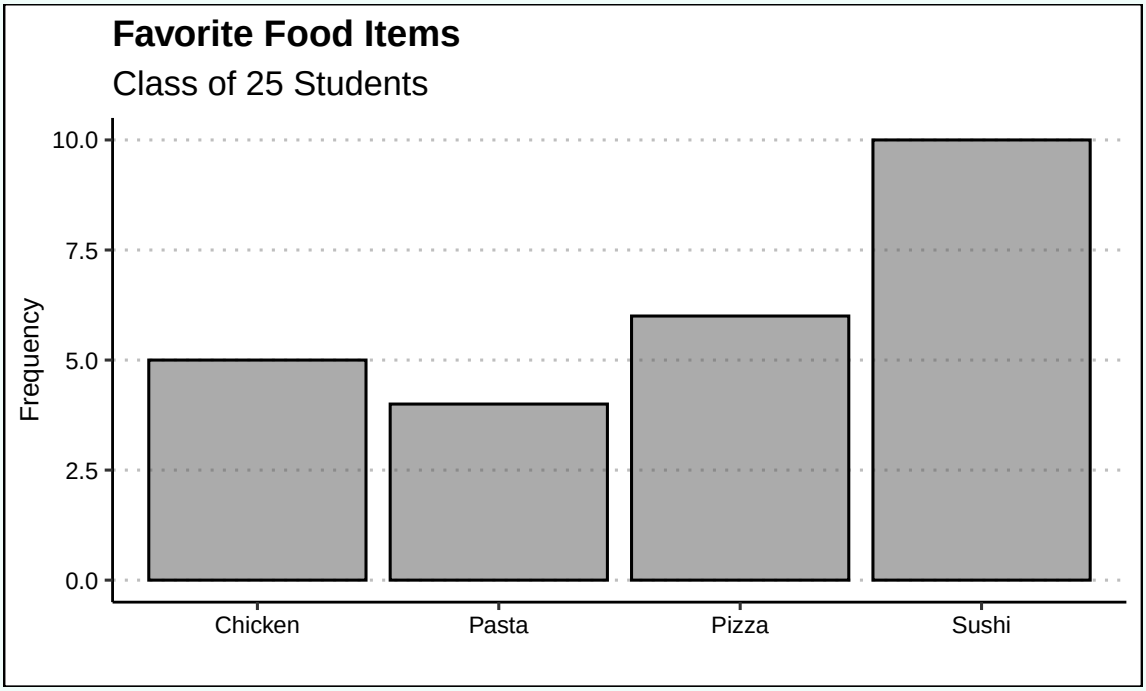
Consider data on students' answers to the question, what is your favorite food? You can see the data below:

PIZZA	SUSHI	SUSHI	CHICKEN	CHICKEN
PASTA	PASTA	PASTA	SUSHI	PASTA
CHICKEN	PIZZA	CHICKEN	SUSHI	PIZZA
SUSHI	SUSHI	SUSHI	SUSHI	PIZZA
PIZZA	CHICKEN	SUSHI	PIZZA	SUSHI

Simply observing raw data can make identifying the most and least popular items challenging. A frequency distribution organizes this information into a clear table, showcasing the popularity of each item. The frequency distribution of the table is displayed below:

Food	Frequency	Relative
Chicken	5	0.20
Pasta	4	0.16
Pizza	6	0.24
Sushi	10	0.40

Each food item is tallied up, and the result is shown in the frequency column. Alternately, we can show the tally as a proportion of the total (i.e., 25). For example, five students liked chicken; out of the 25 students surveyed, this represents 0.2 or 20%. This calculation is shown for each food item in the relative frequency column.
Below, you can see the bar graph showing the frequency distribution of the food items data. Note that the visualization is constructed by showing each food item as a bar with a height equal to the frequency.



2.2 Frequency Distributions (Numerical)

When working with numerical data, building a frequency distributions requires additional steps compared to categorical data. The challenge lies in the absence of predefined categories or classes. To construct a frequency distribution for numerical data, it is essential to determine the

number, width, and limits of the classes. Here are the steps to create a frequency distribution when data is numerical:

1. Determine the Number of Classes: The number of classes can be estimated using the 2^k rule, where k is the smallest integer such that 2^k exceeds the total number of observations by the least amount. This ensures the chosen number of classes provides a reasonable level of granularity for summarizing the data.

EXAMPLE

If a data set has 50 observations, we would choose six classes since $2^6 = 64$ is greater than 50 by the least amount.

Alternatively, you can use **Sturges' rule** to estimate the number of classes. This rule suggests using $k = \lceil 1 + 3.322 \log_{10}(n) \rceil$ where n is the number of observations and $\lceil \cdot \rceil$ denotes the ceiling function (round up to the nearest integer).

EXAMPLE

If a data set has 50 observations, $k = \lceil 1 + 3.322 \times 1.699 \approx 6.64 \rceil = 7$ so we would choose seven classes.

Note that the 2^k rule and Sturges' rule do not always produce the same number of classes. Either method is acceptable — what matters is that you choose one and apply it consistently throughout the rest of the frequency distribution construction.

2. Calculate the Width of Each Class: The width of a class is determined using the formula:

$$Width = \frac{Max - Min}{Number\ of\ Classes}$$

EXAMPLE

: If the data set has 50 observations and the minimum value 20 and the maximum is 78, then the width of each class is $58/6 \approx 9.7$ when using the 6 classes suggested by the 2^k rule. We can round up and use a class width of 10. It is important to note to always round up, as this ensures that all data points are included in an class.

3. Establish Class Limits: The class limits define the range of values in each class. These limits should be chosen such that each data point belongs to only one class.

EXAMPLE

Consider a data set of 50 observations where each class has a width of 10. Set the class limits of the first class to [20,30). Note that the square bracket indicates that the point should be included in the class, whereas) indicates that the point should not be included

in the class. The six classes would be $[20,30)$, $[30,40)$, $[40,50)$, $[50,60)$, $[60,70)$, and $[70,80)$. By choosing these classes, each point belongs to only one class.

Determining the Classes for the Dow Jones Industrial

The **Dow Jones Industrial Average (DJIA)** is one of the most widely followed stock market indexes in the world. It tracks the stock prices of 30 large, well-known U.S. companies — such as Apple, Microsoft, and Goldman Sachs — and summarizes their collective performance into a single number. When people say "the market was up today," they are often referring to the Dow. It serves as a barometer of the overall health of the U.S. economy and is updated continuously during trading hours.

Let's look at a snapshot of the Dow Jones Industrial 30 stock prices. Below you can see the data:

$277	$175	$202	$383	$358
$188	$294	$157	$212	$42
$149	$410	$303	$165	$203
$104	$23	$60	$287	$121
$312	$52	$54	$158	$501
$96	$95	$43	$189	$201

Let's follow the steps to build the frequency distribution.

1. Determine the Number of Classes: Here we choose five classes since $2^5 = 32$ is greater than 30 by the least amount.

2. Calculate the Width of Each Class: The smallest values in the data set is 23 and the maximum is 501. This gives us a range of 478. Now we can just take the range and divide by five to get 95.6. To make things simple we can round to 100 and use a class width of 100.

3. Establish Class Limits: Since we have rounded up we can be flexible with our class limits. The following class limits are suggested $[20,120)$, $[120,220)$, $[220,320)$, $[320,420)$, and $[420,520)$. Note that each class has a width of 100, and that each data point belongs to one single class.

2.3 Frequency Distributions and Bar Graphs in Excel (Categorical)

To construct frequency distributions in Excel, we will use pivot tables. A **pivot table** is an Excel tool that automatically summarizes and counts data, making it ideal for building frequency distributions. It allows you to quickly organize raw data into a structured table without writing any formulas.

Let's use the food data to construct a frequency distribution. Download the data below and follow the steps to build the pivot table.

Download Data

1. Select the Data: Select the data range in Excel while including the headers.

2. Insert a Pivot Table: Go to `Insert → Pivot Chart`. In the dialog box, confirm the data range and choose `Existing Worksheet` to place the pivot table next to the data. Click `OK`.

3. Configure the Pivot Table: In the PivotTable Fields panel on the right, drag the `Favorite Food` field to the `Categories` area and drag it again to the `Values` area. Excel will automatically count the occurrences of each food item, giving you the frequency distribution.

4. Add the Relative Frequency: Drag the `Favorite Food` field to the `Values` area a second time. By default, Excel will show it as a count. To convert it to a percentage, click on the field in the `Values` area and select `Value Field Settings`. Then click on the `Show Values As` tab and choose `% of Column Total` from the dropdown menu. Click `OK`. Excel will automatically display the proportion of each food item relative to the total, giving you the relative frequency distribution directly in the pivot table.

If you need more guidance please refer to the video demonstration below:

Video walkthrough: watch it in the online version of this chapter

2.4 Frequency Distributions and Histograms in Excel (Numerical)

To construct a frequency distribution for numerical data in Excel, we need an extra step compared to the categorical case. Since numerical data has no predefined categories, we must first create the bins manually using a helper column. Once the bins are in place, we can use a pivot table to count the observations in each class.

Let's use the Dow Jones stock price data to construct a frequency distribution. Download the data below and follow the steps.

Download Data

1. Create a Bin Column: In column C, add the header "Bins". In cell C2, enter the following `IFS` formula to assign each stock price to its corresponding class:

```
=IFS(B2<120,"[20,120)",B2<220,"[120,220)",B2<320,"[220,320)",
B2<420,"[320,420)",B2<=520,"[420,520)")
```

Copy the formula down for all 30 rows. Each stock price will now be assigned to one of the five classes defined in Section 2.2. Press `Ctrl + ~` to verify the formulas are correct, then press `Ctrl + ~` again to return to normal view.

2. Insert a Pivot Table: Select the data. Go to `Insert → PivotTable`, confirm the data range, and select `Existing Worksheet`. Click `OK`.

3. Configure the Pivot Table: In the PivotTable Fields panel, drag the `Bins` field to the `Rows` area and drag it again to the `Values` area. Excel will count the number of stock prices in each bin, giving you the frequency distribution. Since the bin labels are text, Excel may sort them alphabetically, placing `[120,220)` before `[20,120)`. To fix this, right-click on the `[20,120)` bin in the pivot table and select `Move → Move to Beginning`. The bins should now appear in the correct numerical order.

4. Add the Cumulative Distribution: Drag the `Bins` field to the `Values` area a second time. Click on the field in the `Values` area and select `Value Field Settings`. Click on the `Show Values As` tab and choose `% Running Total In` from the dropdown menu. Make sure the base field is set to `Bins`. Click `OK`. Excel will display the cumulative percentage for each bin, showing the proportion of stock prices that fall at or below each class. You should obtain the following frequency distribution:

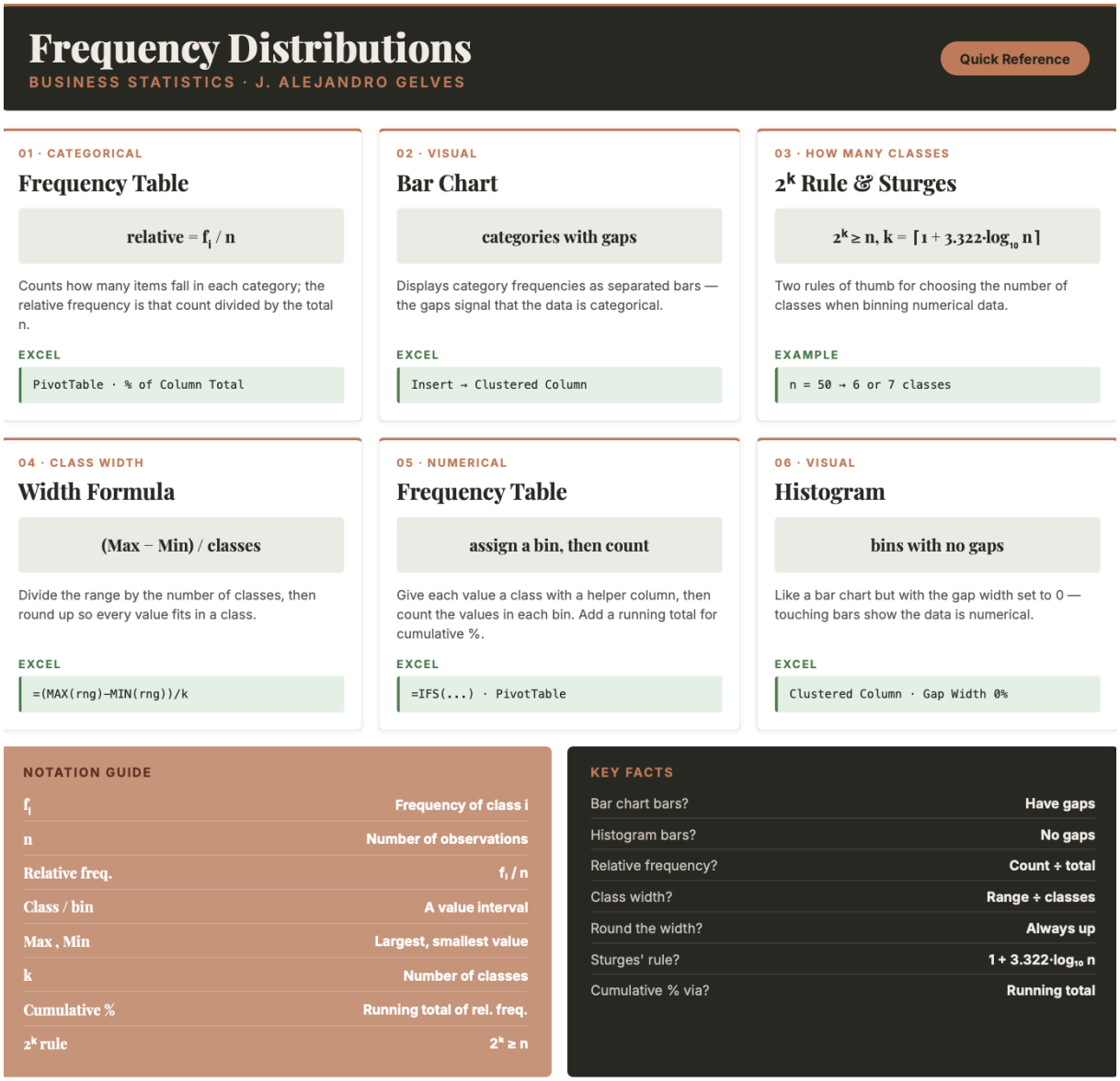
Bin	Frequency	Cumulative %
1. [20,120)	9	30.00%
2. [120,220)	12	70.00%
3. [220,320)	5	86.67%
4. [320,420)	3	96.67%
5. [420,520)	1	100.00%

5. Insert a Histogram: With the pivot table selected, go to `Insert → Charts → Clustered Column`. Excel will generate a bar chart. To convert it into a proper histogram, right-click on any bar and select `Format Data Series`. Set the `Gap Width` to `0%`. This removes the spaces between bars, which is the key visual distinction between a bar chart (used for categorical data) and a histogram (used for numerical data).

The video below walks you through each of these steps:

Video walkthrough: watch it in the online version of this chapter

2.5 Chapter Summary Cheat Sheet



2.6 Exercises

The following exercises will help you practice summarizing data with tables and simple graphs. In particular, the exercises work on:

- Developing frequency distributions for both categorical and numerical data.
- Constructing bar charts, histograms, and line charts.
- Creating contingency tables.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

You will need the **Brain Cancer** data set to answer this question. You can download it here:

Download Data

1. Construct a frequency and relative frequency table of the *Diagnosis* variable. What was the most common diagnosis? What percentage of the sample had this diagnosis?

Answer on page 167 →

2. Construct a bar chart. Summarize the findings.

Answer on page 167 →

3. Construct a contingency table that shows the *Diagnosis* along with the *Status*. Which diagnosis had the highest number of non-survivals (0)? What was the survival rate of this diagnosis?

Answer on page 168 →

Exercise 2

You will need the **Air Quality** data set to answer this question. You can download it here:

Download Data

1. Construct a frequency distribution for the *Temp* variable. Use five classes with widths of $50 < x \leq 60$; $60 < x \leq 70$; etc. Which interval had the highest frequency? How many times was the temperature between 50 and 60 degrees?

Answer on page 168 →

2. Construct a relative frequency, cumulative frequency and the relative cumulative frequency distributions. What proportion of the time was *Temp* between 50 and 60 degrees? How many times was the *Temp* 70 degrees or less? What proportion of the time was *Temp* more than 70 degrees?

Answer on page 168 →

3. Construct the histogram. Is the distribution symmetric? If not, is it skewed to the left or right?

Answer on page 169 →

Exercise 3

You will need the **Portfolio** data set to answer this question. You can download it here:

Download Data

1. Construct a line chart that shows the returns over time for each portfolio (X and Y) by using two lines each with a unique color. Assume the data is for the period 1901 to 2000. Include also a legend that matches colors to portfolios.

Answer on page 169 →

Exercise 4

A company surveyed 40 employees about their primary department. The responses are saved in the Excel file below:

Download Data

1. Create a frequency distribution table for these departments.

Answer on page 169 →

2. Compute the relative frequencies (as percentages).

Answer on page 169 →

3. Create a bar plot of the frequencies. Which department has the highest frequency? What is its relative frequency (percentage)?

Answer on page 170 →

Exercise 5

A class of 50 students took a business statistics exam (out of 100 points). Their scores are recorded in the Excel file below:

Download Data

1. Construct a histogram using five classes with width 10, starting at 50 (bins: [50,60), [60,70), etc.). In which bin is the highest frequency? How many students scored in that range?

Answer on page 170 →

3 Descriptive Statistics III

Understanding where the "center" of a dataset lies is a fundamental step when interpreting and analyzing data. Measures of central location provide different ways to identify the "typical" value in a dataset, each with its unique strengths and limitations. Below I survey some of the most popular measures of central location. These summaries continue the work of describing our data: having seen its overall shape, we now pin down its center — a single representative value that will later anchor the estimates we carry into inference.

3.1 Sample Vs. Population Statistics

In this book you will find formulas for calculating sample statistics and population parameters. In general, you will use population formulas when you have data for the entire group you care about or when you are working with a theoretical distribution and the parameter is known; population formulas use N in denominators and estimate parameters. Use sample formulas when your data are a subset drawn from a larger population and you want unbiased estimators of the population parameters; sample formulas use n and the degrees of freedom correction. In practice most empirical work uses sample formulas because we rarely observe entire populations; below you will see both population and sample versions so you can choose the correct one depending on whether your data represent the full population or a sample.

3.2 The Mean

The **mean** is the average value for a numerical variable. It is a widely understood and straightforward measure to calculate. It incorporates all data points, providing a comprehensive representation of the data set. However, its reliance on every value also makes it sensitive to outliers or skewed distributions, which may cause it to not accurately reflect the true center of the data.

The sample statistic is estimated by:

THE MEAN (SAMPLE)

$$\bar{x} = \frac{\sum x_i}{n}$$

where Σ is the summation operator, x_i is observation i, and n is the number of observations in the sample.

The formula instructs you to add up all of the values in a variable and then divide by the sample size. If you wish to estimate the population parameter you can use:

THE MEAN (POPULATION)

$$\mu = \frac{\sum x_i}{N}$$

where N is the population size.

Let's consider a couple of example to highlight how the formula works.

EXAMPLE 1

Consider the following sample of numbers $x = \{1, 4, 2, 1\}$. The mean would be equal to $\bar{x} = \frac{1+4+2+1}{4}$ or $\bar{x} = 2$.

EXAMPLE 2

Consider the following sample of numbers $x = \{1, 4, 2, 1, 100\}$. The mean is $\bar{x} = 21.6$. Although most of the numbers are in the range $(1, 4)$, the 100 biases the mean to 21.6.

3.3 The Median

The **median** is the value in the middle when data is organized in ascending order. When n is even, the median is the average between the two middle values. The median is resistant to outliers, making it an ideal measure for skewed data or data sets with irregular distributions. It is particularly useful for ordinal data, where precise ranking is important. However, the median does not utilize all data points, which can lead to less precise comparisons compared to other measures like the mean.

EXAMPLE 1

Consider the following numbers $x = \{1, 4, 2, 1\}$. We first sort the data to obtain $x_{sorted} = \{1, 1, 2, 4\}$. Since the number of observations is even $(n = 4)$, the median is the average of the two middle numbers $(1, 2)$. $median_x = \frac{1+2}{2}$ or 1.5.

EXAMPLE 2

Consider the following numbers $x = \{1, 4, 2, 1, 100\}$. We once again sort the number obtaining $x_{sorted} = \{1, 1, 2, 4, 100\}$. Since $n = 5$ in this case we can identify the median as the third value in x_{sorted}. $median_x = 2$. Note that the inclusion of 100 in the data did not change much the measure of central location.

3.4 The Mode

The **mode** is the value with highest frequency from a set of observations. This measure is particularly useful for categorical data as it helps determine popularity of values. It can be applied to both numerical and non-numerical data sets. The mode has its limitations; it may not exist in cases where all values occur with equal frequency, and there may be multiple modes, which can complicate interpretation. Additionally, since the mode focuses only on the most frequent value, it does not account for other data points, limiting its overall utility as a comprehensive measure.

EXAMPLE 1

Consider the following numbers $x = \{1, 4, 2, 1\}$. Since 1 is repeated twice and all other numbers just repeated once, $x_{mode} = 1$.

EXAMPLE 2

Consider the following numbers $x = \{1, 4, 2, 1, 4\}$. Now 4 is also repeated twice. The variable has two modes $x_{mode} = \{1, 2\}$. x is said to be bimodal.

3.5 The Weighted Mean

The weighted mean is useful in scenarios where some data points are more significant than others, such as in financial portfolios, grade point averages, or survey results, as it accounts for variability in importance across observations. This measure requires additional information in the form of weights (w_i), which may not always be available or accurate. It is calculated the sum product of values (x_i) and weights (w_i) and then dividing by the sum of weights. Mathematically, the weighted average is:

THE WEIGHTED MEAN

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

where w_i represents the weight of the ith observation.

Let's consider once more a couple of examples.

EXAMPLE 1

Consider three different stocks $S = \{T, C, X\}$ with stock returns of $R = \{2, 4, 10\}$. Each stock has a weight in the portfolio of $W = \{0.3, 0.2, 0.5\}$. The average return of the portfolio is $\bar{x}_{weighted} = \frac{0.6+0.8+5}{1}$ or $\bar{x}_{weighted} = 6.4$.

EXAMPLE 2

A company evaluates job candidates using three criteria: technical skills, communication, and teamwork. The criteria and their weights are shown in the table below:

Criteria	Weight	Score
Technical Skills	0.50	88
Communication	0.30	76
Teamwork	0.20	82

The weighted mean score for the candidate is:

$$\bar{x}_w = (0.50 \times 88) + (0.30 \times 76) + (0.20 \times 82)$$

$$= 44 + 22.8 + 16.4 = 83.2$$

The candidate's overall score is 83.2 out of 100. Note that a simple average would have given $(88 + 76 + 82)/3 = 82$, a slightly different result. The weighted mean reflects the company's priorities by giving more importance to technical skills.

3.6 The Geometric Mean

The **geometric mean** is a multiplicative average that is less sensitive to outliers relative to the arithmetic mean. It is particularly useful when averaging rates or levels that compound over time. The formula depends on whether the data represents rates or levels.

When working with rates, we add 1 to each value because rates represent changes relative to a base value. The geometric mean is given by:

THE GEOMETRIC MEAN (RATES)

$$\bar{x}_g = \sqrt[n]{(1 + r_1) \times (1 + r_2) \times \ldots \times (1 + r_n)} - 1$$

where r_i are rates of change and n is the number of periods.

When working with levels, the values already reflect standalone quantities rather than relative changes, so no adjustment is needed. The geometric mean simplifies to:

THE GEOMETRIC MEAN (LEVELS)

$$\bar{x}_g = \sqrt[n]{x_1 \times x_2 \times ... \times x_n}$$

where x_i are positive values such as prices, proportions, or index values.

To illustrate the distinction, we work through two examples below.

EXAMPLE 1

A company tracks its market share across four regions: $x = \{0.20, 0.30, 0.10, 0.10\}$. Since market share is a proportion of the whole — a level, not a rate of change — we apply the levels formula directly:

$$\bar{x}_g = \sqrt[4]{0.20 \times 0.30 \times 0.10 \times 0.10}$$
$$= \sqrt[4]{0.0006} = 0.156$$

The geometric mean market share across the four regions is 15.6%.

EXAMPLE 2

An investment fund reports the following annual returns over four years:

Year	Return	Growth Factor
2020	10%	1.10
2021	25%	1.25
2022	-8%	0.92
2023	15%	1.15

Since these are rates of return, we add 1 to each value to obtain the growth factor before applying the formula. The geometric mean is:

$$\bar{x}_g = \sqrt[4]{1.10 \times 1.25 \times 0.92 \times 1.15} - 1$$
$$= \sqrt[4]{1.4539} - 1 = 0.0977$$

The average annual return is 9.77%. Note that the arithmetic mean gives $(10 + 25 - 8 + 15)/4 = 10.5\%$, which overstates the true average. The geometric mean is the appropriate measure here because it accounts for the compounding effect of returns over time.

Date	People	Approval
01/01/24	50	25%
02/01/24	100	25%
03/01/24	30	70%
04/01/24	250	85%

3.7 Measures of Central Location in Excel

Excel has a collection of functions that calculate measures of central location. Let's consider the following data on approval ratings:

You can also download the data here:

Download Data

Mean: To calculate the mean in Excel use `=AVERAGE(C2:C5)`, where column C contains the approval ratings. The mean approval rating is 0.5125.

Median: To calculate the median in Excel use `=MEDIAN(C2:C5)`. The median approval rating is 0.475.

Mode: To calculate the mode in Excel use `=MODE(C2:C5)`. Note that `MODE` only returns one value even if there are multiple modes. For the approval data, the mode is 0.25.

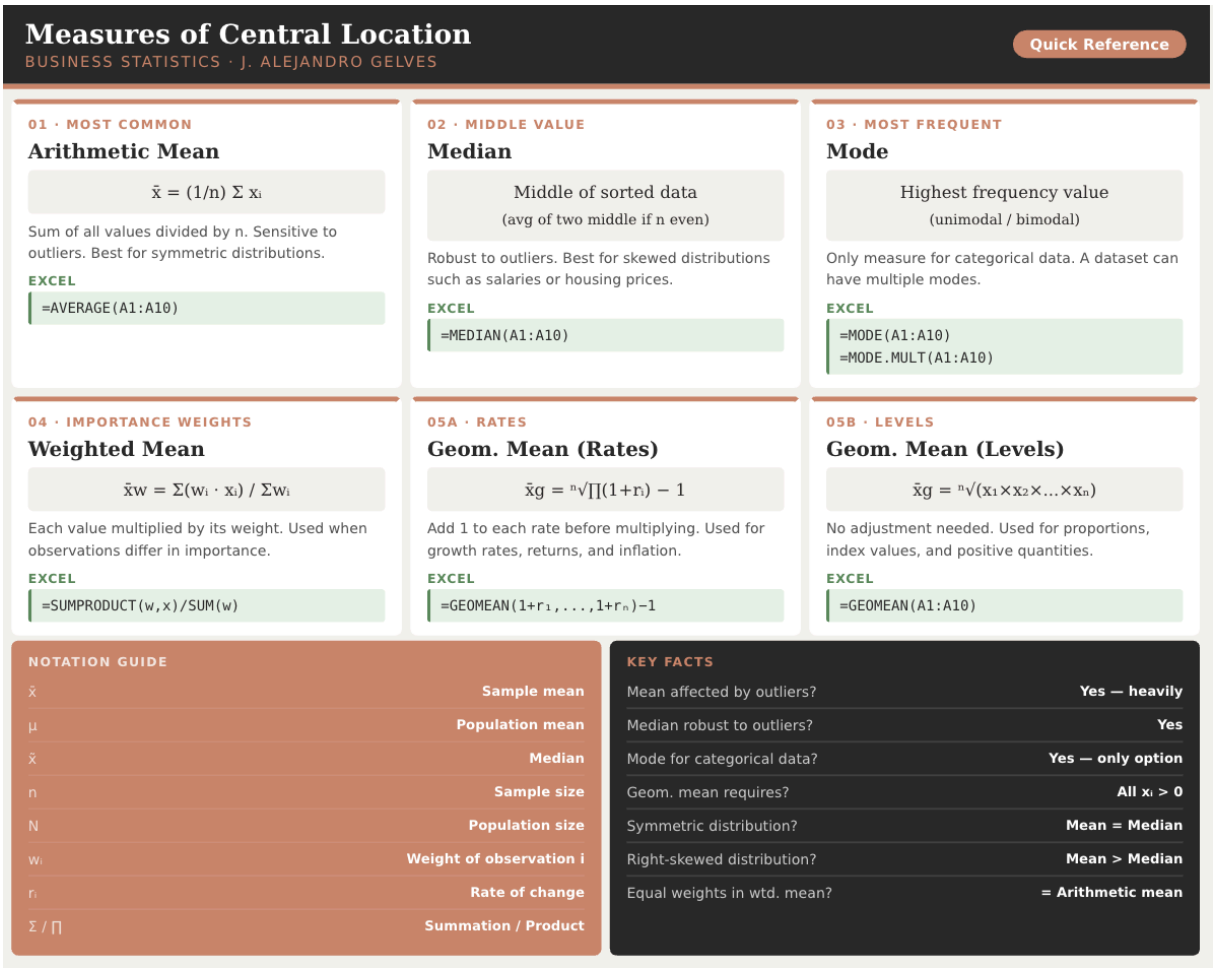
Weighted Mean: To calculate the weighted mean in Excel use `=SUMPRODUCT(B2:B5, C2:C5) / SUM(B2:B5)`, where column B contains the number of people surveyed and column C contains the approval ratings. This formula multiplies each approval rating by its corresponding weight and divides by the total number of people. The weighted mean approval rating is 0.6302326.

Geometric Mean: To calculate the geometric mean in Excel use `=GEOMEAN(C2:C5)`. Note that since approval ratings are already proportions there is no need to add 1 to each value before applying the formula. The geometric mean approval rating is 0.4391.

3.8 Excel Function Summary

- `=AVERAGE()` calculates the mean for a range of numbers.
- `=MEDIAN()` calculates the median for a range of numbers.
- `=MODE()` returns the most frequently occurring value in a range.
- `=SUMPRODUCT()` multiplies corresponding values in two ranges and returns their sum, useful for calculating the weighted mean.
- `=GEOMEAN()` calculates the geometric mean for a range of numbers.

3.9 Chapter Summary Cheat Sheet



3.10 Exercises

The following exercises will help you practice the measures of central location. In particular, the exercises work on:

- Calculating the mean, median, and the mode.
- Calculating the weighted average.
- Applying the geometric mean for growth rates and returns.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using Excel functions when possible.

1. Use the following observations to calculate the mean, the median, and the mode.

8	10	9	12	12

Answer on page 171 →

2. Use following observations to calculate the mean, the median, and the mode.

-4	0	-6	1	-3	-4

Answer on page 171 →

3. Use the following observations, calculate the mean, the median, and the mode.

20	15	25	20	10	15	25	20	15

Answer on page 171 →

Exercise 2

You will need the **OJ** data set to answer this question. You can get the dataset here:

Download Data

1. Find the mean price for Country Hill (*PriceCH*) and Minute Maid (*PriceMM*).

Answer on page 171 →

2. Find the mean price of Country Hill (*PriceCH*) at each store (*StoreID*). Which store provides the better price?

Answer on page 171 →

3. Find the median price paid by Country Hill (*PriceCH*) purchasers (*Purchase*) in all stores? Which store had the better median price?

Answer on page 172 →

Exercise 3

1. Over the past year an investor bought TSLA. She made these purchases on three occasions at the prices shown in the table below. Calculate the average price per share.

Date	Price Per Share	Number of Shares
February	250.34	80
April	234.59	120
Aug	270.45	50

Answer on page 172 →

2. What would have been the average price per share if the investor would have bought equal amounts of shares each month?

Answer on page 173 →

Exercise 4

1. Consider the following observations for the consumer price index (CPI). Calculate the inflation rate (Growth Rate of the CPI) for each period.

1	2	3	4	5
1.0	1.3	1.6	1.8	2.1

Answer on page 173 →

2. What is the average growth rate for the inflation rate?

Answer on page 173 →

Exercise 5

Suppose that you want to invest $1000 dollars in a stock that is predicted to yield the following returns in the next four years. Calculate both the arithmetic mean and the geometric mean. Use the geometric mean to estimate how much money you would have by the end of year 4.

Year	Annual Return
1	17.3
2	19.6
3	6.8
4	8.2

Answer on page 173 →

Exercise 6

You will need the **Hitters** dataset containing data on Major League Baseball players, including their salaries (in thousands of dollars) from 1987. You can get the dataset here:

Download Data

1. Compute the overall mean and median salary for all players. Which measure (mean or median) better represents the typical salary?

Answer on page 173 →

2. filter to only players with at least one home run (HmRun >= 1) and compute the mean and median salary for this subgroup.

Answer on page 173 →

4 Descriptive Stats IV

Measures of dispersion provide insights into how much the values in a data set deviate from the central tendency. In a business context, understanding variability is essential for assessing risk, as it helps decision-makers anticipate fluctuations in revenue, costs, and market conditions. Below, we explore the main statistics that help us quantify variability and use them to make better business decisions. Variability is central to everything that follows: it is precisely because data vary that a sample gives only an imperfect view of the world, so measuring that variation here lays the groundwork for the probability and inference chapters to come.

4.1 The Range

This is the simplest measure of dispersion, defined as the difference between the largest and smallest values in a variable. While straightforward, the range only accounts for the extremes and does not reflect the variability within the variable. The **range** is calculated by:

THE RANGE

$$Range = Maximum - Minimum$$

EXAMPLE

Consider the data $x = \{480, 1050, 1400, 2500, 3200\}$. The range is given by $Range = 3200 - 480 = 2720$.

4.2 The Variance

The variance gives us a more comprehensive look at dispersion by considering how each data point deviates from the mean. It summarizes the squared deviations from the mean by finding their average.The **variance population parameter** is given by:

THE VARIANCE (POPULATION)

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

where μ is the population mean and N is the population size.

The **variance sample statistic** is given by:

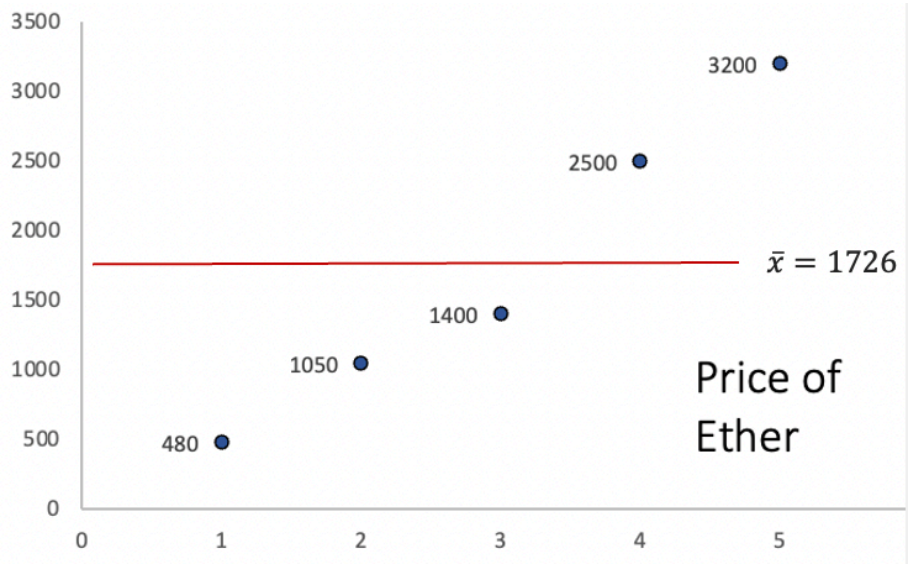
THE VARIANCE (SAMPLE)

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

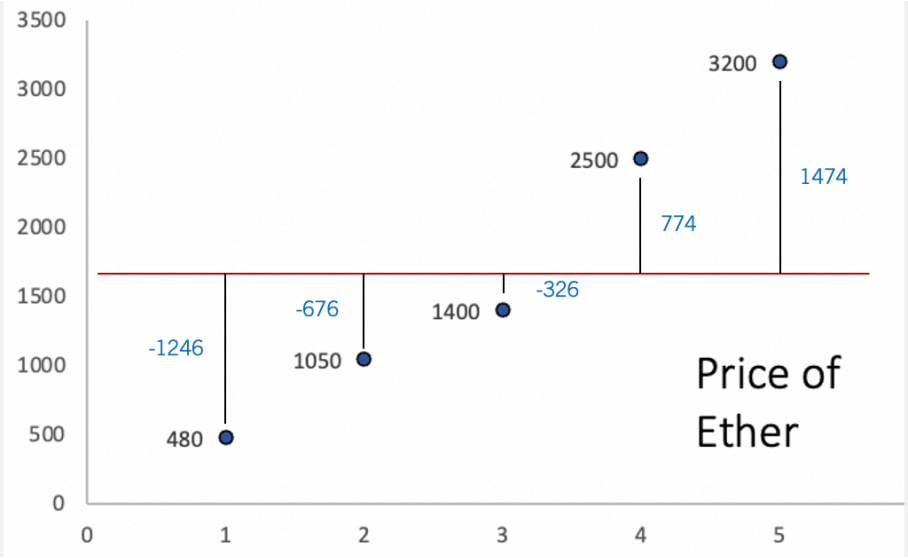
where $\bar{x}$ is the sample mean and n is the sample size.

Variance Intuition

Let's consider a sample of the price of Ether, the native cryptocurrency of the Ethereum blockchain. Ether is one of the largest cryptocurrencies by market capitalization and is widely used to pay for transactions and services on the Ethereum network. Unlike traditional assets such as stocks or bonds, cryptocurrencies like Ether are not backed by earnings or government guarantees, which makes their prices highly sensitive to speculation, regulatory news, and shifts in investor sentiment. As a result, Ether is known for dramatic price swings — sometimes gaining or losing double-digit percentages within a single day — making it an ideal case study for measures of dispersion. Below is a graph of the prices:



The y axis represents the price, while the x axis represents the period of time. The red line is the average price of Ether. The way the variance calculates dispersion, is by first finding the distance between each point and the average. The image below illustrates these distances:



Ideally, our measure of dispersion would simply average the distances from each data point to the mean, giving us a clear indication of how much the data varies around this central point. Since the mean is a measure that balances the distances from the mean, the sum of all deviations is equal to zero. This "balancing" effect is visually apparent in the graph, where positive deviations are exactly offset by negative ones. To address this issue, the variance squares each deviation from the mean before finding the average.

Variance Step By Step Calculation

Consider the data $x = \{480, 1050, 1400, 2500, 3200\}$. Treating this as a sample, let's calculate the variance step by step.

Step 1 — Calculate the sample mean:

$$\bar{x} = \frac{480 + 1050 + 1400 + 2500 + 3200}{5} = \frac{8630}{5} = 1726$$

Step 2 — Calculate the deviation of each observation from the mean:

x_i	$x_i - \bar{x}$
480	$480 - 1726 = -1246$
1050	$1050 - 1726 = -676$
1400	$1400 - 1726 = -326$
2500	$2500 - 1726 = 774$
3200	$3200 - 1726 = 1474$

Step 3 — Square each deviation:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
480	−1246	1,552,516
1050	−676	456,976
1400	−326	106,276
2500	774	599,076
3200	1474	2,172,676

Step 4 — Sum the squared deviations:

$$\sum (x_i - \bar{x})^2 = 1{,}552{,}516 + 456{,}976 + 106{,}276 + 599{,}076 + 2{,}172{,}676 = 4{,}887{,}520$$

Step 5 — Divide by $n - 1$:

$$s^2 = \frac{4{,}887{,}520}{5 - 1} = \frac{4{,}887{,}520}{4} = 1{,}221{,}880$$

The sample variance is $s^2 = 1{,}221{,}880$. Note that the variance is expressed in squared units of the original data, which makes it difficult to interpret directly. This motivates the use of the standard deviation, which brings the measure back to the original units.

4.3 The Standard Deviation

The standard deviation is derived by taking the square root of the variance. Remember, the variance employs squared deviations to eliminate negative values and calculate spread. By taking the square root, the standard deviation reverts the measure of dispersion back to the original units of the variable. This transformation makes the standard deviation more intuitive; it directly quantifies how much each data point deviates from the mean in the same units as the variable itself. Consequently, the standard deviation is a clear and intuitive measure of variability.

To find the **population standard deviation**, take the square root of the variance. For the population parameter use:

THE STANDARD DEVIATION (POPULATION)

$$\sigma = \sqrt{\sigma^2}$$

For the **sample standard deviation** use:

THE STANDARD DEVIATION (SAMPLE)

$$s = \sqrt{s^2}$$

EXAMPLE

Consider once more the price of Ether. That is. $x = \{480, 1050, 1400, 2500, 3200\}$. In the previous section we found that the variance $s^2 = 1221880$. Using the square root we find that $s = \sqrt{1221880} = 1105.387$. If the price of Ether is in dollars, then, the price varies from the mean 1105.39 dollars on average.

4.4 Mean Absolute Deviation

Another measure of data variability is the Mean Absolute Deviation (MAD), which calculates variability using absolute deviations from the mean. This approach not only keeps the measure in the same units as the data but also makes it less sensitive to large deviations. Practically, the **Mean Absolute Deviation** measures the average deviation from the mean, by using absolute deviations. It is calculated by:

THE MEAN ABSOLUTE DEVIATION (POPULATION)

$$MAD = \frac{\sum |x_i - \mu|}{N}$$

where the | | operator indicates absolute value.

You can estimates the mad for a sample with:

THE MEAN ABSOLUTE DEVIATION (SAMPLE)

$$mad = \frac{\sum |x_i - \bar{x}|}{n}$$

Example: Let's consider once more the price of Ether. Recall, that if we calculate the deviations from the mean we obtain the second column in the table below:

Price	Deviations $(x_i - \bar{x})$	Squared Deviations $(x_i - \bar{x})^2$	Absolute Deviations $\lvert x_i - \bar{x} \rvert$
480	-1246	1552516	1246
1050	-676	456976	676
1400	-326	106276	326
2500	774	599076	774
3200	1474	2172676	1474

For the MAD, focus on the fourth column which lists the absolute deviations from the mean. Averaging these gives us the Mean Absolute Deviation (MAD), which equals $mad = 899.2$. This result states that, on average, each data point is roughly 900 dollars away from the mean. Note that the standard deviation is higher (988.73) since the squaring of deviations disproportionately amplifies larger ones, pushing the standard deviation above the MAD.

4.5 Coefficient of Variation

The **Coefficient of Variation** simplifies comparisons of variability across variables with different units or scales, by dividing the standard deviation by the mean. It is calculated by:

THE COEFFICIENT OF VARIATION (CV)

$$CV = \frac{s}{\bar{x}} \times 100$$

EXAMPLE

Consider the table below, that shows information on two different stocks:

Stock	Average Price	s
A	1	1
B	100	1

The third column shows the standard deviation of each stock. One could conclude that both stocks vary the same as they have the same standard deviation of one. Recall that the coefficient of variation considers the variable's scale by incorporating the mean into its calculation. Since one stock is centered around 1 dollar and the other around 100 dollars (second column), it is clear that they do not vary similarly percentage-wise. Stock A varies 100% from the mean, whereas Stock B only varies 1%. Hence, the coefficient of variation would identify Stock A as the more variable stock.

4.6 The Sharpe Ratio

The Sharpe Ratio measures how much excess return an investor receives for the extra volatility (risk) taken on beyond the risk-free rate. It essentially uses the same principle of normalization as the CV but adds the dimension of risk-adjusted performance. If the CV tells you the variability of returns in relation to their mean, the Sharpe Ratio tells you if that variability is worth it by comparing it against what you could safely earn without risk. In essence, while the CV highlights variability, the Sharpe Ratio motivates investment choices by rewarding higher returns per unit of risk taken.

In sum, the **Sharpe ratio** quantifies the excess return of an investment over the risk free return. It is calculated by:

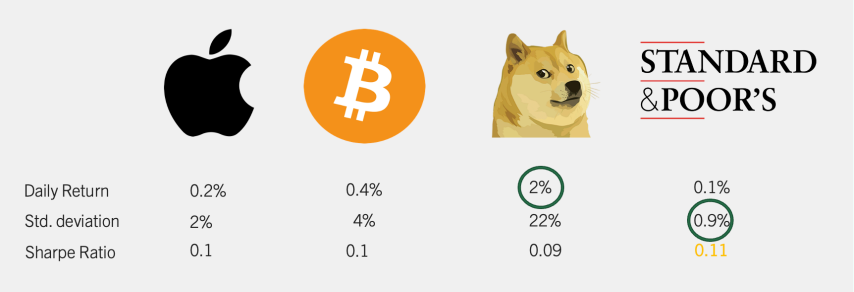
THE SHARPE RATIO

$$\frac{\bar{R_p} - R_f}{s}$$

where $\bar{R_p}$ is the mean return of the portfolio, R_f is the risk free return, and s is the standard deviation.

EXAMPLE

Consider the table below that includes a collection of investments.



The table shows four investments (Apple, Bitcoin, Shiba, and S&P). Just looking at the daily return, it is clear that Shiba provides the best returns. However, the cost for that high return is reflected in variability (22% for Shiba). The S&P is clearly the safest investment with a standard deviation of only 0.9%. The Sharpe Ratio, marries these two metrics showing the return per unit of risk taken over the free rate. If we assume a 0% risk free rate, the investment with the highest Sharpe Ratio is the S&P at 0.11.

4.7 Measures of Dispersion in Excel

Excel has a collection of functions that calculate measures of dispersion. Let's use daily stock return data for three assets — Apple (AAPL), the S&P 500 ETF (SPY), and Tesla (TSLA) — to explore these functions. The data contains daily returns for each stock from January 2020 to July 2021.

Download Data

Count: We begin by finding the number of observations for each stock using `=COUNTIF()`. Assuming the stock names are in column B and returns in column C, the number of observations for AAPL is:

Stock	N
AAPL	394
SPY	394
TSLA	394

Stock	Avg_Return
AAPL	0.1401%
SPY	0.0670%
TSLA	0.3837%

```
=COUNTIF(B:B, "AAPL")
```

The number of daily return observations for each stock is:

Average Return: To calculate the average return we use the geometric mean, since returns compound over time. Using `=GEOMEAN()` with the adjustment for rates and the particular stocks:

```
=GEOMEAN(IF(B2:B1189="AAPL", 1+C2:C1189)) - 1
```

The average daily return for each stock is:

Minimum and Maximum: To find the minimum and maximum return for each stock use `=MINIFS()` and `=MAXIFS()`. Assuming stock names are in column B and returns in column C:

```
=MINIFS(C:C, B:B, "AAPL")
=MAXIFS(C:C, B:B, "AAPL")
```

The minimum and maximum daily returns for each stock are:

Note that TSLA shows the widest range, with a minimum daily return of around -23% and a maximum of around $+18\%$, reflecting its high volatility relative to AAPL and SPY.

Variance and Standard Deviation: To calculate the variance and standard deviation use `=VARPA()` and `=STDEV.S()`. Note that Excel does not have a `=STDEVIF()` function. To calculate a conditional standard deviation use an array formula:

Stock	Min	Max
AAPL	-13.77%	11.32%
SPY	-11.59%	8.67%
TSLA	-23.65%	18.14%

Stock	Variance	Std_Dev
AAPL	0.0659%	2.5667%
SPY	0.0313%	1.7694%
TSLA	0.2524%	5.0239%

Stock	CV	Sharpe
AAPL	14.8284	0.0613
SPY	21.3757	0.0378
TSLA	9.8285	0.0986

```
=VAR(IF(B2:B1189="AAPL", C2:C1189))
=STDEV(IF(B2:B1189="AAPL", C2:C1189))
```

The variance and standard deviation for each stock are:

Coefficient of Variation: The coefficient of variation (CV) standardizes dispersion relative to the mean, allowing comparison across assets. In Excel, divide the standard deviation by the average return:

```
=STDEV(IF(B2:B1189="AAPL",C2:C1189)) / AVERAGE(IF(B2:B1189="AAPL",C2:C1189))
```

Sharpe Ratio: The Sharpe ratio measures the return earned per unit of risk, adjusting for the risk-free rate. Assuming an annual risk-free rate of 4%, the daily risk-free rate is $r_f = 0.04/252$. In Excel:

```
=(AVERAGE(IF(B2:B1189="AAPL",C2:C1189)) - (0.04/252)) /
  STDEV(IF(B2:B1189="AAPL",C2:C1189))
```

The CV and Sharpe ratio for each stock are:

A higher Sharpe ratio indicates better risk-adjusted performance. SPY offers the most consistent return per unit of risk, while TSLA has the highest volatility and the most variable Sharpe ratio — consistent with its reputation as a high-risk, high-reward asset.

4.8 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=COUNTIF(range, criteria)` counts the number of cells in a range that meet a given condition. Used here to count the number of observations for each stock.
- `=MINIFS(min_range, criteria_range, criteria)` returns the minimum value in a range filtered by one or more conditions.
- `=MAXIFS(max_range, criteria_range, criteria)` returns the maximum value in a range filtered by one or more conditions.
- `=VAR.S(range)` calculates the sample variance for a range of numbers.
- `=STDEV.S(range)` calculates the sample standard deviation for a range of numbers. It is the square root of the sample variance.
- `=VAR(IF(criteria_range="condition", value_range))` calculates a conditional variance. Unlike `=AVERAGEIF()`, Excel does not have a native `=VARIF()` function.
- `=STDEV(IF(criteria_range="condition", value_range))` calculates a conditional standard deviation. Like variance, there is no native `=STDEVIF()` in Excel.

4.9 Chapter Summary Cheat Sheet



4.10 Exercises

The following exercises will help you practice the measures of dispersion. In particular, the exercises work on:

- Calculating the range, MAD, variance, and the standard deviation.

- Using R to calculate measures of dispersion.
- Calculating and using the Sharpe ratio to select investments.

Answers are in the Answers to Exercises at the end of the book. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using R functions when possible. Make sure to calculate the deviations from the mean.

1. Use the following observations to calculate the Range, MAD, Variance and Standard Deviation. Assume that the data below is the entire population.

70	68	4	98

Answer on page 174 →

2. Use the following observations to calculate the Range, MAD, Variance and Standard Deviation. Assume that the data below is a sample from the population.

-4	0	-6	1	-3	0

Answer on page 174 →

3. Use the following observations to calculate the Range, MAD, Variance and Standard Deviation. Assume that the data below is a sample from the population.

2	8	4	10	6	6

Answer on page 174 →

Exercise 2

You will need the **Stocks** data set to answer this question. You can find this data here:

Download Data

The data is a sample of daily stock prices for ticker symbols TSLA (Tesla), VTI (S&P 500) and GBTC (Bitcoin).

1. Calculate the standard deviations for each stock. Which stock had the lowest standard deviation?

Answer on page 175 →

2. Calculate the MAD. Does your answer in 1. remain the same?

Answer on page 175 →

3. Finally, calculate the coefficient of variation. Any changes to your conclusions?

Answer on page 175 →

Exercise 3

You will need the **Portfolio** data set to answer this question. The data has 100 records of the returns of two stocks. You can find the data here:

Download Data

1. Calculate the mean and standard deviation for each stock. Which investment has higher returns on average? Which investment is safest as measured by the standard deviation?

Answer on page 175 →

2. Use a Risk Free rate of return of 3.5% to calculate the Sharpe ratio for each stock. Which stock would you recommend?

Answer on page 175 →

Exercise 4

The monthly returns (in percent) for a small project can be found in the Excel sheet below:

Download Data

Compute the sample mean, sample variance, sample standard deviation, coefficient of variation (CV) (in percent), and the mean absolute deviation (MAD). Briefly interpret the CV: what does it tell you about the returns relative to their mean?

Answer on page 176 →

Exercise 5

Prediction markets allow participants to bet on the outcomes of sporting events. A bettor tracks his weekly profit and loss (in dollars) over ten weeks of NFL betting in the spreadsheet below:

Download Data

1. Compute the sample mean, sample variance, sample standard deviation, and MAD. Is the bettor profitable on average?

Answer on page 176 →

2. Compute the Coefficient of Variation (CV). What does it tell you about the reliability of the bettor's weekly results?

Answer on page 177 →

3. Assuming a risk-free weekly return of $0, compute the Sharpe Ratio. Would you consider this a good risk-adjusted return?

Answer on page 177 →

5 Descriptive Stats V

There are statistical measures that describe the shape and distribution of the data beyond measures of central location or dispersion. They provide a view of how data is spread out, where it concentrates, and how it deviates from what might be expected. The tools shown below will help you describe the data's shape, symmetry, and anomalies. With shape, center, and spread all in hand, our description of the data is complete — the foundation on which the coming chapters build models of uncertainty and draw conclusions about the world.

5.1 Quantiles and Percentiles

A **quantile** is a location within a set of ranked numbers (or distribution), below which a certain proportion, q, of that set lie. If we instead express quantiles as a percentage, they are referred to as **percentiles**.

EXAMPLE

Imagine all your data points lined up from smallest to largest. If you say you're looking at the 25th percentile, it means you're finding the value below which 25% of your data falls. If you had 100 test scores, the 25th percentile would be the score where 25 students scored lower than that, and 75 scored higher or equal. It's a way to see where a value stands in relation to the rest of the data in terms of percentage.*

To calculate a percentile we follow the steps below:

1. Sort the data in ascending order. Each point has now a location. The smallest number will be first, the second smallest number will be second, etc.
2. Compute the location of the percentile desired using:

LOCATION OF THE PTH PERCENTILE

$$L_p = 1 + k(n - 1)$$

where L_p\$ is the location (in the sorted data) of the P_{th} percentile, and P is the percentile desired.

3. The data point at location L_p, is the the P_{th} percentile.

EXAMPLE

Let's use the IQ scores for a group of students to find the 25th percentile. $IQ = \{80, 100, 110, 75, 130, 90\}$.

1. We sort the data:

$$IQ_{sorted} = \{75, 80, 90, 100, 110, 130\}$$

2. Find the location of the 25th percentile:

$$L_{25} = 1 + 0.25(6 + 1) = 2.25$$

   The 25th percentile is in the position 2.25 of the sorted data.

3. Retrieve the 25th percentile: Since position 2 is 80 and position 3 is 90, the 25th percentile lies 0.25 of the way between positions 2 and 3. Hence, the 25th percentile is:

$$P_{25} = 80 + 0.25(90 - 80) = 82.5$$

.

5.2 Chebyshev's Theorem

Chebyshev's Theorem is an important theorem, as it helps you form an expectation of the proportion of data that must lie between a given standard deviation from the mean. This offers a baseline to understanding the range and distribution of your data, and aids in detecting outliers. Formally, **Chebyshev's Theorem** states that regardless of the shape of the distribution, at least $(1 - 1/z^2)\%$ of the data lies between z standard deviations from the mean.

CHEBYSHEV'S THEOREM

$$1 - \frac{1}{k^2}$$

where k is the number of standard deviations from the mean and $k > 1$.

EXAMPLE 1

For a given data set, we want to know at least how much of the data lies within two standard deviations of the mean. Substituting $k = 2$ into Chebyshev's formula yields

$1 - \frac{1}{2^2} = 1 - \frac{1}{4} = 0.75$. Hence, at least 75% of the data lies within two standard deviations from the mean.

EXAMPLE 2

A financial analyst is reviewing the annual returns of a mutual fund over several years. The mean annual return is 8% with a standard deviation of 3%. Using Chebyshev's theorem with $k = 3$, at least $1 - 1/9 = 0.889$ or 88.9% of the annual returns lie within three standard deviations of the mean.

5.3 The Empirical Rule

Whereas Chebyshev's theorem holds for any data distribution, the empirical rule is a bit more precise when looking at "bell shaped" data. Formally, the **Empirical Rule** or (68,95,99.7 rule) states that 68%, 95%, and 99.7% of the data lies between 1, 2, and 3 standard deviations from the mean respectively. The rule requires that the data be bell shape (normally) distributed.

EXAMPLE

A company reports that customer satisfaction scores are are in good standing as they have a mean of $\mu = 78$ and a standard deviation of $\sigma = 6$. They claim that 95% of customers have satisfaction scores between 66 and 90. Note that this conclusion is only valid if the data is truly bell-shaped — a histogram of the scores should be inspected to verify this assumption before accepting the claim.

5.4 Outliers (Z-Scores)

Given the boundaries set by both the empirical rule and Chebyshev's theorem, we can classify points as being common (normal) and not common (outliers). Specifically, **outliers** are extreme deviations from the mean. They are values that are not "common" or rarely occurring. Since both the empirical rule and Chebyshev's theorem state that a large proportion of the data is between three standard deviations, it would be uncommon to have a data point that is more that three standard deviations away from the mean.

To identify outliers we use a **z-score**, which is a measure of distance from the mean in units of standard deviations. By definition, z-scores above 3 are suspected to be outliers.

THE Z-SCORE

$$z_i = \frac{x_i - \bar{x}}{s_x}$$

where x_i is the data point, $\bar{x}$ is the sample mean, and s_x is the sample standard deviation.

EXAMPLE

On Jan 22, 2006 Kobe Bryant scored 81 points against the Toronto Raptors. He had averaged 30 point per game with a standard deviation of 4 points. If we calculate the z-score we get:* $z_{81} = \frac{81-30}{4} = 12.5$. This mean that 81 is 12.5 standard deviations away from the mean, making this value extremely rare.

EXAMPLE

On March 10, 2026, Bam Adebayo of the Miami Heat scored 83 points against the Washington Wizards — the second highest single-game scoring performance in NBA history, surpassing Kobe Bryant's iconic 81-point game. Prior to this game, Adebayo had averaged approximately 20 points per game with a standard deviation of 5 points. Calculating the z-score:

$$z_{83} = \frac{83 - 20}{5} = 12.6$$

This means that Adebayo's 83-point performance was 12.6 standard deviations above his mean comparable in statistical terms to Kobe Bryant's 81-point game in 2006, which produced a z-score of 12.5. Both performances illustrate how z-scores allow us to compare exceptional events across different players and eras on a common scale.

5.5 Skew

A measurement of skew, identifies asymmetry in the distribution of data. If most of the data leans towards one side, it's skewed. If it leans to the left, it's left-skewed or negatively skewed, meaning the tail on the left side is longer. If it leans to the right, it's right-skewed or positively skewed, with a longer tail on the right. If the data is evenly distributed, it's not skewed at all, it's symmetric. To determine if the data is skewed, calculate the **Coefficient of Skew**.

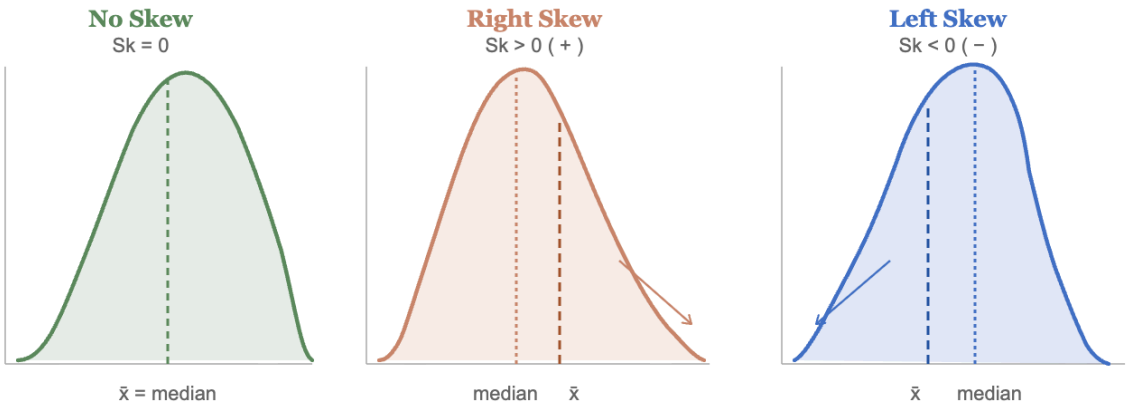
SKEWNESS

$$Sk = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

where n is the sample size, $\bar{x}$ is the sample mean, and s is the sample standard deviation.

$Sk < 0$	$Sk = 0$	$Sk > 0$
Left-skewed	Symmetric	Right-skewed

The image below shows the different types of skew:



The intuition behind the skewness formula lies in the cubed z-scores. Recall that a z-score measures how far each observation is from the mean in standard deviation units. Cubing these z-scores does two things: it preserves the sign and it amplifies large deviations, giving outsized influence to observations in the tails. If the right tail is longer, a few large positive cubed z-scores dominate the sum, producing a positive skewness. If the left tail is longer, large negative cubed z-scores dominate, producing a negative result. A symmetric distribution produces cubed z-scores that cancel out, yielding a skewness of zero.

EXAMPLE

Consider the IQ scores $IQ = \{75, 80, 90, 100, 110, 130\}$ with $n = 6$, $\bar{x} = 97.5$, and $s = 20.43$.

x_i	$\frac{x_i - \bar{x}}{s}$	$\left(\frac{x_i - \bar{x}}{s}\right)^3$
75	−1.101	−1.335
80	−0.857	−0.628
90	−0.367	−0.050
100	0.122	0.002
110	0.612	0.229
130	1.591	4.024

$$Sk = \frac{6}{(5)(4)} \times (2.242) = \frac{6}{20} \times 2.242 = 0.673$$

Since $Sk > 0$ the distribution is right-skewed, which makes intuitive sense — most IQ scores cluster below the mean while the score of 130 pulls the tail to the right.

5.6 Kurtosis

Kurtosis measures the "tailedness" of a distribution — that is, how much of the data is concentrated in the tails relative to a normal distribution. A distribution with heavy tails has more extreme values than expected, while a distribution with light tails has fewer. To measure kurtosis, we use the **Excess Kurtosis** formula, which centers the measure around zero by subtracting 3.

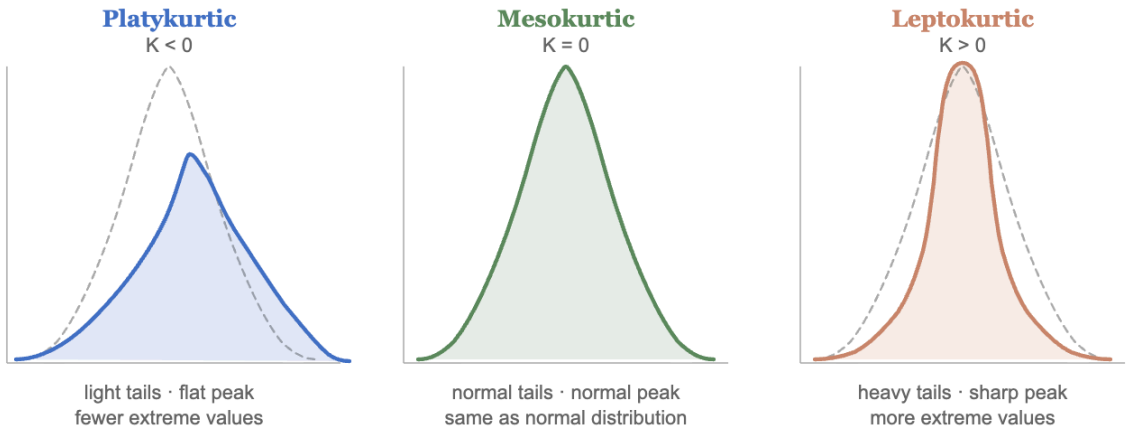
EXCESS KURTOSIS

$$K = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

where n is the sample size, $\bar{x}$ is the sample mean, and s is the sample standard deviation.

$K < 0$	$K = 0$	$K > 0$
Platykurtic (light tails)	Mesokurtic	Leptokurtic (heavy tails)

Below you can see a graph illustrating the different types of kurtosis:



EXAMPLE

Consider once more the IQ scores $IQ = \{75, 80, 90, 100, 110, 130\}$ with $n = 6$, $\bar{x} = 97.5$, and $s = 20.43$.

x_i	$\frac{x_i - \bar{x}}{s}$	$\left(\frac{x_i - \bar{x}}{s}\right)^4$
75	−1.101	1.470
80	−0.857	0.538
90	−0.367	0.018
100	0.122	0.000
110	0.612	0.140
130	1.591	6.401

Applying Excel's bias-corrected formula with $n = 6$:

$$K = \frac{6(7)}{(5)(4)(3)}(8.567) - \frac{3(5)^2}{(4)(3)} = 5.997 - 6.25 = -0.253$$

Since $K < 0$ the distribution is platykurtic — the tails are lighter than a normal distribution.

A distribution is **leptokurtic** ($K > 0$) when it has heavier tails and a sharper peak than a normal distribution — extreme values are more common. A distribution is **platykurtic** ($K < 0$) when it has lighter tails and a flatter peak — extreme values are rare. When $K = 0$ the distribution has the same tail behavior as a normal distribution and is called **mesokurtic**.

EXAMPLE

A risk analyst examines the daily returns of two assets. Asset A has an excess kurtosis of $K = 3.2$ and Asset B has an excess kurtosis of $K = -0.8$. Since $K > 0$, Asset A is leptokurtic — its returns have heavier tails than a normal distribution, meaning extreme gains or losses occur more frequently than expected. Since $K < 0$, Asset B is platykurtic — extreme returns are rare and the distribution is flatter. An investor concerned about tail risk would prefer Asset B, as large unexpected losses are less likely.

5.7 Five Point Summary

A popular way to summarize data is by calculating the minimum, first quartile, median, third quartile and maximum (five point summary). This gives us a good idea of how data is distributed. We can additionally inquire how the middle 50% of the data varies. Recall, that we can use a range to assess dispersion. The **interquartile range (IQR)** quantifies the dispersion of the middle 50% of the data. Formally, the IQR is the difference between the third quartile (75th percentile) and the first quartile (25th percentile).

THE INTERQUARTILE RANGE (IQR)

$$IQR = Q_3 - Q_1$$

where Q_1 is the first quartile (25th percentile) and Q_3 is the third quartile (75th percentile). The IQR measures the spread of the middle 50% of the data.

EXAMPLE

Let's use the IQ scores for a group of students once more. Recall that the data is given by $IQ = \{80, 100, 110, 75, 130, 90\}$. The minimum and the maximum are easily identified as $Max = 130$ and $Min = 75$. The first quartile (P_{25}) was calculated in 1.1 as 78.75. Using the same steps the third quartile (P_{75}) is 115. The median is the average between the third and fourth numbers $Median = 190/2 = 95$. The five point summary is given in the table below:

Min	P_{25}	$Median$	P_{75}	Max
75	78.75	95	115	130

To calculate the interquartile range we find the difference between the 75th and 25th percentiles. $IQR = 115 - 78.75 = 36.25$ which means that the middle 50% of the data has a range of 36.25.

5.8 Outliers (IQR)

An alternate way to identify outliers is by using the interquartile range. This measure and the z-score method can disagree. Any observation falling outside the following limits is considered an outlier:

OUTLIER LIMITS (IQR METHOD)

$$Lower\ Limit = Q_1 - 1.5 \times IQR$$

$$Upper\ Limit = Q_3 + 1.5 \times IQR$$

where Q_1 is the first quartile, Q_3 is the third quartile, and $IQR = Q_3 - Q_1$ is the interquartile range.

If an observation x_i falls below the lower limit or above the upper limit, it is considered an outlier.

eruptions	waiting
3.600	79
1.800	54
3.333	74
2.283	62
4.533	85

EXAMPLE

Consider once more the IQ data.

$$IQ = \{80, 100, 110, 75, 130, 90\}$$

The lower limit for on outlier is given by:

$$LL = Q_1 - 1.5(IQR) = 78.75 - 1.5(36.25) = 24.375$$

The upper limit is given by

$$UL = Q_3 + 1.5(IQR) = 115 + 1.5(36.25) = 169.375$$

Any data point outside the range [24.375,169.375] is considered an outlier. In other words, 200 and 20 would be outliers, but 100 would not.

5.9 Calculating Measures in Excel

Excel provides functions to calculate all the shape and distribution measures covered in this chapter. We will use the same `faithful` dataset from R, which contains data on eruptions of the Old Faithful geyser in Yellowstone National Park. The variable `eruptions` measures the duration of each eruption in minutes.

Assuming the `eruptions` variable is in column A of your Excel spreadsheet, the functions below will replicate the R output.

Outliers in Excel

To identify outliers using the z-score method, calculate the z-score for each observation using `=STANDARDIZE()`. In a new column, enter:

eruptions	waiting	z_eruptions
3.600	79	0.4515
1.800	54	-1.2064
3.333	74	0.2056
2.283	62	-0.7616
4.533	85	1.3109

Min	Q1	Median	Q3	Max	IQR
1.600	2.163	4.000	4.454	5.100	2.292

```
=STANDARDIZE(A2, AVERAGE(A:A), STDEV.S(A:A))
```

Copy the formula down for all observations. Any value greater than 3 or less than −3 in absolute value is a suspected outlier. To filter outliers automatically, use `=IF()`:

```
=IF(ABS(STANDARDIZE(A2, AVERAGE($A:$A), STDEV.S($A:$A))) > 3, "Outlier", "Normal")
```

The z-scores for the first five observations of the `faithful` dataset are:

As confirmed earlier, none of the observations are outliers as all z-scores are within three standard deviations of the mean.

Five Point Summary in Excel

To calculate the five point summary use `=QUARTILE.INC()`. The argument `quart` takes values 0 through 4:

```
=QUARTILE.INC(A:A, 0)   /* Minimum    */
=QUARTILE.INC(A:A, 1)   /* Q1 (P25)   */
=QUARTILE.INC(A:A, 2)   /* Median     */
=QUARTILE.INC(A:A, 3)   /* Q3 (P75)   */
=QUARTILE.INC(A:A, 4)   /* Maximum    */
```

The IQR can then be calculated as:

```
=QUARTILE.INC(A:A, 3) - QUARTILE.INC(A:A, 1)
```

The five point summary for the `eruptions` variable is:

Skewness	Kurtosis
-0.4135	-1.5116

Skewness and Kurtosis in Excel

To calculate skewness use `=SKEW()` and for excess kurtosis use `=KURT()`. Both functions use the bias-corrected formulas presented in this chapter:

```
=SKEW(A:A)    /* Skewness        */
=KURT(A:A)    /* Excess Kurtosis */
```

The skewness and kurtosis for the `eruptions` variable are:

The negative skewness confirms the mild left skew identified earlier in the chapter. The negative excess kurtosis indicates that the distribution is platykurtic (i.e., the eruption durations have lighter tails than a normal distribution), with fewer extreme values than expected.

5.10 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=PERCENTILE.INC(array, k)` returns the kth percentile of a range using the formula $L_p = 1 + k(n-1)$, where k is expressed as a decimal between 0 and 1. This matches the method used in this chapter.
- `=PERCENTILE.EXC(array, k)` returns the kth percentile using the formula $L_p = k(n+1)$.
- `=QUARTILE.INC(array, quart)` returns the five point summary. The argument `quart` takes values 0 (minimum), 1 (Q_1), 2 (median), 3 (Q_3), and 4 (maximum).
- `=STANDARDIZE(x, mean, std_dev)` calculates the z-score for a value `x` given the mean and standard deviation. Alternatively, compute it manually as `=(A1 - AVERAGE(A:A)) / STDEV.S(A:A)`.
- `=SKEW(range)` calculates the skewness of a range using the Fisher-Pearson standardized moment coefficient. Note that Excel does not have a direct equivalent to Pearson's Coefficient of Skew $Sk = 3(\bar{x} - Median)/s_x$ — this must be calculated manually as `=3*(AVERAGE(range) - MEDIAN(range)) / STDEV.S(range)`.
- `=KURT(range)` calculates the excess kurtosis of a range. A positive value indicates leptokurtic (heavy tails) and a negative value indicates platykurtic (light tails).

5.11 Chapter Summary Cheat Sheet

Shape & Distribution

BUSINESS STATISTICS · J. ALEJANDRO GELVES

Quick Reference

01 · LOCATION

Percentiles

$$L_p = 1 + k(n-1)$$

Sort data ascending. Compute location L_p where k is the percentile as a decimal. Interpolate between surrounding values.

EXCEL

```
=PERCENTILE.INC(array, k)
=QUARTILE.INC(array, 0/1/2/3/4)
```

02 · ANY DISTRIBUTION

Chebyshev's Theorem

$$\text{At least } (1 - 1/k^2) \text{ of data within k std. deviations}$$

Holds for ANY distribution. k must be greater than 1.

k=2 → at least 75% **k=3** → at least 88.9%

03 · NORMAL ONLY

Empirical Rule

$$68\% \cdot 95\% \cdot 99.7\%$$

Only for bell-shaped (normal) data. More precise than Chebyshev.

±1σ → 68% **±2σ** → 95% **±3σ** → 99.7%

04 · OUTLIER DETECTION

Z-Score

$$z_i = (x_i - \bar{x}) / s$$

Distance from mean in std. deviation units. Values with $|z| > 3$ are suspected outliers.

EXCEL

```
=STANDARDIZE(x, AVERAGE(r), STDEV.S(r))
```

05 · ASYMMETRY

Skewness

$$Sk = n/[(n-1)(n-2)] \cdot \Sigma[(x_i-\bar{x})/s]^3$$

Cubed z-scores preserve sign and amplify tail values. Positive = right tail longer. Negative = left tail longer.

Sk<0 Left **Sk=0** Symmetric **Sk>0** Right

EXCEL

```
=SKEW(range)
```

06 · TAIL BEHAVIOR

Excess Kurtosis

$$K = n(n+1)/[(n-1)(n-2)(n-3)] \cdot \Sigma[(x_i-\bar{x})/s]^4 - 3(n-1)^2/[(n-2)(n-3)]$$

Fourth power z-scores measure tail heaviness. Subtract 3 to center around normal distribution.

K<0 Platykurtic **K=0** Mesokurtic **K>0** Leptokurtic

EXCEL

```
=KURT(range)
```

07 · FIVE POINT SUMMARY

IQR & Box Plot

$$IQR = Q_3 - Q_1$$

Summarizes data with Min, Q_1, Median, Q_3, Max. The IQR measures the spread of the middle 50% of the data.

EXCEL

```
=QUARTILE.INC(array, 0) → Min
=QUARTILE.INC(array, 1) → Q1
=QUARTILE.INC(array, 2) → Median
=QUARTILE.INC(array, 3) → Q3
=QUARTILE.INC(array, 4) → Max
```

08 · OUTLIER DETECTION

IQR Method

$$LL = Q_1 - 1.5 \times IQR$$
$$UL = Q_3 + 1.5 \times IQR$$

Any observation below LL or above UL is an outlier. Does not assume normality. May disagree with z-score method.

EXCEL

```
LL =QUARTILE.INC(r,1)-1.5*(QUARTILE.INC(r,3)-QUARTILE.INC(r,1))
UL =QUARTILE.INC(r,3)+1.5*(QUARTILE.INC(r,3)-QUARTILE.INC(r,1))
```

NOTATION GUIDE

L_p	Location of pth percentile
k	Percentile as decimal / std. devs.
Q_1 / Q_3	1st / 3rd quartile
IQR	Interquartile range
Sk	Skewness coefficient
K	Excess kurtosis
z_i	Z-score of observation i
LL / UL	Lower / upper outlier limit

KEY FACTS

Chebyshev works for?	Any distribution
Empirical rule requires?	Normal distribution
Z-score outlier threshold?	$\|z\| > 3$
IQR covers middle?	50% of data
Right-skewed: mean vs median?	Mean > Median
Leptokurtic means?	Heavy tails, K > 0
Skew uses which power?	3rd moment
Kurtosis uses which power?	4th moment

5.12 Exercises

The following exercises will help you practice other statistical measures. In particular, the exercises work on:

- Constructing a five point summary and a boxplot.
- Applying Chebyshev's Theorem.
- Identifying skewness.
- Identifying outliers.

Answers are in the Answers to Exercises at the end of the book. Try not to peak until you have formulated your own answer and double checked your work for any mistakes. Each solution is worked by hand and then verified with Excel, assuming the data is entered in column A of your spreadsheet.

Exercise 1

For the following exercises, make your calculations by hand and verify results using Excel functions when possible.

1. Use the following observations to calculate the minimum, the first, second and third quartiles, and the maximum. Are there any outliers? Find the IQR to answer the question.

3	10	4	1	0	30	6

Answer on page 178 →

2. Confirm your finding of an outlier by calculating the z-score. Is 30 an outlier when using a z-Score?

Answer on page 178 →

3. Use Chebyshev's theorem to determine what percent of the data falls between the z-score found in 2.

Answer on page 179 →

Exercise 2

A financial analyst tracks the monthly revenue growth rates (in percent) of a small e-commerce startup over five months:

1	2	3	4	5
2	2	2	6	13

1. Calculate the sample mean and sample standard deviation.

Answer on page 179 →

2. For each observation compute the standardized value $\frac{x_i - \bar{x}}{s}$, its cube, and its fourth power. Organize your results in a table.

Answer on page 179 →

3. Use the results from part 2 to calculate the skewness. Is the distribution left-skewed, right-skewed, or symmetric?

Answer on page 179 →

Part II

Relate

6 Regression I

Measures of association are essential tools in business statistics for analyzing relationships between variables. They help determine whether changes in one variable are linked to changes in another, providing valuable insights into patterns and dependencies. Understanding these relationships is critical for making informed decisions. By quantifying the strength and direction of associations, businesses can better interpret data, optimize strategies, and drive meaningful outcomes. Below we study important measures of association. Association still belongs to the *describe* stage of learning from data — here we characterize how two variables move together — but it also points ahead to inference, where we will ask whether a relationship seen in a sample is real or merely a product of chance.

6.1 The Covariance

The **covariance** is a measure that determines the direction of the relationship between two variables. It is calculated by:

THE COVARIANCE

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

where x_i and y_i are observations, $\bar{x}$ and $\bar{y}$ are the sample means, and n is the sample size.

$s_{xy} > 0$	$s_{xy} = 0$	$s_{xy} < 0$
Direct relationship	No relationship	Inverse relationship

EXAMPLE

Let's consider the following data that captures the price of stocks (SPY) and bonds (BND):

Period	SPY	BND
1	87	316
2	86	380
3	84	416
4	85	430
5	83	440

The idea behind calculating a covariance is to determine whether there is a relationship between the two variables. We start by finding the deviations from the mean for both SPY and BND:

Period	SPY	BND	$SPY - S\bar{P}Y$	$BND - B\bar{N}D$	Product
1	87	316	2	−80	−160
2	86	380	1	−16	−16
3	84	416	−1	20	−20
4	85	430	0	34	0
5	83	440	−2	44	−88

The covariance is the average of the products column using $n - 1$:

$$s_{xy} = \frac{-160 + (-16) + (-20) + 0 + (-88)}{5 - 1} = \frac{-284}{4} = -71$$

Since $s_{xy} < 0$ there is an inverse relationship between SPY and BND. Intuitively, the covariance checks whether on average the products of the deviations from the mean is positive or negative. Whenever x lies below its mean and y lies above its mean, the product is negative — and vice versa — reflecting the inverse relationship between the two variables.

6.2 The Correlation

The **correlation** measures the strength of the linear relationship between two variables. It is calculated by:

THE CORRELATION COEFFICIENT

$$r = \frac{s_{xy}}{s_x \cdot s_y}$$

where s_{xy} is the covariance, s_x is the standard deviation of x, and s_y is the standard deviation of y.
The correlation coefficient lies between $[-1, 1]$:

$r = -1$	$-1 < r < 0$	$r = 0$	$0 < r < 1$	$r = 1$
Perfect inverse	Inverse	No relationship	Direct	Perfect direct

The strength of the relationship is determined by the absolute value of r. Values close to 1 indicate a strong relationship, values close to 0 indicate a weak relationship.

EXAMPLE

Consider the SPY and BND data. The standard deviation of SPY is $s_{SPY} = 1.58$ and that of BND is $s_{BND} = 50.37$. Substituting into the correlation formula:

$$r = \frac{-71}{1.58 \times 50.37} = -0.89$$

This indicates that the relationship between SPY and BND is inverse and strong.

6.3 The Coefficient of Determination (R^2)

The **coefficient of determination** or R^2 measures the percent of variation in y explained by variations in x. It is calculated by:

THE COEFFICIENT OF DETERMINATION

$$R^2 = r^2$$

where r is the correlation coefficient. R^2 ranges from 0 (no relationship) to 1 (perfect relationship).

EXAMPLE

Consider once more the SPY and BND example. The correlation coefficient is $r = -0.89$. Hence:

$$R^2 = (-0.89)^2 = 0.79$$

This indicates that about 79% of the variation in BND can be explained by changes in SPY. The two variables are closely related.

6.4 Measures of Association in Excel

Excel provides straightforward functions to calculate all three measures of association. Let's use the SPY and BND data from the example above.

Download Data

Covariance: To calculate the sample covariance use `=COVARIANCE.S()`. Assuming SPY is in column A and BND is in column B:

```
=COVARIANCE.S(A2:A6, B2:B6)
```

Correlation: To calculate the correlation coefficient use `=CORREL()`:

```
=CORREL(A2:A6, B2:B6)
```

Coefficient of Determination: There is no direct Excel function for R^2 between two variables. Simply square the correlation coefficient:

```
=CORREL(A2:A6, B2:B6)^2
```

The results for the SPY and BND data are:

Covariance	Correlation	R^2
−71.00	−0.8916	0.79

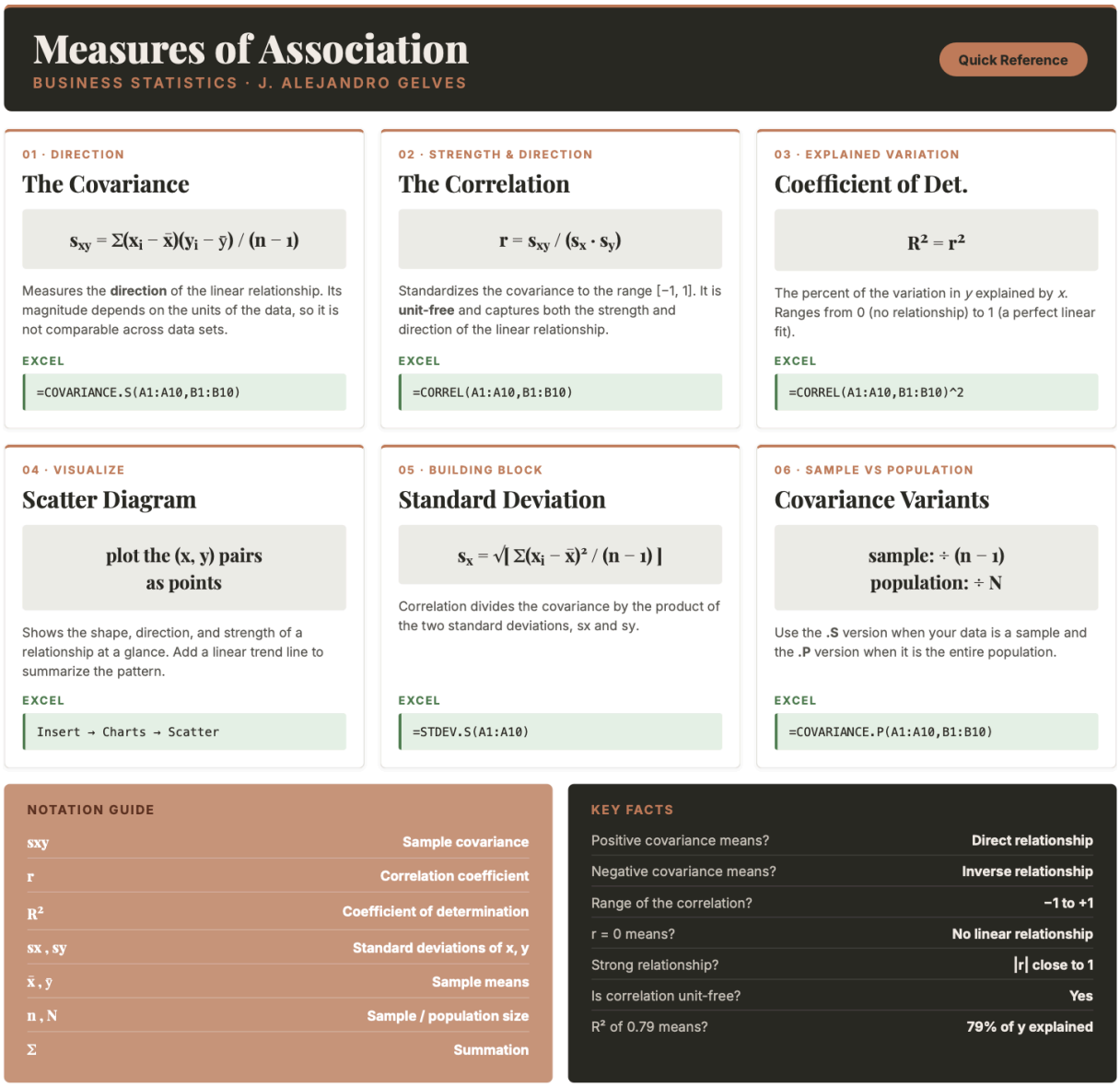
Scatter Plot: To create a scatter plot in Excel, select both columns and go to `Insert → Charts → Scatter`. This will display the relationship between SPY and BND visually. To add a trend line, right-click any point in the scatter plot and select `Add Trendline → Linear`.

6.5 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=COVARIANCE.S(array1, array2)` calculates the sample covariance between two variables. Use `=COVARIANCE.P()` for the population covariance.
- `=CORREL(array1, array2)` calculates the correlation coefficient between two variables. The result is always between −1 and 1.
- `=CORREL(array1, array2)^2` calculates the coefficient of determination R^2 by squaring the correlation coefficient. Excel does not have a standalone R^2 function for two variables.

6.6 Chapter Summary Cheat Sheet



6.7 Exercises

The following exercises will help you understand statistical measures that establish the relationship between two variables. In particular, the exercises work on:

- Calculating covariance and correlation.
- Creating scatter diagrams in Excel.
- Calculating the coefficient of determination.

- Interpreting measures of association in a real-world context.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using Excel functions when possible.

1. Consider the data below. Calculate the covariance and correlation coefficient by finding deviations from the mean. Verify your result using Excel. Is there a direct or inverse relationship between the two variables? How strong is the relationship?

x	20	21	15	18	25
y	17	19	12	13	22

Answer on page 180 →

2. Consider the data below. Calculate the covariance and correlation coefficient by finding deviations from the mean. Verify your result using Excel. Is there a direct or inverse relationship between the two variables? How strong is the relationship?

w	19	16	14	11	18
z	17	20	20	16	18

Answer on page 180 →

Exercise 2

You will need the **mtcars** data set to answer this question. You can download it here:

Download Data

1. Calculate the correlation coefficient between `hp` and `mpg`. Explain the results. Specifically, the direction of the relationship and the strength given the context of the problem.

Answer on page 180 →

2. Create a scatter diagram of the two variables. Is the scatter diagram what you expected after you calculated the correlation coefficient?

Answer on page 180 →

3. Calculate the coefficient of determination. How close is it to one? What else could be explaining the variation in the `mpg`? Let your dependent variable be `mpg`.

Answer on page 181 →

Exercise 3

You will need the **College** data set to answer this question. You can download it here:

Download Data

1. Create a scatter diagram between `GRAD_DEBT_MDN` (Median Debt) on the y-axis and `MD_EARN_WNE_P10` (Median Earnings) on the x-axis. What type of relationship do you observe between the variables?

Answer on page 181 →

2. Calculate the correlation coefficient and the coefficient of determination. According to the data, are higher debts correlated with higher earnings?

Answer on page 181 →

Exercise 4

You will need the **HomePrices** data set to answer this question. You can download it here:

Download Data

1. Create a scatter diagram between `bedrooms` (number of bedrooms) and `Price` (home sale price). What type of relationship do you observe between the two variables? Does the direction make intuitive sense?

Answer on page 181 →

2. Calculate the correlation coefficient between `bedrooms` and `Price`. Explain the direction and strength of the relationship in the context of the housing market.

Answer on page 181 →

3. Calculate the coefficient of determination. What percentage of the variation in `Price` is explained by `bedrooms`? What other factors might account for the remaining variation?

Answer on page 182 →

7 Regression II

Quantifying relationships between variables is an important skill in business analytics. The regression line, representing the best linear fit between two variables, enables businesses to make predictions, identify trends, and optimize strategies using data-driven insights. Understanding how to calculate and interpret a regression line provides the foundation for analyzing business relationships, such as the effect of advertising on sales or customer satisfaction on revenue. Below, we delve into generating and analyzing a regression line. Fitting a line is a descriptive act — a compact summary of how the data behave — but it is also the model whose reliability we will later test in the inference chapters, once we ask how much a different sample might have shifted it.

7.1 The Regression Line

The regression line is calculated to minimize the average distance (or errors) between the line and the observed data points. It is defined by two key components: a **slope** ($\hat{\beta}$) and an **intercept** ($\hat{\alpha}$). Mathematically, the regression line is expressed as:

THE REGRESSION LINE

$$\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$$

where $\hat{y}_i$ are the predicted values of y given x_i, $\hat{\alpha}$ is the intercept, and $\hat{\beta}$ is the slope.

The **slope** determines the steepness of the line and quantifies how much a unit increase in x changes y on average. It is calculated using the covariance between y and x and the variance of x:

THE SLOPE

$$\hat{\beta} = \frac{s_{xy}}{s_x^2}$$

where s_{xy} is the sample covariance between x and y, and s_x^2 is the sample variance of x.

The **intercept** determines where the line crosses the y axis — the predicted value of y when x is zero. Once the slope is estimated, the intercept is calculated by:

THE INTERCEPT

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

where $\bar{y}$ is the mean of y and $\bar{x}$ is the mean of x.

EXAMPLE

Let's examine a data set on Price and Advertisement. In general, one expects that when a company advertises, it can convince consumers to pay more for their product. Below is the data:

Advertisement (x)	Price (y)
2	7
1	3
3	8
4	10

The data shows a clear direct relationship between advertisement and price. Regression allows us to answer two key questions:

1. **Effectiveness:** How much can we increase the price for every additional dollar spent on advertisement?
2. **Prediction:** What is the predicted price if we have a budget of 6 for advertisement?

Step 1 — Calculate the slope: Given that the covariance is $s_{xy} = 3.67$ and the variance of x is $s_x^2 = 1.67$:

$$\hat{\beta} = \frac{3.67}{1.67} = 2.2$$

For every additional dollar spent on advertisement, price increases by 2.2 on average.
Step 2 — Calculate the intercept: Since $\bar{x} = 2.5$ and $\bar{y} = 7$:

$$\hat{\alpha} = 7 - 2.2(2.5) = 1.5$$

If we do not advertise, the predicted price is 1.5.
Step 3 — Write the regression line:

$$\hat{y}_i = 1.5 + 2.2x_i$$

Step 4 — Make a prediction: With an advertisement budget of 6:

$$\hat{y}_i = 1.5 + 2.2(6) = 14.7$$

In conclusion, for every dollar spent on advertisement the price increases by 2.2, and with a budget of 6 the predicted price is 14.7.

7.2 Measures of Goodness of Fit

When analyzing the effectiveness of a regression model, it is crucial to assess how well the model fits the data. Below we revisit the coefficient of determination R^2 and introduce additional measures.

Coefficient of Determination

The **coefficient of determination** or R^2 is the percent of the variation in y that is explained by changes in x. The higher the R^2 the better the explanatory power of the model.

THE COEFFICIENT OF DETERMINATION

$$R^2 = \frac{SSR}{SST}$$

where:

- $SST = \sum(y_i - \bar{y})^2$ — total variation in y
- $SSR = \sum(\hat{y}_i - \bar{y})^2$ — variation explained by the model
- $SSE = \sum(y_i - \hat{y}_i)^2$ — variation unexplained by the model
- $SST = SSR + SSE$

R^2 is always between $[0, 1]$.

EXAMPLE

Consider data on the Weight (y) and Exercise (x) of a particular person:

Weight (y)	Exercise (x)
165	45
170	10
168	25
164	30
165	40

If we used only the mean $\bar{y} = 166.4$ as our prediction, the total squared error would be:

$$SST = (-1.4)^2 + (3.6)^2 + (1.6)^2 + (-2.4)^2 + (-1.4)^2 = 25.2$$

Using the regression line reduces these errors. The mistakes made by the regression line are quantified by:

$$SSE = 0.81 + 0.29 + 0.69 + 5.76 + 0.02 = 7.57$$

The improvement gained by using regression over the mean is:

$$SSR = 5.29 + 9.4 + 0.59 + 0 + 2.35 = 17.63$$

Hence the coefficient of determination is:

$$R^2 = \frac{SSR}{SST} = \frac{17.63}{25.2} \approx 0.70$$

The regression line explains about 70% of the variation in Weight using Exercise.

7.3 Multiple Regression

If we were to predict weight we could use several other variables to get a better prediction. **Multiple regression** is a technique used to predict a variable using more than one independent variable.

THE MULTIPLE REGRESSION LINE

$$\hat{y_i} = \hat{\beta_0} + \hat{\beta_1} x_1 + \hat{\beta_2} x_2 + ... + \hat{\beta_k} x_k$$

where k is the total number of independent variables included in the model.

EXAMPLE

Let's consider an additional variable in our Weight and Exercise example. The table below includes information on Calories (z):

Weight (y)	Exercise (x)	Calories (z)
165	45	1200
170	10	1260
168	25	1220
164	30	1180
165	40	1190

The multiple regression line estimated by computer software is:

$$\hat{y}_i = 83.96 - 0.025x + 0.069z$$

Two conclusions follow from this result. First, as exercise increases weight tends to go down, while more calories increases weight. Second, reducing calorie consumption by 1 unit decreases weight by 0.069 pounds, whereas one additional minute of exercise only reduces weight by 0.025 pounds — making calorie reduction more effective per unit.

Anova

The Anova table helps us understand how well each variable in our regression model explains the dependent variable. It decomposes the *SSR* by variable and tracks the remaining errors.

EXAMPLE

For the Weight, Exercise, and Calories regression, the Anova table is:

Source	Sum Squares
x	17.63
z	6.55
Residuals	1.01

Exercise explains 17.63 of the total variation. Adding Calories reduces the remaining unexplained variation by another 6.55, bringing the total *SSR* to 24.18 and the *SSE* down to 1.01. The R^2 increases from 0.70 to:

$$R^2 = \frac{17.63 + 6.55}{25.2} = 0.96$$

Adjusted R^2

The **adjusted** R^2 penalizes a model for including additional explanatory variables, preventing artificial inflation of R^2.

THE ADJUSTED

R^2

$$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-k-1}$$

where k is the number of explanatory variables and n is the sample size.

EXAMPLE

For the Weight example with both Exercise and Calories ($k = 2$, $n = 5$, $R^2 = 0.96$):

$$\bar{R}^2 = 1 - (1 - 0.96)\frac{5-1}{5-2-1} = 0.92$$

Residual Standard Error

The **Residual Standard Error** estimates the average dispersion of the data points around the regression line.

THE RESIDUAL STANDARD ERROR

$$s_e = \sqrt{\frac{SSE}{n-k-1}}$$

where k is the number of independent variables and n is the sample size.

EXAMPLE

For the Weight and Exercise model ($k = 1$):

$$s_e = \sqrt{\frac{7.57}{5-1-1}} = 1.59$$

Adding Calories ($k = 2$) reduces this to:

$$s_e = \sqrt{\frac{1.01}{5-2-1}} = 0.71$$

The smaller residual standard error confirms that adding Calories improves the model's fit.

7.4 Regression with Dummy Variables

So far every predictor has been numerical. But many of the variables we care about in business are **categorical** — the brand of a product, whether a customer churned, the gender of a respondent, or the instructor teaching a class. Regression can use these variables too, once we encode them as numbers. The tool for doing so is the **dummy variable**.

THE DUMMY VARIABLE

$$d_i = \begin{cases} 1 & \text{if observation } i \text{ is in the category} \\ 0 & \text{otherwise} \end{cases}$$

A dummy (or *indicator*) variable equals 1 when an observation belongs to a particular category and 0 when it does not.

A single dummy recovers a difference of means. Suppose we regress a numerical variable y on a single dummy d, giving $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 d$. Because d takes only two values, the line makes only two predictions. When $d = 0$ the prediction is $\hat{\beta}_0$; when $d = 1$ it is $\hat{\beta}_0 + \hat{\beta}_1$. Those two predictions are exactly the two group averages:

A DUMMY AND THE GROUP MEANS

$$\hat{\beta}_0 = \bar{y}_0 \qquad \hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$$

The intercept is the mean of the **baseline** group (the one coded 0), and the slope is the *difference* between the two group means.

In other words, regressing a variable on a single dummy is just another way of computing a difference of means — the slope *is* that difference.

More than two categories. When a categorical variable has k categories, we create $k-1$ dummies and leave one category out to act as the **baseline** (or reference). Each remaining dummy's coefficient measures the gap between its category's mean and the baseline mean. We never include a dummy for *every* category alongside an intercept — doing so makes the columns perfectly collinear (the **dummy variable trap**), and the regression cannot be estimated.

EXAMPLE

The **Spin** data set records 90 spin classes taught by three instructors — Olivia, Sam, and Ben — along with `Total_kj`, the total energy (in kilojoules) burned during each class. We want each instructor's average energy output.
Because there are three instructors, we build two dummies and let **Olivia** be the baseline:

- $Sam = 1$ if the class was taught by Sam, 0 otherwise.
- $Ben = 1$ if the class was taught by Ben, 0 otherwise.

An Olivia class therefore has $Sam = 0$ and $Ben = 0$. Regressing `Total_kj` on the two dummies gives:

$$\widehat{Total_kj} = 197.03 - 75.67\, Sam + 27.90\, Ben$$

Reading off each instructor's predicted (average) energy:

Instructor	*Sam*	*Ben*	Prediction	Mean `Total_kj`
Olivia (baseline)	0	0	197.03	197.03
Sam	1	0	$197.03 - 75.67 = 121.37$	121.37
Ben	0	1	$197.03 + 27.90 = 224.93$	224.93

The predictions land exactly on the three group averages. The intercept is Olivia's mean; the coefficient on *Sam* (-75.67) says Sam's classes burn about 75.67 kJ *less* than Olivia's on average, and the coefficient on *Ben* ($+27.90$) says Ben's burn about 27.90 kJ *more*. Regression with dummy variables has reproduced the group means — and, as a bonus, reported how far each group sits from the baseline.

Dummy Variables in Excel

Download Data

With `Instructor` in column A and `Total_kj` in column C, we build the two dummies with `IF`. Rather than creating helper columns, `HSTACK` lets us assemble both dummies *inside* the `LINEST` call:

```
=LINEST(C2:C91, HSTACK(IF(A2:A91="Sam",1,0), IF(A2:A91="Ben",1,0)), TRUE, TRUE)
```

Remember that `LINEST` returns the coefficients **in reverse order**. Because we stacked *Sam* first and *Ben* second, the first row of the output reads:

Position	Row 1 value	Meaning
Col 1	27.90	coefficient on *Ben*
Col 2	-75.67	coefficient on *Sam*
Col 3	197.03	intercept (Olivia)

If you prefer helper columns, put the two `IF` formulas in columns D and E and pass `D2:E91` as the x range — the result is identical.

To predict each instructor's average directly, use `TREND` with a new x row for each combination:

```
=TREND(C2:C91, HSTACK(IF(A2:A91="Sam",1,0), IF(A2:A91="Ben",1,0)), {0,0})  → 197.03
=TREND(C2:C91, HSTACK(IF(A2:A91="Sam",1,0), IF(A2:A91="Ben",1,0)), {1,0})  → 121.37
=TREND(C2:C91, HSTACK(IF(A2:A91="Sam",1,0), IF(A2:A91="Ben",1,0)), {0,1})  → 224.93
```

You can confirm each group mean directly with `AVERAGEIF`:

```
=AVERAGEIF(A2:A91, "Olivia", C2:C91)   → 197.03
=AVERAGEIF(A2:A91, "Sam", C2:C91)      → 121.37
=AVERAGEIF(A2:A91, "Ben", C2:C91)      → 224.93
```

7.5 Regression in Excel

Excel estimates regressions through three worksheet functions. `LINEST` fits the model and reports every diagnostic we need; `TREND` produces predictions; and `HSTACK` assembles the independent variables into the single block that `LINEST` requires. All three work for simple *and* multiple regression, so one set of tools covers the whole chapter.

Download Data

Reading the LINEST Output

`LINEST` is an **array function**. Written with both optional arguments set to `TRUE`, it returns a block of 5 rows by $k + 1$ columns:

```
=LINEST(y_range, x_range, TRUE, TRUE)
```

The third argument asks Excel to estimate an intercept; the fourth asks for the full diagnostics. In current versions of Excel the result *spills* automatically — type the formula in one cell and press `Enter`. The block is laid out as follows, for a model with k independent variables:

Row	Column 1	Column 2	Remaining columns
1	$\hat{\beta}_k \ldots \hat{\beta}_1$, then $\hat{\beta}_0$		coefficients
2	standard error of each coefficient		
3	R^2	s_e	`#N/A`
4	F statistic	$df = n - k - 1$	`#N/A`
5	SSR	SSE	`#N/A`

Two details matter enormously in practice.

Row 1 runs backwards. The coefficients appear in reverse order of your x columns — the *last* variable first and the intercept *last*. With one predictor the row is simply $\{\hat{\beta}, \hat{\alpha}\}$; with two it is $\{\hat{\beta}_2, \hat{\beta}_1, \hat{\beta}_0\}$. Misreading this is the most common `LINEST` error.

The x range must be one contiguous block. `LINEST` will not accept scattered columns. When your independent variables are not side by side, `HSTACK` joins them into a single array:

```
=LINEST(B2:B6, HSTACK(C2:C6, F2:F6), TRUE, TRUE)
```

To pull a single number out of the block instead of spilling all of it, wrap the call in `INDEX(array, row, column)`.

Simple Regression (Weight and Exercise)

With `Weight` in column A and `Exercise` in column B:

```
=LINEST(A2:A6, B2:B6, TRUE, TRUE)
```

Reading the block, and recalling that row 1 is reversed:

- `=INDEX(LINEST(A2:A6,B2:B6,TRUE,TRUE),1,1)` $\rightarrow \hat{\beta} = -0.153$
- `=INDEX(LINEST(A2:A6,B2:B6,TRUE,TRUE),1,2)` $\rightarrow \hat{\alpha} = 171.00$
- `=INDEX(LINEST(A2:A6,B2:B6,TRUE,TRUE),3,1)` $\rightarrow R^2 = 0.70$
- `=INDEX(LINEST(A2:A6,B2:B6,TRUE,TRUE),3,2)` $\rightarrow s_e = 1.59$
- `=INDEX(LINEST(A2:A6,B2:B6,TRUE,TRUE),5,1)` $\rightarrow SSR = 17.63$
- `=INDEX(LINEST(A2:A6,B2:B6,TRUE,TRUE),5,2)` $\rightarrow SSE = 7.57$

Since $SST = SSR + SSE$, the total variation is $17.63 + 7.57 = 25.2$, matching the value we computed by hand. (`=DEVSQ(A2:A6)` returns the same 25.2 as a check.)

Multiple Regression (Weight, Exercise, and Calories)

With `Calories` in column C, the two independent variables are adjacent, so `B2:C6` works directly. Using `HSTACK` makes the intent explicit and keeps the formula correct even if the columns are later moved apart:

```
=LINEST(A2:A6, HSTACK(B2:B6, C2:C6), TRUE, TRUE)
```

The first row of the output is $\{0.069,\ -0.025,\ 83.96\}$ — reading **right to left**, that is an intercept of 83.96, a coefficient of -0.025 on Exercise, and 0.069 on Calories, giving

$$\hat{y}_i = 83.96 - 0.025x + 0.069z$$

The remaining diagnostics come from the same block:

- `=INDEX(LINEST(A2:A6,HSTACK(B2:B6,C2:C6),TRUE,TRUE),3,1)` → $R^2 = 0.96$
- `=INDEX(LINEST(A2:A6,HSTACK(B2:B6,C2:C6),TRUE,TRUE),3,2)` → $s_e = 0.71$
- `=INDEX(LINEST(A2:A6,HSTACK(B2:B6,C2:C6),TRUE,TRUE),4,2)` → $df = n-k-1 = 2$
- `=INDEX(LINEST(A2:A6,HSTACK(B2:B6,C2:C6),TRUE,TRUE),5,1)` → $SSR = 24.19$
- `=INDEX(LINEST(A2:A6,HSTACK(B2:B6,C2:C6),TRUE,TRUE),5,2)` → $SSE = 1.01$

Building the Anova Table

`LINEST` reports the *total SSR* for a model, not the contribution of each variable. To split *SSR* by variable, run the models in sequence and take the difference:

```
=INDEX(LINEST(A2:A6, B2:B6, TRUE, TRUE), 5, 1)                    → 17.63   (x alone
=INDEX(LINEST(A2:A6, HSTACK(B2:B6,C2:C6), TRUE, TRUE), 5, 1)      → 24.19   (x and z
```

The difference, $24.19 - 17.63 = 6.55$, is what Calories adds once Exercise is already in the model. The residual $SSE = 1.01$ comes from row 5, column 2 of the larger model. Those three numbers are exactly the Anova table above.

Adjusted R^2

`LINEST` does not report $\bar{R}^2$, but every ingredient is in the block. With R^2 in row 3 column 1 and $df = n - k - 1$ in row 4 column 2:

```
=1-(1-INDEX(LINEST(A2:A6,HSTACK(B2:B6,C2:C6),TRUE,TRUE),3,1))
     *(COUNT(A2:A6)-1)/INDEX(LINEST(A2:A6,HSTACK(B2:B6,C2:C6),TRUE,TRUE),4,2)
```

which returns 0.92, matching the hand calculation.

Making Predictions

`TREND` applies the fitted model to new values of the independent variables. It accepts the same x block as `LINEST`, so it works for one predictor or many:

```
=TREND(y_range, x_range, new_x_range, TRUE)
```

For the Advertisement and Price example, with a budget of 6:

```
=TREND(B2:B5, A2:A5, 6)   → 14.7
```

For the Weight model, predicting a person who exercises 35 minutes and eats 1210 calories:

```
=TREND(A2:A6, HSTACK(B2:B6, C2:C6), HSTACK(35, 1210))
```

The `new_x` block must have the same number of columns, in the same order, as the x block used to fit the model. Omitting `new_x` returns the fitted value for every observation in the original data.

7.6 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=LINEST(y_range, x_range, TRUE, TRUE)` estimates the regression and returns a $5 \times (k+1)$ block containing the coefficients, their standard errors, R^2, s_e, the F statistic, the degrees of freedom, *SSR* and *SSE*. Works for any number of independent variables. **Row 1 is in reverse order**, ending with the intercept.
- `=INDEX(LINEST(...), row, column)` extracts a single value from the `LINEST` block. Common picks: `(1,1)` the slope of the last x, `(3,1)` R^2, `(3,2)` s_e, `(4,2)` df, `(5,1)` *SSR*, `(5,2)` *SSE*.
- `=HSTACK(range1, range2, ...)` joins ranges side by side into a single array, so that non-adjacent independent variables can be passed to `LINEST` or `TREND`. It also lets dummy variables be built inline without helper columns.
- `=TREND(y_range, x_range, new_x_range, TRUE)` returns predicted values of y. Handles simple and multiple regression; `new_x_range` must match the column layout of `x_range`.
- `=IF(cell="Category", 1, 0)` builds a dummy variable, coding each row 1 when it belongs to a category and 0 otherwise.
- `=AVERAGEIF(range, criteria, average_range)` returns the mean of a numerical column for the rows matching a category — a quick way to verify the group means produced by a dummy-variable regression.
- `=DEVSQ(y_range)` returns *SST* directly, as a check on *SSR* + *SSE*.

> **Note**
>
> **A note on versions.** `HSTACK` requires Microsoft 365 or Excel 2024. In earlier versions, place the independent variables in adjacent columns and pass the block directly to `LINEST` and

`TREND` — every other formula in this chapter is unchanged. `LINEST` and `TREND` are array formulas: in versions before dynamic arrays, select the full output range first and confirm with `Ctrl+Shift+Enter`.

7.7 Chapter Summary Cheat Sheet

The Regression Line

BUSINESS STATISTICS · J. ALEJANDRO GELVES

Quick Reference

01 · THE LINE

Regression Line

$$\hat{y} = \hat{\alpha} + \hat{\beta}x$$

The best linear fit between two variables. Estimate it in one step with the Data Analysis ToolPak.

EXCEL

```
Data → Data Analysis → Regression
```

02 · STEEPNESS

The Slope

$$\hat{\beta} = s_{xy} / s^2_x$$

How much y changes, on average, for a one-unit increase in x.

EXCEL

```
=SLOPE(y_range, x_range)
```

03 · STARTING POINT

The Intercept

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

The predicted value of y when x equals zero.

EXCEL

```
=INTERCEPT(y_range, x_range)
```

04 · GOODNESS OF FIT

Coefficient of Det.

$$R^2 = SSR / SST$$

The percent of the variation in y explained by the model. Ranges 0 to 1.

EXCEL

```
=RSQ(y_range, x_range)
```

05 · TYPICAL ERROR

Residual Std. Error

$$s_e = \sqrt{[\, SSE / (n - k - 1) \,]}$$

The average distance of the data points from the regression line.

EXCEL

```
=STEYX(y_range, x_range)
```

06 · PREDICTION

Forecasting

$$\hat{y} = \hat{\alpha} + \hat{\beta}x$$

Plug an x value into the fitted line to predict y.

EXCEL

```
=FORECAST.LINEAR(x, y_range, x_range)
```

NOTATION GUIDE

$\hat{y}$	Predicted value of y
$\hat{\alpha}$	Intercept
$\hat{\beta}$	Slope
SST	Total variation
SSR	Explained variation
SSE	Unexplained variation
$\bar{R}^2$	Adjusted R^2
s_e	Residual standard error
n , k	Sample size, # predictors

KEY FACTS

SST equals?	SSR + SSE
Range of R^2?	0 to 1
Higher R^2 means?	Better fit
Adjusted R^2 does what?	Penalizes extra predictors
Multiple regression?	More than one x
ToolPak returns?	Coefficients, R^2, ANOVA
Slope formula uses?	Covariance ÷ variance of x

7.8 Exercises

The following exercises will help you get practice on Regression Line estimation and interpretation. In particular, the exercises work on:

- Estimating the slope and intercept.
- Calculating measures of goodness of fit.
- Prediction using the regression line.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results using Excel functions when possible.

1. Consider the data below. Calculate the deviations from the mean for each variable and use the results to estimate the regression line. Verify your result using Excel. On average, by how much does y increase per unit increase of x?

x	20	21	15	18	25
y	17	19	12	13	22

Answer on page 182 →

2. Calculate SST, SSR, and SSE. What is the R^2? What is the Standard Error estimate? Is the regression line a good fit for the data?

Answer on page 182 →

3. Assume that x is observed to be 32. What is your prediction of y? How confident are you in this prediction?

Answer on page 183 →

Exercise 2

You will need the **Education** data set to answer this question. You can download it here:

Download Data

The data shows the years of education ($Education$) and the annual salary in thousands ($Salary$) for a sample of 100 people.

1. Estimate the regression line using Excel. By how much does an extra year of education increase the annual salary on average? What is the salary of someone without any education?

Answer on page 183 →

2. Confirm that the regression line is a good fit for the data. What is the estimated salary of a person with 16 years of education?

Answer on page 183 →

Exercise 3

You will need the **FoodSpend** data set to answer this question. You can download it here:

Download Data

1. Remove any observations with missing values. Create a dummy variable that is equal to 1 if an individual owns a home and 0 if the individual does not. Find the mean of your dummy variable. What proportion of the sample owns a home?

Answer on page 183 →

2. Run a regression with *Food* as the dependent variable and your dummy variable as the independent variable. What is the interpretation of the intercept and slope?

Answer on page 184 →

3. Now run a regression with your dummy variable as the dependent variable and *Food* as the independent variable. What is the interpretation of the intercept and slope? Hint: you might want to plot the scatter diagram and the regression line.

Answer on page 184 →

Exercise 4

You will need the **Population** data set to answer this question. You can download it here:

Download Data

1. Run a regression of *Population* on *Year* for Japan. How well does the regression line fit the data?

Answer on page 184 →

2. Create a prediction for Japan's population in 2030. What is your prediction?

Answer on page 185 →

3. Create a scatter diagram and include the regression line. How confident are you of your prediction after looking at the diagram?

Answer on page 185 →

Part III

Uncertainty

8 Probability I

Decision-making under uncertainty is at the core of business strategy, risk management, and forecasting. This section will study tools and concepts that will make us smarter when making data-driven decisions. Mastering probability in finance, marketing, supply chain, or operations helps professionals analyze risks, optimize strategies, and stay competitive in uncertain environments. We start with the basic concepts and ideas of probability below. Probability marks a turning point in the book. Until now we have described data we already collected; from here we build the mathematics of uncertainty — the bridge that will later let us reason from an imperfect sample back to the world that produced it.

8.1 Frequentist Vs. Bayesian Statistics

There are two interpretations of probability. The **frequentist** interpretation assumes that probabilities represent proportions of specific events occurring over infinitely identical trials. In contrast, the **Bayesian** interpretation assumes that probabilities are subjective beliefs about the relative likelihood of events. Despite their philosophical differences, both approaches adhere to the core rules of probability —including the Addition Rule, Multiplication Rule, and Bayes' Theorem— which provide a consistent mathematical foundation for making probabilistic inferences.

EXAMPLE

A company might use frequentist statistics to determine the average return rate of a product, but apply Bayesian methods to adjust sales forecasts based on changing consumer behavior.

8.2 Experiments and Sets

The following concepts form the foundation of probability theory. They provide a structure that is useful when analyzing uncertainty and randomness.

- An **experiment** is a process that leads to one of several outcomes. *Ex: Tossing a Die, Tossing a Coin, Drawing a Card, etc.*
- An **outcome** is the result of an experiment. *Ex: A coin landing on heads, drawing the ace of spades.*
- The **sample space** (S) of an experiment contains all possible outcomes of the experiment. *Ex:* $S = \{1, 2, 3, 4, 5, 6\}$ is the sample space for tossing a die.

- An **event** is a subset of the sample space. *Ex:* $A = \{2, 4, 6\}$ is the event of tossing an even number when rolling a die. $B = \{1A, 1D, 1S, 1H\}$ is the event of drawing an ace from a deck of cards.

8.3 Basic Probability Concepts

A **probability** is a numerical value that measures the likelihood that an event occurs. To calculate **probabilities**, we find the ratio between favorable outcomes and total outcomes. Hence, the probability of an event is given by:

PROBABILITY OF AN EVENT

$$P(A) = \frac{Number\ of\ outcomes\ in\ A}{Number\ of\ outcomes\ in\ Sample\ Space}$$

The probability of A satisfies $0 \leq P(A) \leq 1$. Since A is a subset of S, the number of elements in A is at most that of the sample space S. Hence, if A=S, then P(A)=1. If A is a null event (i.e., has no elements) P(A)=0 and is impossible. We can conclude that P(A) ranges from 0 to 1, inclusive.

In many cases, we analyze how multiple events interact within the sample space, particularly when they either exclude each other or collectively cover all possible outcomes.**Mutually exclusive** events do not share any common outcomes. The occurrence of one event precludes the occurrence of others. **Exhaustive** events include all outcomes in the sample space.

Given these concepts if follows that the sum of the probabilities of a list of **mutually exclusive** and **exhaustive** events equals 1. Formally, we state: $\sum P(x_i) = 1$, where x_i is outcome i.

EXAMPLE

Let's consider a standard deck of 52 playing cards to illustrate the key probability concepts. If we randomly draw one card from a standard deck, the sample space S consists of all 52 cards. Suppose we define event A as *drawing a red card*. Since there are 26 red cards, the probability is:

$$P(A) = \frac{\text{Number of red cards}}{\text{Total number of cards}} = \frac{26}{52} = 0.5$$

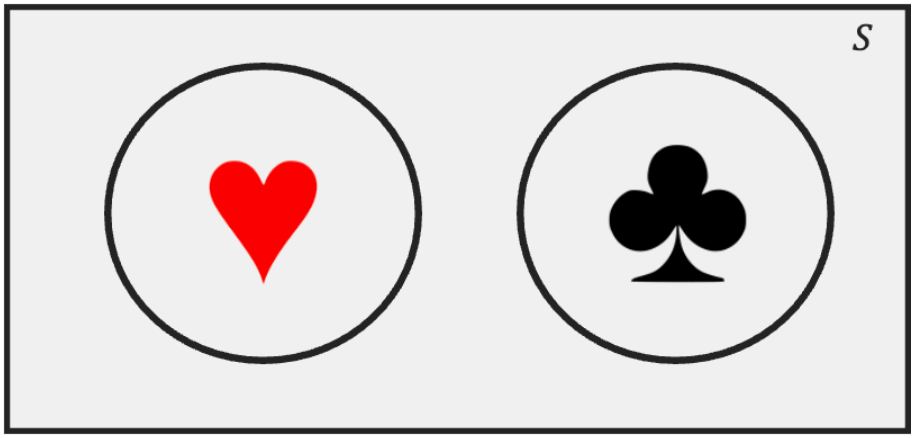
Since $(0 \leq P(A) \leq 1)$, this follows the basic probability rule. Next let's consider mutually exclusive events. Let event B be drawing a heart and event C be drawing a spade. Since a single drawn card cannot be both a heart and a spade, these events are mutually exclusive. The four suits (*hearts, diamonds, clubs, spades*) form an exhaustive set, because every card belongs to exactly one suit. The sum of their probabilities equals 1:

$$P(\text{hearts}) + P(\text{diamonds}) + P(\text{clubs}) + P(\text{spades}) = \frac{13}{52} + \frac{13}{52} + \frac{13}{52} + \frac{13}{52} = 1$$

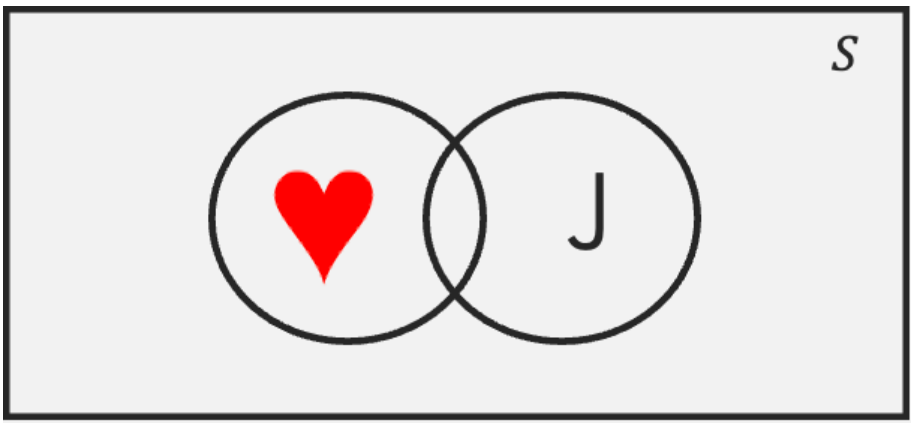
Note that this also satisfies the rule $\sum P(x_i) = 1$, where x_i represents each possible outcome (suit).

8.4 Venn Diagram

A Venn diagram is a visual representation of sets and their relationships. It consists of overlapping circles, each representing a set. Below is an example:



The rectangle enclosing the circles represents the sample set (i.e., the 52 playing cards). Each circle represents a set or event The heart circle contains all 13 hearts; the same holds for the clubs circle. The area outside the circle that has hearts represents its complement, meaning elements that are clubs, spades, or diamonds. Since the two sets do not overlap, they are mutually exclusive (i.e., if a card is a heart, it can't be a spade). The total area covered by both sets represents the **union** denoted by $\cup$. This is the set that contains all elements that belong to at least one of the sets. Formally, we would represent this set as $(Hearts \cup Spades)$.



The overlapping region of two sets contains elements that belong to both sets. This region is called the **intersection**. The intersection contains elements that are Hearts and Jacks (i.e., the jack of hearts). Formally, we denote the intersection by $(Hearts \cap Jacks)$.

8.5 Probability Rules

Probability rules help us quantify uncertainty, avoid cognitive biases, and make rational predictions based on available data. These rules allow us to make reliable decisions. Below we introduce some useful rules.

Complement Rule

The **Complement Rule** states that:

THE COMPLEMENT RULE

$$P(A^c) = 1 - P(A)$$

where A^c is the complement of A.

This rule is useful when calculating probabilities for complex scenarios, where finding the probability of the complement is easier than computing the event itself.

EXAMPLE

The probability of drawing a spade, diamond or clubs can be estimated by just calculating the probability of drawing a heart. Since the probability of drawing a heart is $P(Heart) = \frac{13}{52}$, then using the complement rule, $P(Not\text{Heart}) = 1 - \frac{13}{52} = \frac{39}{52}$.

De Morgan's Laws

De Morgan's Laws relate the complement of a union to the complement of an intersection. They are handy for rewriting a complicated event as an easier-to-compute one.

DE MORGAN'S LAWS

$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

The complement of a union is the intersection of the complements, and the complement of an intersection is the union of the complements.

EXAMPLE

Roll a die with $A = \{1,2,3\}$ and $B = \{2,4,6\}$. Their union is $A \cup B = \{1,2,3,4,6\}$, so the complement is $(A \cup B)^c = \{5\}$. De Morgan's Law gives the same answer from the complements: $A^c \cap B^c = \{4,5,6\} \cap \{1,3,5\} = \{5\}$.

Addition Rule

The **Addition Rule** states:

THE ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

where $\cap$ is intersection and $\cup$ is union.

This formula prevents double-counting when events overlap by including the intersection of events.

EXAMPLE

Let's calculate the probability of drawing either a heart or a face card. If we let event A be drawing a heart, and event B be drawing a face card, the probability is given by:

$$P(A \cup B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52}$$

Multiplication Rule

The **Multiplication Rule** states:

THE MULTIPLICATION RULE

$$P(A \cap B) = P(A|B)P(B)$$

where $P(A|B)$ is the conditional probability.

Conditional probability measures the likelihood of an event occurring given that another event has already happened. If we reorganize the formula above we can get:

CONDITIONAL PROBABILITY

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Note that this formula focuses on cases where Event B has already occurred. We can then find the fraction of the cases where Event A also happens.

EXAMPLE

Suppose we randomly draw two cards from a standard deck of 52 cards, without replacement. The probability that both cards are aces is given by:

$$P(Ace \cap Ace) = P(Ace|Ace) \times P(Ace) = \frac{4}{52} \times \frac{3}{51}$$

Lastly, if events A and B are independent, we can simplify the multiplication rule to:

INDEPENDENCE

$$P(A \cap B) = P(A)P(B)$$

Hence, we can define independence as: events A and B are independent if the occurrence (or non occurrence) of one does not affect the probability of the other occurring.

EXAMPLE

Suppose we randomly draw a card from a standard deck, replace it, and then draw another card. The probability of drawing a heart in the first draw is 13/52, and the probability of drawing a spade in the second draw is also 13/52.

EXAMPLE

Suppose that the weather is sunny; this has no effect on the outcome of a lottery. Hence, the weather and lottery outcomes are independent.

Law of Total Probability

The **Law of Total Probability** states:

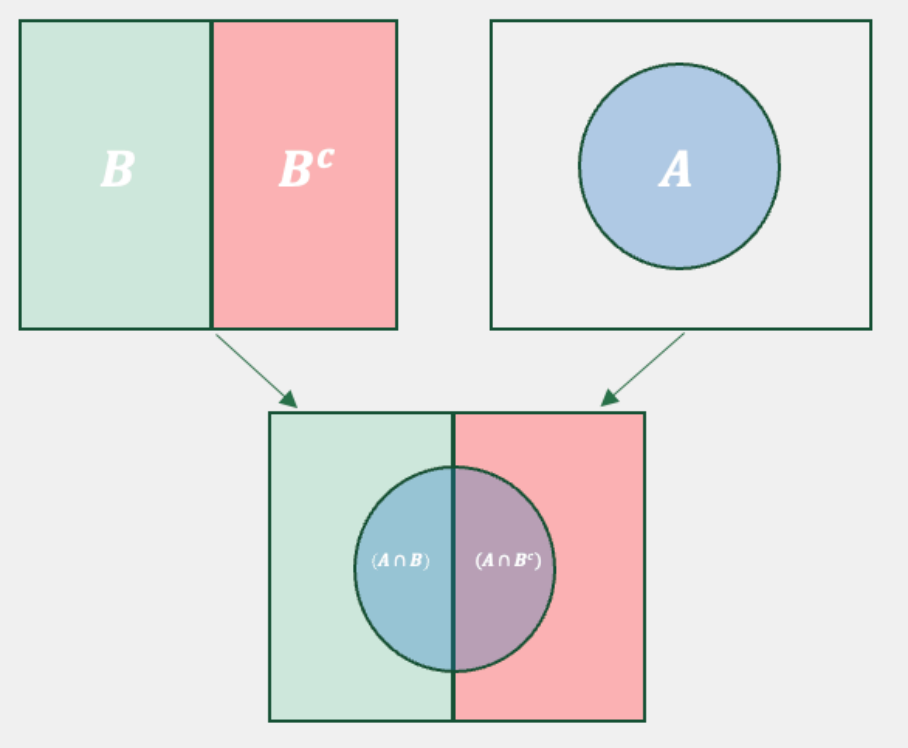
LAW OF TOTAL PROBABILITY

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

which, using the multiplication rule, can also be written as

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

The formula mainly states that if an event A can occur in multiple mutually exclusive scenarios (or conditions), then the total probability of A is the sum of its probabilities under each scenario, weighted by how often those scenarios occur. The Venn diagram helps us visualize the law of total probability.



The image shows how event A can occur. Essentially, when B occurs A is possible in the left semicircle. When B^c, then A is possible in the right semicircle. The addition of both semicircles, complete set A.

EXAMPLE

Consider the sets face cards and hearts. The entire set of hearts can be completed by taking face cards that are hearts, and non face cards (the complement of face cards) that are hearts.

The same idea extends to **any** number of scenarios. If $B_1, B_2, \dots, B_k$ are mutually exclusive and together cover the whole sample space, the total probability of A is found by summing across all of them:

LAW OF TOTAL PROBABILITY (SEVERAL EVENTS)

$$P(A) = \sum_{i=1}^{k} P(A \cap B_i) = \sum_{i=1}^{k} P(A|B_i)\, P(B_i)$$

where the events $B_1, B_2, \ldots, B_k$ are mutually exclusive and exhaustive (they partition the sample space).

EXAMPLE

A factory makes a part on three machines. Machine 1 produces 50% of the parts with a 2% defect rate, Machine 2 produces 30% with a 3% defect rate, and Machine 3 produces 20% with a 5% defect rate. The overall probability that a randomly chosen part is defective is

$$P(D) = (0.50)(0.02) + (0.30)(0.03) + (0.20)(0.05) = 0.029$$

so about 2.9% of all parts are defective.

Bayes' Theorem

Bayes' Theorem states:

BAYES' THEOREM

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

In essence, Bayes' theorem is a way of updating probabilities in light of new evidence. The theorem tells us that the probability of an event A given that we have observed some evidence is equal to the prior probability $(P(A))$ times a factor that measures the informativeness of the observed evidence $P(B|A)/P(B)$.

EXAMPLE

Assume that a rare form of cancer affects 1% of the population. That is, 1 out of 100 people will likely be affected by it. There is a test for the cancer, but it is imperfect. In particular, if you don't have cancer, the test says that you do 10% of the time (false positive rate). Also, if you do have cancer, it will tell you that you don't have it 5% of the time. Let's say you get tested, and your doctor calls you and says you have tested positive. How likely is it that you have this type of cancer?

We can use the Law of Total Probability and Bayes' theorem to answer this question. Let's first identify what we want to estimate. Since we need to figure out the likelihood of having cancer, given that we tested positive, we are looking for $P(C|+)$. Now, we can set up Bayes'

rule to see what additional information we need to find our desired value. Bayes' rule states:

$$P(C|+) = \frac{P(+|C)P(C)}{P(+)}$$

Since the problem does not provide the $P(+)$ we must estimate it with the Law of Total Probability, mainly:

$$P(+) = P(+|C)P(C) + P(+|NC)P(NC) = (0.95 \times 0.01 + (0.1 \times 0.99) = 0.1085$$

We can now use this probability to find our new belief that we have cancer:

$$P(C|+) = \frac{(0.95 \times 0.01)}{0.1085} = 0.0875$$

In conclusion, if we before thought we had a 1% chance of having the rare form of cancer, once we were told that we tested positive, we should update our belief to about 8.75%.

8.6 Contingency Tables

A **contingency table** (or cross-tabulation) summarizes the relationship between two or events (or variables). It is particularly helpful when analyzing frequencies of events and there relationship to other events.

The contingency table below shows the frequency of event A and B occurring/not occurring.

Table 8.1: Contingency table: frequencies of A and B (n = 100)

	B	B^c	Total
A	26	34	60
A^c	14	26	40
Total	40	60	100

We can see that event A occurs 60 times and that event B occurs 40 times. Most importantly, we can also see how the two events are not mutually exclusive, since A and B occur 26 times. The table can be easily translated to probabilities, by divinding each entry by the total amount of outcomes (i.e., 100). We call the table that includes probabilities the **joint probability table**.

Table 8.2: Joint probability table: joint probabilities inside, marginal probabilities in the Total row and column

	B	B^c	Total
A	0.26	0.34	0.60
A^c	0.14	0.26	0.40
Total	0.40	0.60	1

In the middle of the table we have the joint probabilities or intersections of events (i.e., $A \cap B$, $A \cap B^c$, etc.). The values in the totals columns or rows are the probabilities of each event happening (i.e., $P(A)$, $P(B^c)$, etc.). We call these probabilities **marginal probabilities** since they reside at the margin of the table.

8.7 Counting Rules

Counting rules are useful in probability as they allow us to determine the number of ways events can occur without listing out every possibility. Essentially, they facilitate the enumeration of possible outcomes for large sets.

The Fundamental Counting Principle

The most basic counting rule — and the one all the others build on — is the **Fundamental Counting Principle** (also called the Multiplication Principle). It states that if a task is completed in separate stages, the total number of ways to complete it is the product of the number of choices available at each stage.

THE FUNDAMENTAL COUNTING PRINCIPLE

$$m_1 \times m_2 \times \cdots \times m_k$$

where m_i is the number of choices at stage i, for a task made up of k stages.

EXAMPLE

A restaurant offers 3 appetizers, 4 entrées, and 2 desserts. The number of different three-course meals is $3 \times 4 \times 2 = 24$.

Factorial

To illustrate the factorial, consider the number of ways that we can organize a group of three people (Ann, Beth, Charlie). The enumeration of all the possible orderings in this case is not that tedious and it can be shown in a table:

Ordering	Order
1	(Ann, Beth, Charlie)
2	(Ann, Charlie, Beth)
3	(Beth, Ann, Charlie)
4	(Beth, Charlie, Ann)
5	(Charlie, Ann, Beth)
6	(Charlie, Beth, Ann)

The table shows that there are six possible ways to organize a group of three people. This is really just the Fundamental Counting Principle at work: there are 3 choices for the first spot, then 2 remaining choices for the second, and only 1 left for the third, giving $3 \times 2 \times 1 = 6$. The **Factorial** captures exactly this pattern — it counts how many ways a group of N objects can be organized. Formally, we write:

THE FACTORIAL

$$Factorial = N!$$

where the operator ! means to take the product of all positive integers from N down to 1. Note that 0! is equal to one.

If we then try to find in how many ways we can organize a group of three people, we can avoid enumeration by calculating $3! = 3 \times 2 \times 1 = 6$.

EXAMPLE

Let's assume that a manager has locked down the first five batters in a given game, but is undecided on the batting order. The number of ways that he can organize the five batters is given by $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Combinations

The **Combination** function counts the number of ways to choose x objects from a total of n objects. The order in which the x objects are listed does not matter. If repetition is not allowed use:

COMBINATIONS (NO REPETITION)

$$C(n,x) = \frac{n!}{(n-x)!x!}$$

EXAMPLE

Let's imagine visiting the Mexican restaurant Chipotle. We are choosing a bowl for lunch and we can choose from the following items: beans, chicken, rice, pico, guac, and beef. If we are only allowed to select four items out of the six, in how many ways can we construct a bowl? Below we use the combination formula to show that there are 15 possible bowls.

$$C(6,4) = \frac{6!}{(6-4)!4!} = \frac{6 \times 5}{2} = 15$$

There are situations when we wish to count all possible combinations inclusive of repetition of elements. If repetition is allowed we use the following formula:

COMBINATIONS (WITH REPETITION)

$$C(n+x-1,x) = \frac{(x+n-1)!}{(n-1)!x!}$$

EXAMPLE

Imagine you visit the ice cream store. You want a three scoop dish of ice cream. In how many ways can you choose your dish if the flavors are vanilla, chocolate, smores, brownie, and chips? Below we use the combination formula with repetition to show that the answer is 35.

$$C(7,3) = \frac{(5+3-1)!}{(5-1)!3!} = 7 \times 5 = 35$$

Permutations

The **Permutation** function also counts the number of ways to choose x objects from a total of n objects. However, the order in which the x objects are listed does matter. Note that since order matters, the permutation will always have a higher number of possible outcomes relative to the combination. When repetition is not allowed, permutations can be counted by:

PERMUTATIONS (NO REPETITION)

$$P(n,x) = \frac{n!}{(n-x)!}$$

EXAMPLE

The Olympic medal finals for the 100 meters sprint has eight runners competing for three medals. How many medal podiums are possible? There are a total of 336. Below you can see the calculation.

$$P(8,3) = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$$

As with combinations, there might be situations where we wish to count when repetition is allowed. If this is the case, use:

PERMUTATIONS (WITH REPETITION)

$$P(n,x) = n^x$$

EXAMPLE

You are trying to guess the combination of a lock that has three possible numbers. Since there are 10 numbers to choose from, there are $10^3 = 1000$ possible permutations.

8.8 Probability in Excel

Excel has a collection of functions that make counting and probability calculations straightforward.

Contingency tables: Use a **PivotTable** (`Insert → PivotTable`) to build a contingency table from raw data — place one variable in the Rows area, the other in the Columns area, and any field in the Values area as a count. To turn the counts into a joint probability table, use `Value Field Settings → Show Values As → % of Grand Total`. You can also count a specific combination of conditions directly with `=COUNTIFS(range1, criteria1, range2, criteria2)`.

Factorial: The `=FACT(number)` function returns the factorial of a number. For example, `=FACT(5)` returns 120.

Combinations: The `=COMBIN(number, number_chosen)` function counts the number of combinations without repetition. Use `=COMBINA(number, number_chosen)` when repetition is allowed.

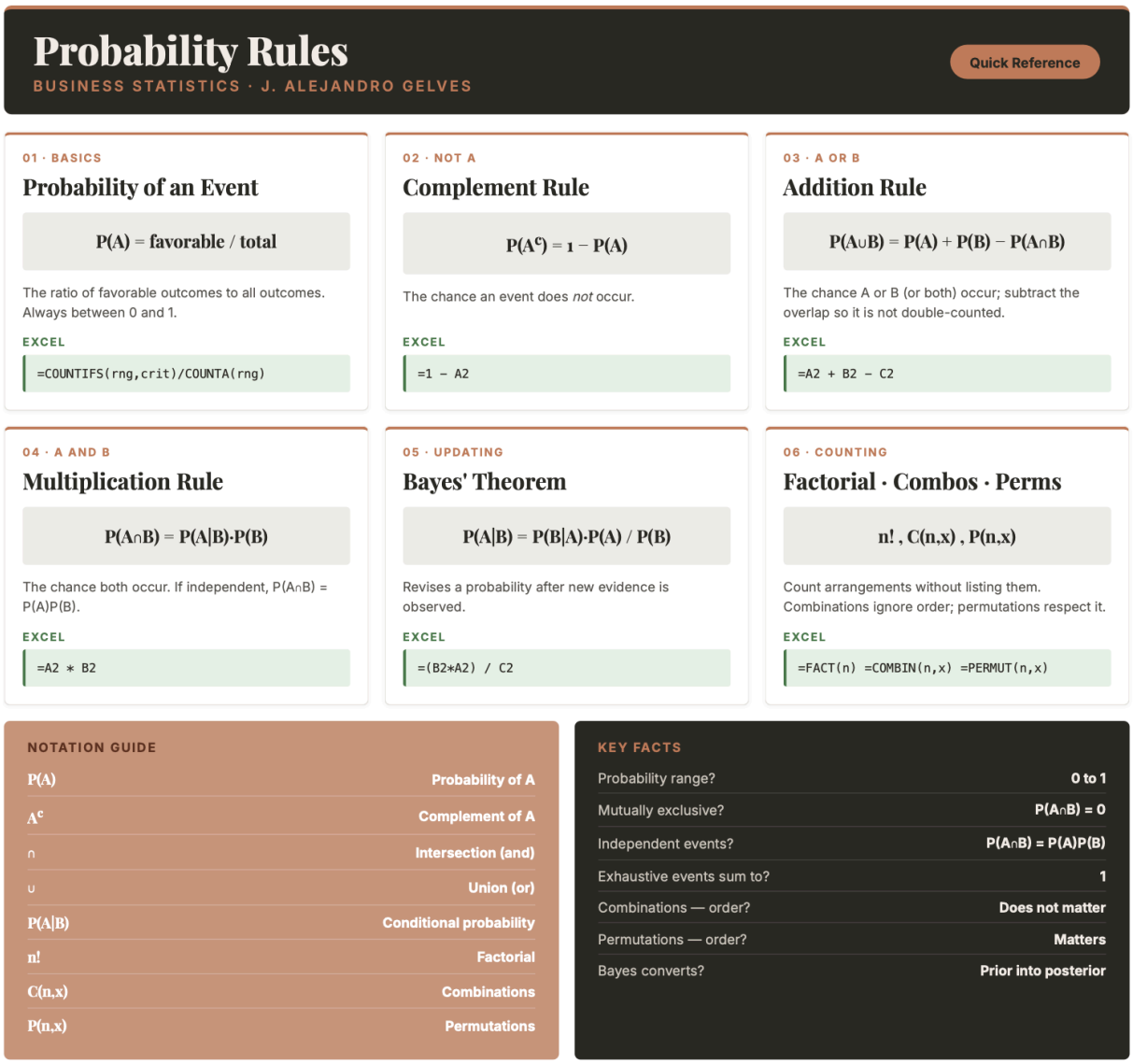
Permutations: The `=PERMUT(number, number_chosen)` function counts the number of permutations without repetition. Use `=PERMUTATIONA(number, number_chosen)` when repetition is allowed.

8.9 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=COUNTIFS(range1, criteria1, ...)` counts the observations that satisfy several conditions at once, useful for building contingency tables.
- `=FACT(number)` returns the factorial of a number.
- `=COMBIN(number, number_chosen)` returns the number of combinations without repetition; `=COMBINA(number, number_chosen)` allows repetition.
- `=PERMUT(number, number_chosen)` returns the number of permutations without repetition; `=PERMUTATIONA(number, number_chosen)` allows repetition.

8.10 Chapter Summary Cheat Sheet



8.11 Exercises

The following exercises will help you practice some probability concepts and formulas. In particular, the exercises work on:

- Calculating simple probabilities.
- Applying probability rules.
- Using counting rules.

Answers are in the Answers to Exercises at the end of the book. Try not to peak until you have a formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results with a calculator or Excel.

1. A sample space S yields five equally likely events, A, B, C, D, and E. Find $P(D)$, $P(B^c)$, and $P(A \cup C \cup E)$.

Answer on page 185 →

2. Consider the roll of a die. Define A as {1,2,3}, B as {1,2,3,5,6}, C as {4,6}, and D as {4,5,6}. Are the events A and B mutually exclusive, exhaustive, both or none? What about events A and D?

Answer on page 185 →

3. A recent study suggests that 33.1% of the adult U.S. population is overweight and 35.7% obese. What is the probability that a randomly selected adult in the U.S. is either obese or overweight? Are the events mutually exclusive and exhaustive? What is the probability that their weight is normal?

Answer on page 185 →

Exercise 2

For the following exercises, make your calculations by hand and verify results with a calculator or Excel.

1. Let $P(A) = 0.65$, $P(B) = 0.3$, and $P(A|B) = 0.45$. Calculate $P(A \cap B)$, $P(A \cup B)$, and $P(B|A)$.

Answer on page 185 →

2. Let $P(A) = 0.4$, $P(B) = 0.5$, and $P(A^c \cap B^c) = 0.24$. Calculate $P(A^c|B^c)$, $P(A^c \cup B^c)$, and $P(A \cup B)$.

Answer on page 186 →

3. Stock A will rise in price with a probability of 0.4, stock B will rise with a probability of 0.6. If stock B rises in price, then A will also rise with a probability of 0.5. What is the probability that at least one of the stocks will rise in price? Prove that events A and B are (are not) mutually exclusive (independent).

Answer on page 186 →

Exercise 3

1. Create a joint probability table from the contingency table below. Find $P(A)$, $P(A \cap B)$, $P(A|B)$, and $P(B|A^c)$. Determine whether the events are independent or mutually exclusive.

	B	B^c
A	26	34
A^c	14	26

Answer on page 186 →

Exercise 4

You will need the **Crash** data set to answer this question. The data shows information on several car crashes. Specifically, whether the crash was Head-On or Not Head-On and whether there was Daylight or No Daylight. You can download it here:

Download Data

1. Create a contingency table.

Answer on page 187 →

2. Find the probability that a) a car crash is Head-On, b) a car crash is in daylight c) a car crash is Head-On given that there is daylight.

Answer on page 187 →

3. Show that Crashes and Light are dependent.

Answer on page 188 →

Exercise 5

1. Use Bayes' Theorem in the following question. Let $P(A) = 0.7$, $P(B|A) = 0.55$, and $P(B|A^c) = 0.10$. Find $P(A^c)$, $P(A \cap B)$, $P(A^c \cap B)$, $P(B)$, and $P(A|B)$.

Answer on page 188 →

2. Some find tutors helpful when taking a course. Julia has a 40% chance to fail a course if she does not have a tutor. With a tutor, the probability of failing is only 10%. There is a 50% chance that Julia finds an available tutor. What is the probability that Julia will fail the course? If she ends up failing the course, what is the probability that she had a tutor?

Answer on page 188 →

Exercise 6

1. Calculate the following values and verify your results using Excel. a) 3!, b) 4!, c) C_6^8, d) P_6^8.

Answer on page 188 →

2. There are 10 players in a local basketball team. If we chose 5 players to randomly start a game, in how many ways can we select the five players if order doesn't matter? What if order matters?

Answer on page 188 →

9 Probability II

Studying discrete probability distributions is essential for understanding and predicting outcomes in uncertain scenarios, particularly in business, finance, and data science. These distributions model events with distinct, countable outcomes, such as customer purchases, defect counts, or market trends, enabling decision-makers to quantify risks and optimize strategies. Below, we introduce some popular distributions and their business application. Each distribution is a model of how an uncertain outcome behaves — the kind of model that, in the inference chapters, will let us judge how far a sample statistic might stray from the truth.

9.1 Random Variables

The study of probability distributions begins with a random variable. A **random variable** assigns a numerical value to each possible experimental outcome. The table below presents a collection of experiments along with their corresponding random variables.

Table 9.1: Experiments and their random variables

Experiment	Random variable (x)	Possible values
Contact five customers	Number of customers who place an order	0, 1, 2, 3, 4, 5
Operate a restaurant for one day	Number of customers	0, 1, 2, 3, ...
Sell a car	Result of sale	0 = no sale; 1 = sale

For restaurant owners, the number of customers they attract is a crucial variable to monitor. Tracking this variable and its possible values helps in making informed business decisions. Below we will study how this type of experiment and several others behave, so that we can make better decisions. Note that since the outcomes are integer values, the resulting distribution is discrete. The **probability mass function** (PMF) summarizes the possible values generated by a discrete probability distribution.

9.2 Expected Value and Variance

When summarizing a random variable, we are mostly interested in the variable's central tendency (Expected Value) and dispersion (Variance).

The **expected value** (mean) is a measure of central location. For a discrete random variable it is given by:

THE EXPECTED VALUE

$$E(x) = \mu = \sum xf(x)$$

where $f(x)$ is the probability mass function. For a continuous random variable it is $E(x) = \int_{-\infty}^{\infty} xf(x)\,dx$, where $f(x)$ is the probability density function.

The **variance** summarizes the deviation of the values of the random variable from the mean. It is calculated by:

THE VARIANCE

$$var(x) = E[(x - E(x))^2] = E(x^2) - [E(x)]^2$$

Note that this formula can be used for both discrete and continuous random variables.

EXAMPLE

Let's consider an incentive pay scheme for a company. The table below summarizes the scheme:

Bonus, x (in \$1,000)	Performance type	Probability, $f(x)$
10	Superior	0.15
6	Good	0.25
3	Fair	0.40
0	Poor	0.20

If a worker receives a "Superior" rating, they earn a \$10,000 bonus. Based on years of employee data, the company has determined that a "Superior" rating occurs 15% of the time. It also lists all of the possible ratings of its employees and their likelihood on the table.

To summarize this distribution we can calculate the mean bonus paid. Using the expected value formula we find that the company pays 4,200 dollars on average in compensation.

$$E(x) = (10 \times 0.15) + (6 \times 0.25) + (3 \times 0.4) + (0 \times 0.2) = 4.2$$

The variability of the compensation is given by:

$$var(x) = (100 \times 0.15) + (36 \times 0.25) + (9 \times 0.4) + (0 \times 0.2) - (4.2)^2 = 9.96$$

The expected value and the variance has helped us summarize the results of the company's incentive scheme. Mainly, the company pays on average an incentive of 4,200 dollars give or take 3,156 dollars.

9.3 Discrete Uniform Distribution

The **discrete uniform distribution** is a probability distribution that assigns equal probability to each outcome in a finite set of possible outcomes. In other words, each outcome in the set is equally likely to occur.

THE DISCRETE UNIFORM DISTRIBUTION

$$f(x) = 1/n$$

$$E(x) = \frac{\sum x_i}{n}$$

$$var(x) = \frac{\sum (x_i - E(x))^2}{n}$$

where n is the number of possible outcomes and x_i are the possible values.

EXAMPLE

Consider tossing a fair die. Since all outcomes are equally likely, the probability of the die landing on 6 is $f(x) = 1/6$. The expected value on a roll of a die is $E(x) = \frac{1+2+3+4+5+6}{6} = 3.5$ and the variance is $var(x) = \frac{(-2.5)^2+(-1.5)^2+(-0.5)^2+0.5^2+1.5^2+2.5^2}{6} \approx 2.92$.

9.4 Binomial Distribution

The binomial distribution is a probability distribution that describes the outcome of a sequence of n independent Bernoulli trials. In a Bernoulli trial, there are only two possible outcomes: "success" and "failure". The probability of success is denoted by p, and the probability of failure is denoted by $q = 1 - p$. In a sequence of n independent Bernoulli trials, the number of successes (x) is a random variable that follows a binomial distribution.

THE BINOMIAL DISTRIBUTION

$$f(x) = C_x^n(p^x)(1-p)^{n-x}$$

$$E(x) = np \qquad var(x) = np(1-p)$$

where n is the number of trials, x is the number of successes, p is the probability of success, and C_x^n is the number of ways there can be x successes in n trials.

EXAMPLE

Consider the following scenario. Imagine your favorite team in the NBA Finals, trailing by 2 points with seconds left. Your star player, a 90% free-throw shooter, gets three shots. What's the chance your team takes the lead?
Since shooting free-throws can be thought as a binomial experiment, we can find the probability using the binomial probability mass function. To take the lead, the star player needs to make all three shots. We can substitute three into the function:

$$f(3) = C_3^3(0.9^3)(0.1)^0 = 0.9^3 = 0.729$$

The function indicates that there is a 72.9% chance of taking the lead.

9.5 The Hypergeometric Distribution

The **hypergeometric distribution** is a probability distribution that describes the outcome of drawing a sample from a population without replacement. It is used to calculate the probability of drawing a certain number of successes (x) in a sample of a given size (n), where the success or failure of each individual draw is dependent on the success or failure of other draws.

The **hypergeometric** experiment differs from the binomial since:

- trials are not independent.
- the probability of success changes from trial to trial.

THE HYPERGEOMETRIC DISTRIBUTION

$$f(x) = \frac{C_x^r C_{n-x}^{N-r}}{C_n^N}$$

$$E(x) = n\frac{r}{N} \qquad var(x) = n\frac{r}{N}\left(1 - \frac{r}{N}\right)\left(\frac{N-n}{N-1}\right)$$

where n is the sample size, x is the number of successes in the sample, r is the number of successes in the population, and N is the number of elements in the population.

EXAMPLE

Picture a box of 12 electric fuses from the manufacturer Ontario Electric. Five fuses are known to be defective. An inspector randomly picks 3 fuses to test. What is the probability that exactly one of the three chosen is defective?
We can use the hypergeometric probability mass function to solve this problem. From the problem we know that $N = 12$, $n = 3$, $r = 5$, and $x = 1$. Substituting these values into the PMF yields:

$$f(1) = \frac{C_1^5 C_4^7}{C_3^{12}} = 0.4772$$

There is a 47.72% chance that the inspector finds a defective fuse in his sample of three.

9.6 Poisson Distribution

The **Poisson distribution** estimates the number of successes (x) over a specified interval of time or space.

THE POISSON DISTRIBUTION

$$f(x) = \frac{\mu^x e^{-\mu}}{x!}$$

where μ is the expected number of successes in any given interval (and also the variance), and e is Euler's number (2.71828...).

An experiment satisfies a Poisson process if:

- The number of successes with a specified time or space interval equals any integer between zero and infinity.
- The number of successes counted in non-overlapping intervals are independent.
- The probability of success in any interval is the same for all intervals of equal size and is proportional to the size of the interval.

EXAMPLE

Imagine a fast-food restaurant's drive-thru where, on average, 10 cars arrive in a 15-minute span. What's the likelihood of exactly 5 cars showing up in 15 minutes?

We can use the Poisson PMF function for this problem. The arrival rate is 10 cars every 15 minutes. So there are 10 successes in the 15 minute interval. We can substitute values into the PMF:

$$f(5) = \frac{10^5 e^{-10}}{5!} = 0.0378$$

There is a 3.78% chance of five cars arriving within a 15 minute interval.

9.7 Discrete Distributions in Excel

Excel has built-in functions for the binomial, Poisson, and hypergeometric distributions. The discrete uniform distribution does not have a dedicated function, but its probabilities are easy to compute by hand.

Probability mass function (PMF). Set the last (cumulative) argument to `FALSE` to get the probability of an exact value.

- Binomial: `=BINOM.DIST(x, n, p, FALSE)`. For the basketball example, `=BINOM.DIST(3, 3, 0.9, FALSE)` → 0.729.
- Poisson: `=POISSON.DIST(x, mean, FALSE)`. For the drive-thru example, `=POISSON.DIST(5, 10, FALSE)` → 0.0378.
- Hypergeometric: `=HYPGEOM.DIST(x, n, r, N, FALSE)`, where `x` is the number of successes in the sample, `n` is the sample size, `r` is the number of successes in the population, and `N` is the population size. For the fuse example, `=HYPGEOM.DIST(1, 3, 5, 12, FALSE)` → 0.4772.
- Discrete uniform: the probability of any single value is `=1/n`. For the die, `=1/6` → 0.167.

Cumulative probabilities. Set the last argument to `TRUE` to get $P(X \le x)$.

- Die $P(X \le 3)$: for the discrete uniform the cumulative probability is $(x - a + 1)/n$, so `=(3-1+1)/6` → 0.5.
- Basketball $P(\text{at least 2 shots}) = 1 - P(X \le 1)$: `=1 - BINOM.DIST(1, 3, 0.9, TRUE)` → 0.972.
- For any upper-tail probability, subtract the cumulative from 1.

Quantiles. For the binomial, `=BINOM.INV(n, p, alpha)` returns the smallest value x such that $P(X \le x) \ge alpha$. Excel has no inverse function for the Poisson, hypergeometric, or discrete uniform distributions; for those, build a column of cumulative probabilities and read off the smallest value whose cumulative probability reaches the desired percentile.

Generating random numbers. Use the Analysis ToolPak: `Data → Data Analysis → Random Number Generation`. Choose the distribution (for example, Binomial), enter the parameters

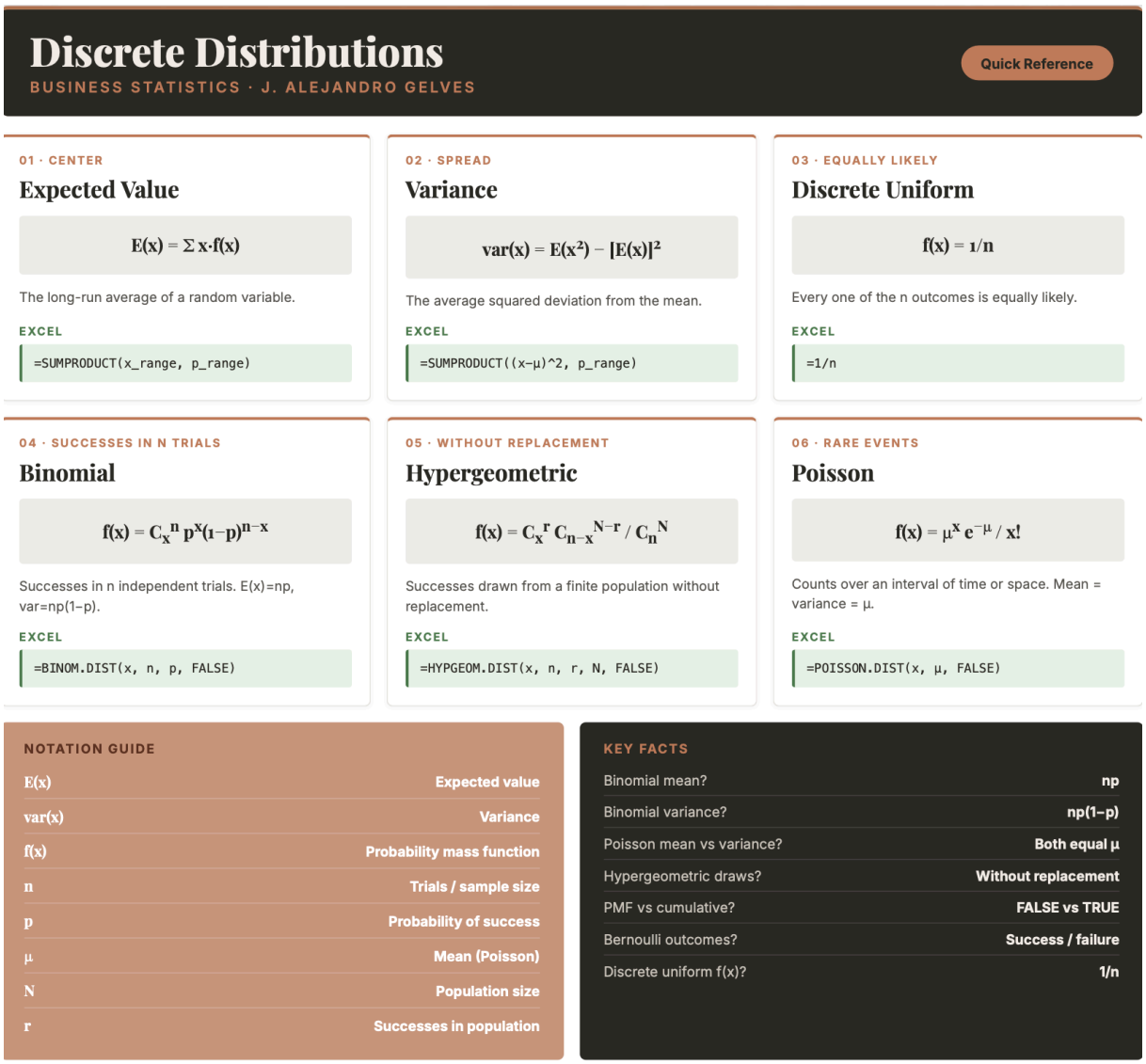
(number of trials and probability of success), specify how many values to generate, and Excel fills a column with random draws. You can then summarize them with a PivotTable or a histogram.

9.8 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=BINOM.DIST(x, n, p, cumulative)` returns the binomial probability. Use `FALSE` for the PMF and `TRUE` for the cumulative distribution.
- `=BINOM.INV(n, p, alpha)` returns the smallest value of x for which the cumulative binomial probability is at least `alpha`.
- `=POISSON.DIST(x, mean, cumulative)` returns the Poisson probability (`FALSE` for the PMF, `TRUE` for the cumulative distribution).
- `=HYPGEOM.DIST(sample_s, number_sample, population_s, number_pop, cumulative)` returns the hypergeometric probability.
- **Data Analysis ToolPak → Random Number Generation** draws random values from a chosen distribution.

9.9 Chapter Summary Cheat Sheet



9.10 Exercises

The following exercises will help you practice some probability concepts and formulas. In particular, the exercises work on:

- Calculating probabilities for discrete random variables.
- Calculating the expected value and standard deviation.
- Applying the binomial, Poisson and hypergeometric probability distributions.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results with a calculator or Excel.

1. Consider the table below. Calculate the mean and standard deviation. What is the probability that $x < 15$?

x	5	10	15	20
$P(X = x)$	0.35	0.3	0.2	0.15

Answer on page 188 →

2. Consider the table below. Calculate the mean and standard deviation. What is the probability that $x \geq -9$?

y	-23	-17	-9	-3
$P(Y = y)$	0.5	0.25	0.15	0.1

Answer on page 189 →

3. The returns on a couple of funds depends on the state of the economy. The economy is expected to be Good with a probability of 20%, Fair with probability of 50% and Poor with probability of 30%. Which fund would you choose if you want to maximize your return? What would you choose if you really dislike risk?

State of Economy	Fund 1	Fund 2
Good	20	40
Fair	10	20
Poor	-10	-40

Answer on page 189 →

Exercise 2

1. Use the table below. A portfolio has 200,000 dollars invested in Asset X and 300,000 dollars in asset Y. If the correlation coefficient between the two investments is 0.4, what is the expected return and standard deviation of the portfolio?

Measure	X	Y
Expected Return (%)	8	12
Standard Deviation (%)	12	20

Answer on page 189 →

Exercise 3

1. Let Z be a binomial random variable with $n = 5$ and $p = 0.35$ use the binomial formula to find $P(Z = 1)$, $P(Z \geq 2)$. What is the expected value and standard deviation of Z?

Answer on page 189 →

2. Let W be a binomial random variable with $n = 200$ and $p = 0.77$ use the binomial formula to find $P(W > 160)$, $P(155 \leq W \leq 165)$. What is the expected value and standard deviation of W?

Answer on page 189 →

3. Sixty percent of a firm's employees are men. Suppose four of the firm's employees are randomly selected. What is more likely, finding three men and one woman, or two men and one woman? Does your answer change if the proportion falls to 50%?

Answer on page 190 →

Exercise 4

1. Assume that S is a Poisson process with mean of $\mu = 1.5$. Calculate $P(S = 2)$ and $P(S \geq 2)$. What is the mean and standard deviation of S?

Answer on page 190 →

2. Assume that T is a Poisson process with mean of $\mu = 20$. Calculate $P(T = 14)$ and $P(18 \leq T \leq 23)$.

Answer on page 190 →

3. A local pharmacy administers on average 84 Covid-19 vaccines per week. The vaccines shots are evenly administered across all days. Find the probability that the number of vaccine shots administered on a Wednesday is more than eight but less than 12.

Answer on page 190 →

Exercise 5

1. Assume that X is a hypergeometric random variable with $N = 25$, $S = 3$, and $n = 4$. Calculate $P(X = 0)$, $P(X = 1)$, and $P(X \leq 1)$.

Answer on page 190 →

2. Compute the probability of at least eight successes in a random sample of 20 items obtained from a population of 100 items that contains 25 successes. What are the expected value and standard deviation of the number of successes?

Answer on page 191 →

3. For 1 dollar a player gets to select six numbers for the base game of Powerball. In the game, five balls are randomly drawn from 59 consecutively numbered white balls. One ball, called the Powerball, is randomly drawn from 39 consecutively numbered red balls. What is the probability that a player is able to match two out of five randomly drawn white balls? What is the probability of winning the jackpot?

Answer on page 191 →

10 Probability III

Studying continuous probability distributions is crucial for understanding and predicting outcomes in uncertain scenarios where variables can take on any value within a range, rather than being limited to distinct, countable outcomes. These distributions are widely used in fields such as engineering, economics, natural sciences, and machine learning to model phenomena like time to failure, stock prices, rainfall amounts, or human heights. By mastering continuous probability distributions, analysts and decision-makers can assess uncertainties, estimate probabilities of events, and develop data-driven strategies to address real-world challenges. Below, we introduce some popular continuous distributions and their practical applications. The normal distribution in particular is the workhorse that connects probability to inference: it describes how sample means behave, and so becomes the foundation for the estimates and tests developed in the chapters that follow.

10.1 Continuous Random Variables

Continuous random variables are characterized by their probability density function $f(x)$. The probability density function does not directly provide probabilities!

The probability of a continuous random variable assuming a single value is zero. Instead, probabilities are defined for intervals. These are calculated by areas under the PDF curve (integral).

10.2 Uniform Distribution

The **uniform** distribution assigns equal density across an interval $[a, b]$:

THE UNIFORM DISTRIBUTION

$$f(x) = \frac{1}{b-a} \quad (a \leq x \leq b)$$

$$E(x) = \frac{a+b}{2} \qquad var(x) = \frac{(b-a)^2}{12}$$

where a is the lower limit and b is the upper limit; $f(x) = 0$ outside $[a, b]$.

EXAMPLE

Consider the travel time between NY and CHI. Typically, this trips can take anywhere from 120-140 minutes. The specific time of a particular flight is unpredictable, but assumed to be anywhere between this interval. We assume a uniform distribution with upper limit of 140 and lower limit of 120. The probability that the flight take 122 minutes is zero since the interval has an infinite amount of possible values. The probability of the flight taking 130 minutes or less is 0.5.

$$f(x <= 130) = \frac{130 - 120}{20} = 0.5$$

The uniform distribution has no dedicated Excel function, but its cumulative probability follows directly from the formula. In Excel, `=(130-120)/(140-120)` returns 0.5.

10.3 Normal Distribution

The **normal** PDF is given by

THE NORMAL DISTRIBUTION

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}(\frac{x-\mu}{\sigma})^2}$$

where μ is the mean, σ is the standard deviation, $\pi \approx 3.1416$, and $e \approx 2.7183$.

The normal distribution has the following properties:

- It is symmetrical about the mean μ.
- The mean is at the middle and divides the area of the distribution into halves.
- The total area under the curve is equal to 1.
- The distribution is completely determined by its mean and standard deviation.

The **standard normal** has a mean of 0 and a standard deviation of 1. Otherwise, it has the exact same properties as the normal distribution.

EXAMPLE

Dr Tires is planning on offering a mileage guarantee on each set of tires they sell. They are considering a 40,000 mile guarantee. Given that the mean tire mileage is 36,500 miles with a standard deviation of 5,000 miles, we can estimate that the probability that a tire lasts more than 40,000 miles is 24.2%. In Excel, use `=1 - NORM.DIST(40000, 36500, 5000, TRUE)`, which returns 0.242.

10.4 Exponential Distribution

The **exponential distribution** is useful in computing probabilities for the time it takes to complete a task. It describes the time between events in a Poisson process.

The probability density function is given by:

THE EXPONENTIAL DISTRIBUTION

$$f(x) = \frac{1}{\mu} e^{-x/\mu} = \lambda e^{-\lambda x}$$

where μ is the mean and $\lambda = 1/\mu$ is the rate.

EXAMPLE

Let x represent the loading time for a truck at the Dock Dash loading dock. If the average loading time is 15 minutes, the probability that loading a truck will take 6 minutes or less is 32.97%. In Excel, the exponential functions use the rate $\lambda = 1/\mu = 1/15$: `=EXPON.DIST(6, 1/15, TRUE)` returns 0.3297.
We can also estimate that the probability that it takes between 6 and 18 minutes is 36.91%: `=EXPON.DIST(18, 1/15, TRUE) - EXPON.DIST(6, 1/15, TRUE)` returns 0.3691.

10.5 Triangular Distribution

The **triangular distribution** is characterized by a single mode (the peak of the distribution) and two boundaries. It is often used in situations where the lower and upper bounds of a potential outcome are known, but the exact likelihood of the outcome is uncertain.

The probability density function, expected value, and variance are:

THE TRIANGULAR DISTRIBUTION

$$f(x) = \begin{cases} \dfrac{2(x-a)}{(b-a)(c-a)} & a \le x < c \\ \dfrac{2}{b-a} & x = c \\ \dfrac{2(b-x)}{(b-a)(b-c)} & c < x \le b \\ 0 & \text{otherwise} \end{cases}$$

$$E(x) = \frac{a+b+c}{3} \qquad var(x) = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$$

where a is the minimum, b is the maximum, and c is the mode.

EXAMPLE

Bite Bliss is planning a new store in Williamsburg. It is estimated that the minimum weekly sales are 1000 and the maximum is 6000. They also estimate that the most likely outcome is around 3000. The probability that future sales will be between 2000 and 2500 is 12.5%. Excel has no triangular distribution function, but the cumulative probability follows from its formula. For a value x in the lower part of the range ($a \le x < c$), $F(x) = \frac{(x-a)^2}{(b-a)(c-a)}$. In Excel:

```
=(2500-1000)^2/((6000-1000)*(3000-1000)) - (2000-1000)^2/((6000-1000)*(3000-1000))
```

This returns $0.225 - 0.10 = 0.125$.

10.6 Continuous Distributions in Excel

Excel provides functions for the normal and exponential distributions. The uniform and triangular distributions do not have dedicated functions, but their probabilities follow directly from their formulas.

Uniform. The density is $f(x) = 1/(b-a)$ and the cumulative probability is $P(X \le x) = (x-a)/(b-a)$. Both are computed with ordinary arithmetic.

Normal. Use `=NORM.DIST(x, mean, standard_dev, cumulative)` — set `cumulative` to `TRUE` for $P(X \le x)$ and `FALSE` for the density. For the standard normal ($\mu = 0$, $\sigma = 1$) use `=NORM.S.DIST(z, cumulative)`. To find a value given a probability (a quantile), use `=NORM.INV(probability, mean, standard_dev)` or `=NORM.S.INV(probability)`.

Exponential. Use `=EXPON.DIST(x, lambda, cumulative)`, where `lambda` is the rate $\lambda = 1/\mu$. Set `cumulative` to `TRUE` for $P(X \le x)$ and `FALSE` for the density.

Triangular. Compute the cumulative probability from its formula: $F(x) = \frac{(x-a)^2}{(b-a)(c-a)}$ for $a \le x < c$, and $F(x) = 1 - \frac{(b-x)^2}{(b-a)(b-c)}$ for $c \le x \le b$.

10.7 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=NORM.DIST(x, mean, standard_dev, cumulative)` returns the normal probability (`TRUE` for the cumulative distribution, `FALSE` for the density).
- `=NORM.S.DIST(z, cumulative)` returns the standard normal probability.
- `=NORM.INV(probability, mean, standard_dev)` and `=NORM.S.INV(probability)` return the value (quantile) corresponding to a given cumulative probability.

- `=EXPON.DIST(x, lambda, cumulative)` returns the exponential probability, where `lambda` is the rate $1/\mu$.
- The **uniform** and **triangular** distributions have no built-in functions; compute their probabilities directly from the formulas above.

10.8 Chapter Summary Cheat Sheet

Continuous Distributions

BUSINESS STATISTICS · J. ALEJANDRO GELVES

Quick Reference

01 · CONSTANT DENSITY

Uniform

$$f(x) = 1/(b - a)$$

Equal density across [a, b]. Cumulative: (x−a)/(b−a).

EXCEL

```
=(x-a)/(b-a)
```

02 · BELL CURVE

Normal

$$f(x) = (1/\sigma\sqrt{2\pi})\, e^{-\frac{1}{2}((x-\mu)/\sigma)^2}$$

Symmetric, fully described by its mean μ and standard deviation σ.

EXCEL

```
=NORM.DIST(x, μ, σ, TRUE)
```

03 · STANDARDIZED

Standard Normal

$$z = (x - \mu) / \sigma$$

The normal distribution with mean 0 and standard deviation 1.

EXCEL

```
=NORM.S.DIST(z, TRUE)
```

04 · WAITING TIME

Exponential

$$f(x) = \lambda e^{-\lambda x}$$

Time between events in a Poisson process; rate λ = 1/μ.

EXCEL

```
=EXPON.DIST(x, λ, TRUE)
```

05 · THREE POINTS

Triangular

$$F(x) = (x-a)^2 / (b-a)(c-a)$$

Used when only the minimum, most-likely, and maximum are known.

EXCEL

```
=(x-a)^2/((b-a)*(c-a))
```

06 · QUANTILES

Inverse Normal

$$\text{value} \leftarrow \text{probability}$$

Go from a probability back to a value, such as a percentile or cutoff.

EXCEL

```
=NORM.INV(p, μ, σ)
```

NOTATION GUIDE

f(x)	Probability density
μ	Mean
σ	Standard deviation
a , b	Lower, upper limits
c	Mode (triangular)
λ	Rate (exponential)
z	Standard normal value

KEY FACTS

P(single value)?	Zero
Normal shape?	Symmetric bell
Standard normal?	μ = 0, σ = 1
Exponential rate λ?	1/μ
TRUE argument gives?	Cumulative P(X≤x)
Triangular needs?	min, mode, max
NORM.INV returns?	A value from a probability

10.9 Exercises

The following exercises will help you practice some probability concepts and formulas. In particular, the exercises work on:

- Calculating probabilities for continuous random variables.
- Calculating the expected value and standard deviation.
- Applying the uniform, normal, and exponential distributions.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

For the following exercises, make your calculations by hand and verify results with a calculator or Excel.

1. A random variable X follows a continuous uniform distribution with minimum of -2 and maximum of 4. Determine the height of the density function $f(x)$, the mean, the standard deviation, and calculate $P(X \leq -1)$.

Answer on page 191 →

2. Your internet provider will arrive sometime between 10:00 am and 12:00 pm. Suppose you have to run a quick errand at 10:00 am. If it takes 15 minutes to run the errand, what is the probability that you will be back before the internet provider arrives? What if you take 30 minutes?

Answer on page 191 →

Exercise 2

1. A random variable Z follows a standard normal distribution. Find $P(-0.67 \leq Z \leq -0.23)$, $P(0 \leq Z \leq 1.96)$, $P(-1.28 \leq Z \leq 0)$ and $P(Z > 4.2)$.

Answer on page 192 →

2. Let Y be normally distributed with $\mu = 2.5$ and $\sigma = 2$. Find $P(Y > 7.6)$, $P(7.4 \leq Y \leq 10.6)$, a y such that $P(Y > y) = 0.025$, and a y such that $P(y \leq Y \leq 2.5) = 0.4943$.

Answer on page 192 →

3. Assume that football game times are normally distributed with a mean of 3 hours and a standard deviation of 0.4 hour. What is the probability that the game lasts at most 2.5 hours? Find the maximum value for a game to be in the bottom 1% of the distribution.

Answer on page 192 →

Exercise 3

1. Random variable S is exponentially distributed with mean of 0.1. What is the standard deviation of S? What is $P(0.10 \leq S \leq 0.2)$?

Answer on page 192 →

2. A tollbooth operator has observed that cars arrive randomly at a rate of 360 cars per hour. What is the mean time between car arrivals? What is the probability that the next car will arrive within ten seconds?

Answer on page 192 →

Part IV

Infer

11 Inference I

Statistical inference is a cornerstone of modern business decision-making. It enables companies to extract meaningful insights from data amid uncertainty and complexity. Businesses can predict customer behavior, optimize marketing strategies, and assess risks by drawing reliable conclusions from sample data. In this chapter, we explore how we can draw insights about a population using sample data. This is the stage the whole book has been building toward: moving from describing a sample to reasoning about the population behind it, using probability to keep our uncertainty honest.

11.1 Statistical Inference

The power of statistical inference lies in allowing us to learn about a population by studying a sample. We estimate a **sample statistic**, such as the sample mean $\bar{x}$, and use it as a guess for the corresponding **population parameter**, such as the population mean μ — as long as the sample is drawn randomly. Certain properties of the sample mean make this a sound guess. Below we study three of them: unbiasedness, consistency, and precision, and then the shape of the sampling distribution through the Central Limit Theorem.

11.2 Unbiasedness

Unbiasedness states that the expected value of the sample mean equals the population mean.

UNBIASEDNESS

$$E[\bar{X}] = \mu$$

where $\bar{X}$ is the sample mean and μ is the population mean.

The difference between a sample mean and the population mean is called the **sampling error**.

SAMPLING ERROR

$$\text{Sampling Error} = \bar{x} - \mu$$

Sampling error is not a mistake — it is the natural consequence of sampling variability. It would be zero if the sample included the entire population, and it tends to shrink as the sample size grows.

EXAMPLE

Consider the four numbers below as the entire population:

x_1	x_2	x_3	x_4
3	12	18	30

The population mean is $\mu = \frac{3+12+18+30}{4} = 15.75$.
In practice we rarely see the whole population. Suppose instead we take a random sample of three elements. The table lists all possible samples of size three and their sample means:

Sample	Outcome	$\bar{x}$
1	{3,12,18}	11
2	{3,12,30}	15
3	{3,18,30}	17
4	{12,18,30}	20

None of the individual sample means equals $\mu = 15.75$ — each carries some sampling error. Yet the **mean of the sample means** is exactly the population mean:

$$\frac{11 + 15 + 17 + 20}{4} = \frac{63}{4} = 15.75 = \mu$$

This confirms $E[\bar{X}] = \mu$: although any single sample mean deviates from μ, on average the sample mean lands on the true population mean.

11.3 Consistency

Consistency states that as the sample size increases, the sample mean converges to the population mean. Larger samples produce more accurate estimates, so the probability that the sample mean deviates from μ by more than any fixed amount approaches zero as n grows.

EXAMPLE

Using the same population $\{3, 12, 18, 30\}$ with $\mu = 15.75$, we compare the average absolute sampling error for samples of size two, three, and four.
For all samples of size two:

Sample	Outcome	$\bar{x}$	$\bar{x} - \mu$
1	{3,12}	7.5	−8.25
2	{3,18}	10.5	−5.25
3	{3,30}	16.5	0.75
4	{12,18}	15	−0.75
5	{12,30}	21	5.25
6	{18,30}	24	8.25

The average absolute error for size two is $\frac{8.25+5.25+0.75+0.75+5.25+8.25}{6} = 4.75$.
For size three (means $11, 15, 17, 20$) the average absolute error falls to $\frac{4.75+0.75+1.25+4.25}{4} = 2.75$.
For size four (the entire population) the error is $\bar{x} - \mu = 15.75 - 15.75 = 0$.
As the sample size increases from two to four, the average error shrinks from 4.75 to 0.

i Note

To explore how the sampling error changes with the sample size, visit https://jagelves.shinyapps.io/Sampling/.

11.4 Precision

Precision means that as the sample size grows, the sample mean becomes a more tightly clustered estimate of μ. This is because the standard deviation of the sample mean — the **standard error** — is smaller than the population standard deviation.

THE STANDARD ERROR OF THE MEAN

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

where σ is the population standard deviation and n is the sample size. The sample size n inversely affects the spread: larger samples give smaller standard errors.

When the population is small relative to the sample, a **finite population correction** is applied: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$, where N is the population size. Most populations are large enough that this correction can be ignored.

EXAMPLE

For the population $\{3, 12, 18, 30\}$, the population variance is

$$\sigma^2 = \frac{(3-15.75)^2 + (12-15.75)^2 + (18-15.75)^2 + (30-15.75)^2}{4} = 96.1875$$

For a sample of size $n = 3$ from this finite population of size $N = 4$, the variance of the sample mean is

$$\frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{96.1875}{3} \cdot \frac{4-3}{4-1} = 10.6875$$

We can verify this directly from the four sample means $(11, 15, 17, 20)$:

$$Var(\bar{x}) = \frac{(11-15.75)^2 + (15-15.75)^2 + (17-15.75)^2 + (20-15.75)^2}{4} = 10.6875$$

The variance of the sample mean (10.6875) is far smaller than the population variance (96.1875) — and it shrinks further as n grows.

11.5 Asymptotic Normality and the Central Limit Theorem

To build confidence intervals or test hypotheses, we need the *shape* of the sampling distribution, not just its center and spread. This is where the **Central Limit Theorem** (**CLT**) becomes one of the most useful results in statistics. It states that, regardless of the population's distribution (provided it has finite variance), the standardized sample mean becomes approximately normal as n grows large.

CENTRAL LIMIT THEOREM (MEAN)

$$\frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \approx N(0, 1)$$

This lets us use the normal (z) distribution for probability statements even when the data are skewed or discrete. In summary, for the sample mean $\bar{X}$:

- If the population is normal, the sampling distribution of $\bar{X}$ is exactly normal for any n.
- If the population is non-normal, the sampling distribution of $\bar{X}$ is approximately normal for large n (a common rule of thumb is $n > 30$).

11.6 Inference for Proportions

Inference also applies to **proportions** — for example, the proportion of left-handed people or of customers who churn. A proportion is the outcome of a binomial process: the number of successes x in n trials, where x/n is the proportion of successes. To estimate the population proportion p, we use the **sample proportion** $\hat{p} = x/n$, which is unbiased ($E[\hat{p}] = p$). Its standard error is:

THE STANDARD ERROR OF THE PROPORTION

$$se(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

where p is the proportion and n is the sample size.

By the Central Limit Theorem, the sampling distribution of $\hat{p}$ is also approximately normal for a large enough sample.

CENTRAL LIMIT THEOREM (PROPORTION)

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx N(0,1)$$

Because the underlying population is binary, the normal approximation is only good when there are enough expected successes and failures — the usual rule is $np \geq 5$ and $n(1-p) \geq 5$ (some texts use a more conservative threshold of 10). In practice we check these conditions using the estimated $\hat{p}$.

> **Note**
>
> To explore the Central Limit Theorem and how the sample size affects the sampling distribution press here.

11.7 Sampling in Excel

Excel can both generate random samples and compute the normal probabilities used in inference. There is no need for a programming loop — you fill a block of cells with formulas, and each column can serve as a separate sample.

Generating random numbers. `=RAND()` returns a random decimal between 0 and 1, and `=RANDBETWEEN(min, max)` returns a random integer. To draw from a continuous uniform distribution on $[a, b]$, use `=a + (b - a)*RAND()`. To draw from a normal distribution, invert a random probability with `=NORM.INV(RAND(), mean, standard_dev)`. In modern Excel,

`=RANDARRAY(rows, cols, min, max, FALSE)` fills an entire block at once — for example, a block with one column per sample.

```
=RANDARRAY(10, 1000, 100, 200, FALSE)
```

This single formula creates 1000 samples of size 10, each drawn from a uniform distribution on $[100, 200]$.

Freezing a sample. `RAND`, `RANDBETWEEN`, and `RANDARRAY` recalculate every time the sheet changes (or when you press `F9`). To lock a sample in place, copy the block and use `Paste Special → Values`.

Computing probabilities. Use `=NORM.DIST(x, mean, standard_dev, TRUE)` for $P(X \leq x)$, `=1 - NORM.DIST(...)` for the upper tail, and the difference of two calls for an interval. The standard error is entered directly as the `standard_dev` argument, for example `=NORM.DIST(85, 80, 14/SQRT(100), TRUE)`.

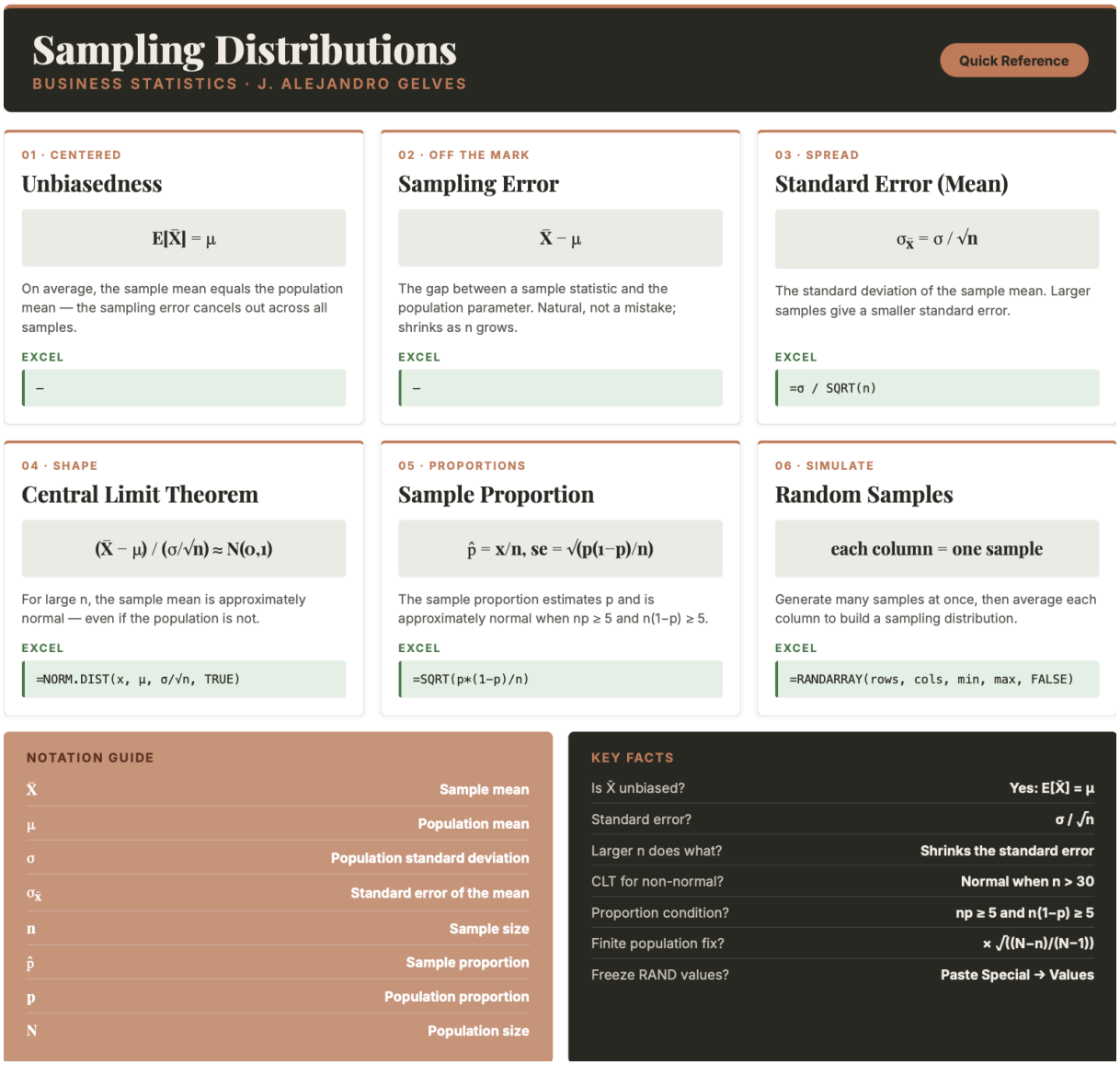
Summarizing samples. Use `=AVERAGE(range)` for the mean of a sample and `=STDEV.S(range)` for its standard deviation.

11.8 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=RAND()` returns a random decimal between 0 and 1.
- `=RANDBETWEEN(bottom, top)` returns a random integer in a range.
- `=RANDARRAY(rows, cols, min, max, integer)` fills a block with random numbers — ideal for generating many samples at once.
- `=NORM.INV(probability, mean, standard_dev)` returns a value from a probability; paired with `RAND()` it generates normal random numbers.
- `=NORM.DIST(x, mean, standard_dev, cumulative)` returns a normal probability (`TRUE` for the cumulative distribution).
- `=AVERAGE(range)` and `=STDEV.S(range)` return the mean and standard deviation of a sample.

11.9 Chapter Summary Cheat Sheet



11.10 Exercises

The following exercises will help you test your knowledge of inference. In particular, the exercises work on:

- The Central Limit Theorem.
- The sampling distribution of the mean.
- The sampling distribution of the proportion.

Answers are in the Answers to Exercises at the end of the book. Try not to peek until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

In this exercise we will simulate the Central Limit Theorem in Excel. The idea is to generate many samples of random numbers, where **each column is one sample**, and then study the distribution of the sample means.

1. Using the uniform distribution with a minimum of 100 and a maximum of 200, generate 1000 samples of size 10. Fill a block of 10 rows and 1000 columns with random numbers so that each column is one sample. What are the mean and standard deviation of this population?

Answer on page 193 →

2. Compute the mean of each sample (each column). Then find the mean and the standard deviation of the 1000 sample means. How do they compare to the population mean and to $\sigma / \sqrt{n}$?

Answer on page 193 →

3. Create a histogram of the sample means. Is the distribution uniform or normal? What is the probability that a sample mean falls between 140 and 160?

Answer on page 193 →

Exercise 2

1. A random sample of $n = 100$ is taken from a population with mean $\mu = 80$ and standard deviation $\sigma = 14$. Calculate the expected value and standard error of the sampling distribution of the sample mean. What is the probability that the sample mean falls between 77 and 85?

Answer on page 193 →

2. Assume that miles-per-gallon of combustion cars are normally distributed with a mean of 33.8 and a standard deviation of 3.5. What is the probability that the mean mpg of four randomly selected cars is more than 35? What is the probability that all four selected cars have mpg greater than 35?

Answer on page 194 →

Exercise 3

1. A random sample of $n = 200$ is taken from a population with a proportion of $p = 0.75$. Calculate the expected value and standard error of the sampling distribution of the proportion. What is the probability that the sample proportion is between 0.7 and 0.8?

Answer on page 194 →

2. Twenty-three percent of employees at a fintech firm work from home. If we take a sample of 50 employees, what is the probability that more than 20% of them work from home? What if the sample increases to 200? Why does the probability change?

Answer on page 194 →

Exercise 4

1. A production process for energy drinks is being evaluated. The machine that fills the cans is calibrated so that each can has 350ml of drink with a standard deviation of 10ml. Every hour, ten cans are sampled and the average amount of drink is recorded (see table below). Is the machine working properly?

1	2	3	4	5	6	7	8
$\bar{x} = 310$	$\bar{x} = 315$	$\bar{x} = 325$	$\bar{x} = 330$	$\bar{x} = 328$	$\bar{x} = 347$	$\bar{x} = 339$	$\bar{x} = 350$

Answer on page 194 →

2. The production of Good Guy dolls has a 1% defective rate. A quality inspector takes five samples of size 1000. The proportions are shown in the table below. Is the production process under control?

1	2	3	4	5
$\hat{p} = 0.009$	$\hat{p} = 0.012$	$\hat{p} = 0.008$	$\hat{p} = 0.011$	$\hat{p} = 0.0102$

Answer on page 194 →

12 Inference II

A single number is rarely the whole story. When a business estimates average customer spending or the proportion of defective units from a sample, the estimate is almost never exactly right — it carries sampling error. Reporting a lone **point estimate** hides this uncertainty. Instead, we surround the estimate with a range of plausible values and attach a level of confidence to it. This chapter shows how to build these **confidence intervals** for means and proportions, how the sample size and confidence level affect their width, and how to construct them in Excel. A confidence interval is the natural next step in learning from data: having accepted that a single sample gives only an imperfect view of the world, we report a range of plausible values for the population instead of pretending our estimate is exact.

12.1 Confidence Intervals

A **confidence interval** (**CI**) is a range of plausible values for a population parameter, together with a **confidence level** that states how often the procedure captures the true parameter. A 95% confidence interval means that if we repeated the sampling process many times, about 95% of the intervals produced would contain the true population parameter.

The confidence level and the **significance level** α are two sides of the same coin: a 95% confidence level corresponds to $\alpha = 0.05$, the risk that the interval misses the parameter. Every confidence interval has the same shape — a point estimate plus or minus a **margin of error**:

THE CONFIDENCE INTERVAL

$$\text{point estimate} \pm \text{margin of error}$$

The margin of error is a **critical value** (from the normal or t distribution) multiplied by the **standard error** of the estimate.

Two forces control the width of the interval. A **higher confidence level** widens it (more certainty requires a bigger net), and a **larger sample** narrows it (more data gives a more precise estimate).

> **Note**
>
> A confidence interval describes the *procedure*, not a single interval. Once computed, a specific interval either contains the parameter or it does not — the 95% refers to how often the method succeeds over many samples.

12.2 Confidence Intervals for the Mean

When the population standard deviation σ is known and the sampling distribution is normal (by the Central Limit Theorem), the confidence interval for the mean uses a z critical value:

CONFIDENCE INTERVAL FOR THE MEAN (

σ known)

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

where $\bar{x}$ is the sample mean, $z_{\alpha/2}$ is the normal critical value (e.g., 1.96 for 95%), σ is the population standard deviation, and n is the sample size.

EXAMPLE

A sample of $n = 50$ gives an average life expectancy of $\bar{x} = 78.1$ years, and the population standard deviation is $\sigma = 4.5$. For a 90% confidence interval, $z_{0.05} \approx 1.645$. The standard error is

$$\frac{\sigma}{\sqrt{n}} = \frac{4.5}{\sqrt{50}} \approx 0.637$$

so the interval is

$$78.1 \pm 1.645 \times 0.637 \approx 78.1 \pm 1.05 = [77.05,\ 79.15]$$

We are 90% confident that the true mean life expectancy is between 77.05 and 79.15 years.

12.3 The t-Distribution

In practice we almost never know the population standard deviation σ. When we estimate it with the sample standard deviation s, we add a second source of uncertainty: not only is $\bar{x}$ a noisy estimate of the mean, but s is a noisy estimate of the spread. Pretending s equals the true σ would make our intervals too narrow and overstate our confidence. The t-distribution corrects for this by accounting for the extra variability that comes from estimating σ from the same small sample.

Properties of the t-distribution. The t-distribution is bell-shaped and symmetric around zero, just like the standard normal, but it differs in three important ways. It is *flatter in the middle and heavier in the tails*, which means extreme values are more likely and, in turn, the critical values are larger — this is what makes a t-based interval wider than a z-based one. It is a *family* of curves rather than a single fixed shape: each is indexed by its degrees of freedom. And it is *asymptotically normal* — as the degrees of freedom increase, the tails thin out and the curve becomes almost indistinguishable from the standard normal. In practice, by about $n = 30$ the two are close enough that the choice rarely matters.

Why degrees of freedom? The degrees of freedom count the number of independent pieces of information available to estimate variability. To compute the sample standard deviation s, we first need the sample mean $\bar{x}$. Once $\bar{x}$ is fixed, the deviations from it must sum to zero, so the last deviation is completely determined by the other $n-1$: only $n-1$ of them are free to vary. We therefore "spend" one degree of freedom estimating the mean, leaving $n-1$ to estimate the spread. With fewer degrees of freedom (a small sample), s is a shakier estimate of σ, so the t-distribution has fatter tails to compensate; with more degrees of freedom, s becomes reliable and the distribution tightens toward the normal.

> **Note**
>
> Think of the degrees of freedom as a measure of how much the sample can be trusted to pin down the spread. A sample of $n = 5$ (4 degrees of freedom) leaves a lot of room for s to be off, so the interval must be wide; a sample of $n = 100$ (99 degrees of freedom) estimates σ well, and the t critical value is barely larger than z.

CONFIDENCE INTERVAL FOR THE MEAN (

σ unknown)

$$\bar{x} \pm t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation and $t_{\alpha/2}$ is the t critical value with $n-1$ degrees of freedom.

EXAMPLE

A sample of $n = 20$ customer wait times has a mean of $\bar{x} = 8.4$ minutes and a sample standard deviation of $s = 2.1$. For a 95% confidence interval with $n - 1 = 19$ degrees of freedom, $t_{0.025} \approx 2.093$. The margin of error is

$$2.093 \times \frac{2.1}{\sqrt{20}} \approx 0.98$$

so the interval is $8.4 \pm 0.98 = [7.42,\ 9.38]$ minutes. Because we estimated σ from the sample, the t critical value (2.093) is slightly larger than the corresponding z value (1.96), making the interval a little wider.

12.4 Confidence Intervals for a Proportion

For a population proportion p, the sample proportion $\hat{p} = x/n$ is approximately normal by the Central Limit Theorem, so the interval uses a z critical value:

CONFIDENCE INTERVAL FOR A PROPORTION

$$\hat{p} \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where $\hat{p}$ is the sample proportion and n is the sample size.

EXAMPLE

A random sample of 100 yields 40 successes, so $\hat{p} = 0.4$. For a 90% confidence interval, $z_{0.05} \approx 1.645$. The standard error is

$$\sqrt{\frac{0.4 \times 0.6}{100}} \approx 0.049$$

so the interval is

$$0.4 \pm 1.645 \times 0.049 \approx 0.4 \pm 0.081 = [0.319,\ 0.481]$$

We are 90% confident that the true population proportion is between 31.9% and 48.1%.

12.5 Determining the Sample Size

Because the margin of error shrinks as n grows, we can turn the question around: how large a sample do we need to achieve a desired margin of error E? Solving the margin-of-error expression for n gives:

REQUIRED SAMPLE SIZE

$$\text{Mean:}\quad n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2 \qquad\qquad \text{Proportion:}\quad n = \hat{p}(1-\hat{p})\left(\frac{z_{\alpha/2}}{E}\right)^2$$

Always round n **up** to the next whole number. If $\hat{p}$ is unknown, use $\hat{p} = 0.5$ for the most conservative (largest) sample size.

EXAMPLE

To estimate a mean within $E = 2$ units at 95% confidence when $\sigma = 10$:

$$n = \left(\frac{1.96 \times 10}{2}\right)^2 = (9.8)^2 = 96.04 \to 97$$

A sample of 97 observations is needed.

12.6 Confidence Intervals in Excel

Excel can produce the margin of error directly, so a confidence interval is just the sample mean plus or minus that margin.

Critical values. For a z critical value use `=NORM.S.INV(1 - alpha/2)` — for example, `=NORM.S.INV(0.975)` returns 1.96. For a t critical value use `=T.INV.2T(alpha, deg_freedom)`, which returns the positive two-tailed value directly — for example, `=T.INV.2T(0.05, 19)` returns 2.093.

Margin of error for a mean. Excel has two dedicated functions that return the margin of error in one step. Use `=CONFIDENCE.NORM(alpha, standard_dev, size)` when σ is known, and `=CONFIDENCE.T(alpha, standard_dev, size)` when it is estimated by s. The `alpha` argument is the significance level (`0.05` for 95%).

```
=CONFIDENCE.T(0.05, 2.1, 20)
```

This returns 0.98, the margin of error for the wait-time example. The interval is then `=8.4 - 0.98` and `=8.4 + 0.98`.

Margin of error for a proportion. There is no dedicated function, so build it from the standard error:

```
=NORM.S.INV(1 - 0.05/2) * SQRT(0.4*(1-0.4)/100)
```

This returns 0.096, the 95% margin of error for the proportion example.

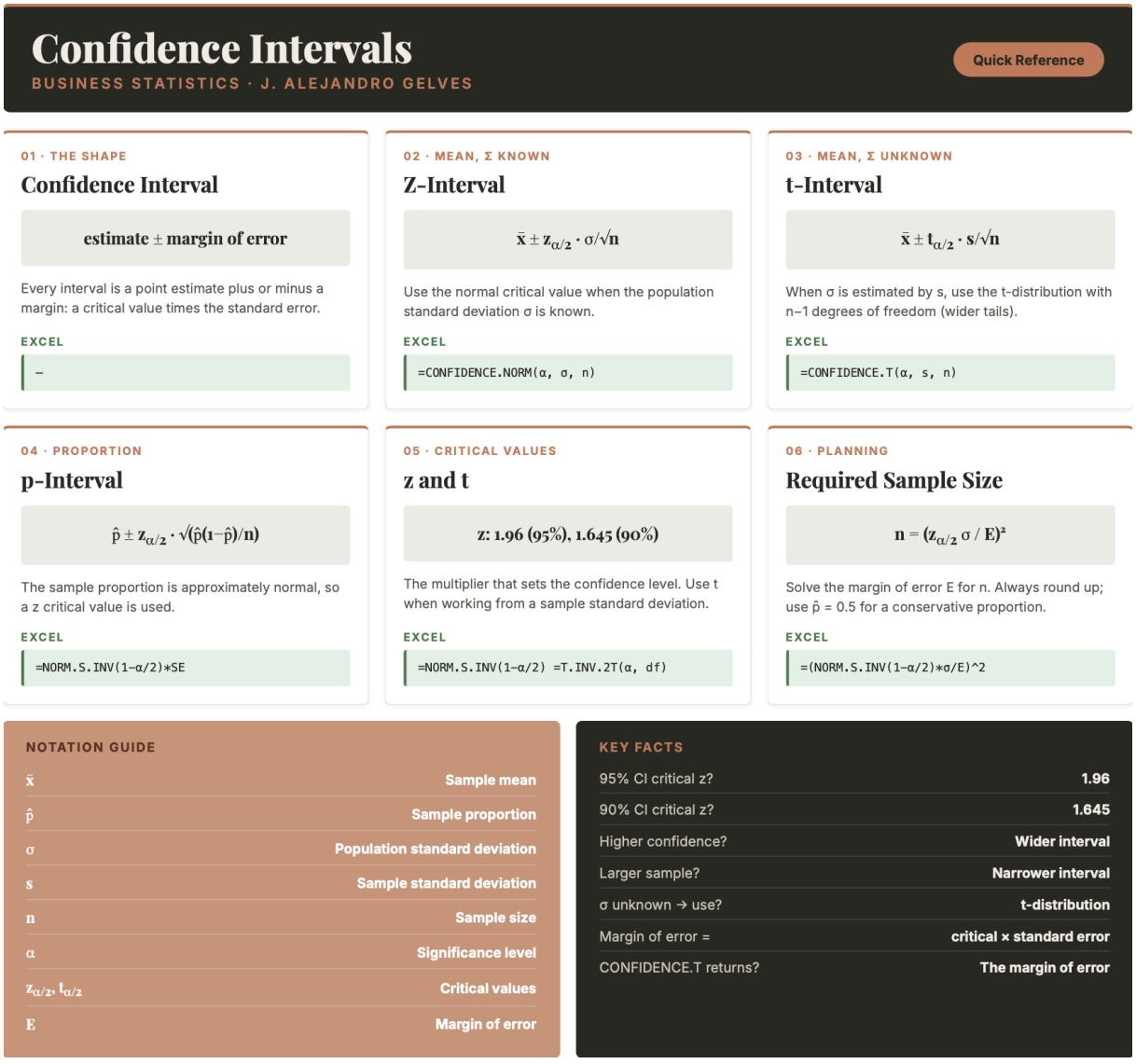
12.7 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=CONFIDENCE.NORM(alpha, standard_dev, size)` returns the margin of error for a mean using the normal distribution (when σ is known).

- `=CONFIDENCE.T(alpha, standard_dev, size)` returns the margin of error for a mean using the t-distribution (when σ is unknown).
- `=NORM.S.INV(probability)` returns a z critical value; `=T.INV.2T(probability, deg_freedom)` returns a two-tailed t critical value.
- `=AVERAGE(range)` and `=STDEV.S(range)` return the sample mean and sample standard deviation.

12.8 Chapter Summary Cheat Sheet



12.9 Exercises

The following exercises will help you test your knowledge of confidence intervals. In particular, the exercises work on:

- Simulating confidence intervals.
- Estimating confidence intervals for means using the t-distribution.
- Estimating confidence intervals for proportions.

Try not to peek at the answers until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

In this exercise you will simulate confidence intervals in Excel. The idea is to draw many samples, build a confidence interval around each one, and see how often the interval captures the population mean.

1. Using the exponential distribution with a rate of 0.02, generate 10,000 samples of size 50. Arrange the data so that **each row is one sample**. What are the population mean and standard deviation?

Answer on page 195 →

2. Compute the mean of each sample (each row). What are the mean and standard deviation of the 10,000 sample means?

Answer on page 195 →

3. Using the standard error from part 2, construct a 90% confidence interval around the first sample mean. Does it include the population mean of 50?

Answer on page 195 →

4. Build a 90% confidence interval around every sample mean and count how many include the population mean. About how many of the 10,000 intervals do you expect to succeed?

Answer on page 195 →

Exercise 2

1. A random sample of 24 observations is used to estimate the population mean. The sample mean is 104.6 and the standard deviation is 28.8. The population is normally distributed. Construct a 90% and a 95% confidence interval for the population mean. How does the confidence level affect the size of the interval?

Answer on page 196 →

2. A random sample from a normally distributed population yields a mean of 48.68 and a standard deviation of 33.64. Compute a 95% confidence interval assuming (a) the sample size is 16 and (b) the sample size is 25. What happens to the interval as the sample size increases?

Answer on page 196 →

Exercise 3

You will need the **sleep** data set for this problem. It records the extra hours of sleep (*extra*) for ten students under each of two sleep-inducing drugs (*group*). Construct a 95% confidence interval for each group. Which drug appears more effective at increasing sleeping time?

Download Data

Answer on page 196 →

Exercise 4

1. A random sample of 100 observations results in 40 successes. Construct a 90% and a 95% confidence interval for the population proportion. Can we conclude at either confidence level that the population proportion differs from 0.5?

Answer on page 196 →

2. You will need the **HairEyeColor** data set for this problem. It records the hair and eye color of 592 statistics students. Construct a 95% confidence interval for the proportion of students with Hazel eyes.

   Download Data

Answer on page 197 →

13 Inference III

Business decisions are often framed as claims to be tested: a new supplier promises fewer defects, a marketing team believes a campaign lifted sales, an analyst suspects average spending has changed. **Hypothesis testing** is the formal procedure for deciding whether the evidence in a sample is strong enough to support such a claim, or whether the difference we see could just be sampling error. In this chapter you will learn to state null and alternative hypotheses, compute a test statistic, weigh it against a chosen level of risk, and reach a defensible conclusion — all in Excel. This is where the cycle closes: we began by wanting to learn something about the world, and hypothesis testing is the formal step of asking whether the data we collected actually support the idea we started with.

13.1 Hypothesis Testing Intuition

The **null hypothesis** (H_0) is a statement about a population parameter — usually the status quo, or "no effect." It always contains some form of the equality sign ($=$, $\geq$, or $\leq$). The **alternative hypothesis** (H_a) directly contradicts the null and states the effect we are looking for, using a strict inequality ($\neq$, $>$, or $<$).

Suppose we hypothesize that the average annual income in the U.S. is \$50,000. When we take a large sample, the sample mean behaves like a normal random variable (by the Central Limit Theorem) centered at the hypothesized value. Our job is to decide whether the observed sample mean is "close enough" to \$50,000 to have plausibly come from that distribution, or "far enough" that we should doubt the null and favor the alternative.

EXAMPLE

We test whether the average income is \$50,000. A sample of 100 people has a mean of \$48,000. Is \$48,000 close to \$50,000, or far? It depends entirely on the **standard error** of the sampling distribution:

- If the standard error were small (say \$1,000), then \$48,000 is two standard errors below \$50,000 — strong evidence *against* the null.
- If the standard error were large (say \$15,000), then \$48,000 is only a fraction of a standard error away — perfectly *consistent* with the null.

Distance alone tells us nothing; distance measured in standard errors is what matters.

We capture that idea with the **test statistic**, which measures how far the sample mean sits from the hypothesized value in standard-error units.

TEST STATISTIC FOR A MEAN

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

with degrees of freedom $df = n - 1$, where $\bar{x}$ is the sample mean, μ_0 is the hypothesized mean, s is the sample standard deviation, and n is the sample size.

The larger the test statistic in absolute value, the stronger the evidence against the null hypothesis.

13.2 The Null and Alternative Hypotheses

Every test starts by writing the two competing hypotheses. The direction of the alternative determines whether the test is one-tailed or two-tailed.

For a mean:

- $H_0 : \mu \leq \mu_0$ vs. $H_a : \mu > \mu_0$ — right-tailed test
- $H_0 : \mu \geq \mu_0$ vs. $H_a : \mu < \mu_0$ — left-tailed test
- $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$ — two-tailed test

For a proportion:

- $H_0 : p \leq p_0$ vs. $H_a : p > p_0$ — right-tailed test
- $H_0 : p \geq p_0$ vs. $H_a : p < p_0$ — left-tailed test
- $H_0 : p = p_0$ vs. $H_a : p \neq p_0$ — two-tailed test

13.3 Two Approaches to a Decision

There are two equivalent ways to decide whether the observed difference is too large to be explained by sampling error. Both always reach the same conclusion at the same significance level α.

The critical value (rejection region) approach. Choose a significance level α, look up the **critical value** from the t or z distribution for that α and the direction of the test, and reject H_0 if the test statistic falls beyond it. For a two-tailed test at $\alpha = 0.05$ with $df = 99$, the critical values are approximately ± 1.984; a test statistic more extreme than these lands in the outer 5% of the distribution.

The p-value approach. Compute the **p-value** — the probability, assuming the null is true, of observing a test statistic at least as extreme as the one obtained. Then apply the decision rule:

reject H_0 if p-value $\leq \alpha$, and fail to reject it otherwise. A small p-value (say 0.003) means the data would be very unlikely if the null were true, so it is evidence *against* H_0; a large p-value (say 0.42) means the data are consistent with H_0.

13.4 Type I and Type II Errors

Because a decision is made from a sample, it can be wrong in two ways.

	H_0 is true	H_0 is false
Reject H_0	Type I error (α)	Correct decision (power)
Fail to reject H_0	Correct decision	Type II error (β)

A **Type I error** is rejecting a true null hypothesis (a false positive); its probability is the significance level α. A **Type II error** is failing to reject a false null hypothesis (a false negative); its probability is β. The **power** of a test, $1 - \beta$, is the chance of correctly detecting a real effect. Lowering α guards against false positives but makes false negatives more likely, so the choice of α balances the two risks.

13.5 Steps to Perform a Hypothesis Test

1. **State the hypotheses.** Write H_0 and H_a using the forms above.
2. **Choose the significance level** α. Common choices are 0.10, 0.05, and 0.01, corresponding to confidence levels of 90%, 95%, and 99%.
3. **Compute the test statistic.** For a mean, use the t statistic given above. For a proportion, use:

   TEST STATISTIC FOR A PROPORTION

   $$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

   where $\hat{p}$ is the sample proportion, p_0 is the hypothesized proportion, and n is the sample size. Note that the hypothesized p_0 is used in the standard error.

4. **Find the p-value** (replacing t with z for proportions):
   - Right-tailed test: p-value $= P(T \geq t)$
   - Left-tailed test: p-value $= P(T \leq t)$

- Two-tailed test: p-value $= 2 \times P(T \geq |t|)$

5. **Decide.** Reject H_0 if p-value $< \alpha$; otherwise fail to reject it. Remember that failing to reject H_0 is *not* proof that H_0 is true — it only means the sample did not provide enough evidence against it.

13.6 Hypothesis Testing in Excel

Excel does not have a one-click one-sample test, but every piece is a single formula. First build the test statistic from the sample, then convert it to a p-value.

The test statistic. With the data in a range, the t statistic for a mean is:

```
=(AVERAGE(A2:A21) - 100) / (STDEV.S(A2:A21) / SQRT(COUNT(A2:A21)))
```

p-values from the t-distribution. Use `=T.DIST(t, df, TRUE)` for a left-tailed test, `=T.DIST.RT(t, df)` for a right-tailed test, and `=T.DIST.2T(ABS(t), df)` for a two-tailed test. For the critical value use `=T.INV.2T(alpha, df)` (two-tailed) or `=T.INV(alpha, df)` (one-tailed).

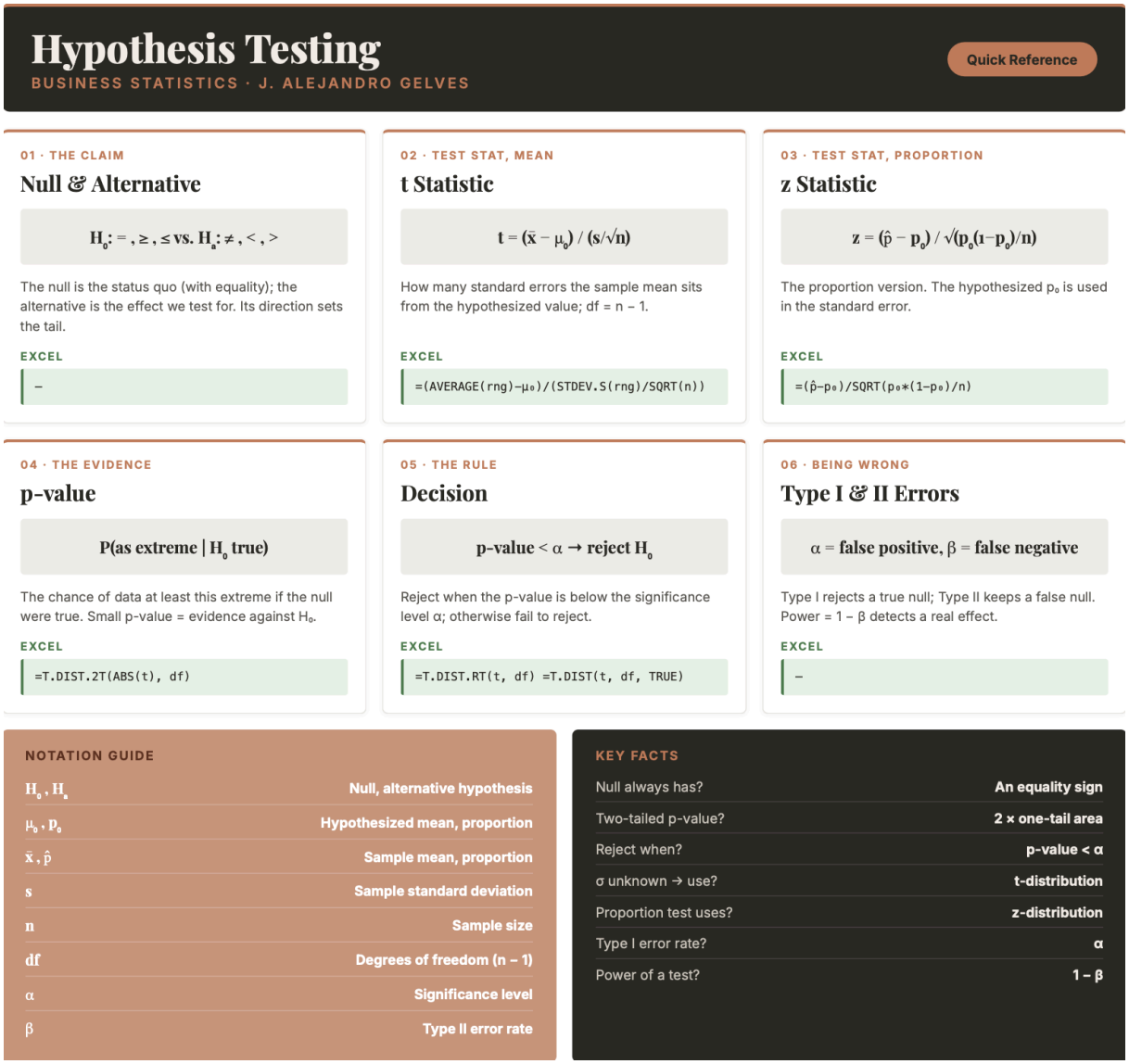
p-values for a proportion. The proportion test statistic is a z value, so use the standard normal: `=NORM.S.DIST(z, TRUE)` for a left-tailed test, `=1 - NORM.S.DIST(z, TRUE)` for a right-tailed test, and `=2*(1 - NORM.S.DIST(ABS(z), TRUE))` for a two-tailed test.

13.7 Excel Function Summary

Below is a list of the Excel functions used in this section:

- `=T.DIST(x, deg_freedom, TRUE)` returns the left-tail probability of the t-distribution.
- `=T.DIST.RT(x, deg_freedom)` returns the right-tail probability, and `=T.DIST.2T(x, deg_freedom)` returns the two-tailed probability (with a positive `x`).
- `=T.INV.2T(probability, deg_freedom)` returns the two-tailed t critical value.
- `=NORM.S.DIST(z, TRUE)` returns the standard normal cumulative probability, used for proportion tests.
- `=AVERAGE`, `=STDEV.S`, `=COUNT`, and `=COUNTIF` build the sample mean, standard deviation, size, and counts that go into the test statistic.

13.8 Chapter Summary Cheat Sheet



13.9 Exercises

The following exercises will help you test your knowledge of hypothesis testing. In particular, the exercises work on:

- Stating null and alternative hypotheses.
- Testing hypotheses about a population mean.
- Testing hypotheses about a population proportion.

Try not to peek at the answers until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

1. Consider the following hypothesis: $H_0 : \mu = 50$, $H_a : \mu \neq 50$. A sample of 16 observations yields a mean of 46 and a standard deviation of 10. Calculate the test statistic. At a 5% significance level, does the population mean differ from 50?

Answer on page 197 →

2. Consider the following hypothesis: $H_0 : \mu \geq 100$, $H_a : \mu < 100$. A sample from a normally distributed population yields the values in the table below. Test the hypothesis at a 1% significance level.

96	102	93	87	92	82

Answer on page 197 →

3. Consider the following hypothesis: $H_0 : \mu \leq 210$, $H_a : \mu > 210$. A sample from a normally distributed population yields the values in the table below. Test the hypothesis at a 10% significance level.

210	220	299	220	290	280	233	221	292	299

Answer on page 197 →

Exercise 2

According to nps.gov, the time between Old Faithful's eruptions is on average 92 minutes. Use the **faithful** data set and a two-tailed test to determine whether this claim is true.

Download Data

Answer on page 198 →

Exercise 3

1. To test whether the population proportion differs from 0.4, a scientist draws a random sample of 100 observations and obtains a sample proportion of 0.48. State the competing hypotheses. At a 5% significance level, does the population proportion differ from 0.4?

Answer on page 198 →

2. A sample of 320 observations yields 128 successes. Test $H_0 : p \geq 0.45$ against $H_a : p < 0.45$ at a 5% significance level.

Answer on page 198 →

3. Determine whether more than 50% of the observations in a population are below 10, using the sample data below. Conduct the test at a 1% significance level.

8	12	5	9	14	11	9	3	7	12

Answer on page 198 →

Exercise 4

According to worldatlas.com, 5% of the population has hazel eyes. Use the **HairEyeColor** data set and a two-tailed test to determine whether this claim holds for the 592 surveyed students.

Download Data

Answer on page 198 →

14 Regression and Inference

This final chapter brings hypothesis testing and regression together. You will learn to judge whether a correlation is real or just noise, whether two groups truly differ, and whether a regression coefficient carries any explanatory weight. These are the tools that turn a descriptive relationship into a defensible conclusion — letting you evaluate market trends, compare performance across groups, and decide which predictors actually matter in a model. It is a fitting close to the book: it unites the descriptive tool of regression with the inferential habit of asking whether a pattern is real, turning a fitted line into a tested claim about the world.

14.1 Correlation Significance

A sample correlation coefficient is almost never exactly zero, even when the two variables are unrelated. To decide whether an observed correlation reflects a real relationship in the population, we test the population correlation ρ:

- $H_0 : \rho \geq 0$ vs. $H_a : \rho < 0$ — left-tailed
- $H_0 : \rho \leq 0$ vs. $H_a : \rho > 0$ — right-tailed
- $H_0 : \rho = 0$ vs. $H_a : \rho \neq 0$ — two-tailed

CORRELATION TEST STATISTIC

$$t = \frac{r_{xy}\sqrt{n-2}}{\sqrt{1-r_{xy}^2}}$$

with degrees of freedom $df = n - 2$, where r_{xy} is the sample correlation coefficient and n is the sample size.

EXAMPLE

A sample of $n = 30$ shows a correlation of $r = 0.5$ between advertising and sales. Testing $H_0 : \rho = 0$ against $H_a : \rho \neq 0$:

$$t = \frac{0.5\sqrt{30-2}}{\sqrt{1-0.5^2}} = \frac{0.5 \times 5.29}{0.866} = 3.06$$

With $df = 28$, the two-tailed p-value is about 0.005. Because $0.005 < 0.05$, we reject the null and conclude the correlation is statistically significant.

14.2 Difference of Means Tests

We often want to know whether two populations have different means — for example, whether two divisions post different sales, or whether a treatment changes an outcome. The test statistic depends on how the two samples relate.

DIFFERENCE OF MEANS TEST STATISTICS

$$\text{Unpaired, unequal variances:} \quad t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$\text{Unpaired, equal variances:} \quad t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\text{Paired:} \quad t = \frac{\bar{d} - d_0}{s/\sqrt{n}}$$

where d_0 is the hypothesized difference, s_p^2 is the pooled variance, and $\bar{d}$ and s are the mean and standard deviation of the paired differences.
The degrees of freedom differ across the three tests:

Test	Degrees of freedom
Unpaired, unequal variances	$df = \dfrac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$ (rounded down)
Unpaired, equal variances	$df = n_1 + n_2 - 2$
Paired	$df = n - 1$, where n is the number of **pairs**, not the number of observations

A **paired** test is used when the two samples are naturally linked — the same subjects measured twice (before and after), for instance. An **unpaired** (independent) test is used when the two samples come from separate groups.

EXAMPLE (UNPAIRED).

A company evaluates a training program. Six employees who took the program score a mean of $\bar{x}_1 = 33.17$ on a productivity assessment (with $s_1 = 7.55$), while six *different* employees who did not take it score a mean of $\bar{x}_2 = 30.00$ (with $s_2 = 7.18$). Testing $H_0 : \mu_1 - \mu_2 = 0$ against $H_a : \mu_1 - \mu_2 \neq 0$ with unequal variances:

$$t = \frac{(33.17 - 30.00) - 0}{\sqrt{\frac{7.55^2}{6} + \frac{7.18^2}{6}}} = \frac{3.17}{4.25} = 0.74$$

The Welch formula gives about 10 degrees of freedom, and the two-tailed p-value is roughly 0.47. Because $0.47 > 0.05$, we fail to reject the null: the 3.17-point gap is swamped by the large variation *between* employees, so we cannot conclude the program helped.

EXAMPLE (PAIRED).

Now suppose those same six employees were each measured *before and after* the program, so every "after" score is matched to that person's own "before" score:

Employee	Before	After	Difference
1	20	23	3
2	32	36	4
3	25	27	2
4	40	43	3
5	28	31	3
6	35	39	4

The group means are identical to the unpaired case (30.00 before, 33.17 after), but now we work with each person's change. The individual gains are remarkably consistent: $\bar{d} = 3.17$ with $s = 0.75$. Testing $H_0 : \mu_d = 0$ against $H_a : \mu_d \neq 0$:

$$t = \frac{\bar{d} - 0}{s/\sqrt{n}} = \frac{3.17}{0.75/\sqrt{6}} = 10.30$$

With $df = n - 1 = 6 - 1 = 5$ (six *pairs*, not twelve observations) the two-tailed p-value is about 0.0001. Because $0.0001 < 0.05$, we strongly reject the null: the program clearly raised productivity.

The two examples use the **same numbers** yet reach opposite conclusions, and the reason is the design. The unpaired test must carry the large variation *between* employees — scores range from 20 to 40 — and that noise hides the modest improvement. The paired test cancels that person-to-person variation by looking only at each individual's change, leaving a small, consistent difference that stands out sharply (s drops from about 7.5 to 0.75). Whenever the data are naturally matched, pairing is the more powerful choice.

14.3 Regression Inference

After fitting a regression, two kinds of tests tell us whether the model has explanatory power.

Testing a single coefficient. Each coefficient is tested against zero — is this predictor related to y, holding the others fixed?

TEST FOR A SINGLE COEFFICIENT

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_a : \beta_j \neq 0, \qquad t = \frac{b_j}{se(b_j)}$$

where b_j is the estimated coefficient and $se(b_j)$ is its standard error, with $df = n - k - 1$.

Testing joint significance. The F-test asks whether *all* the slope coefficients are zero at once — is the model, as a whole, better than no predictors?

JOINT SIGNIFICANCE (F-TEST)

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0, \qquad F = \frac{SSR/k}{SSE/(n-k-1)} = \frac{MSR}{MSE}$$

where k is the number of predictors and n is the sample size.

The ANOVA table organizes these pieces:

Source	df	SS	MS	F
Regression	k	SSR	$MSR = \frac{SSR}{k}$	$F = \frac{MSR}{MSE}$
Residual	$n-k-1$	SSE	$MSE = \frac{SSE}{n-k-1}$	
Total	$n-1$	SST		

14.4 Regression and Inference in Excel

Correlation significance. Compute the correlation with `=CORREL(x_range, y_range)`, then turn it into a test statistic and p-value:

```
=(r*SQRT(n-2)) / SQRT(1 - r^2)
```

followed by `=T.DIST.2T(ABS(t), n-2)` for a two-tailed test (or `=T.DIST` / `=T.DIST.RT` for one tail).

Difference of means. The `=T.TEST(array1, array2, tails, type)` function returns the p-value directly. The `tails` argument is 1 (one-tailed) or 2 (two-tailed); the `type` argument is 1 (paired), 2 (two-sample, equal variances), or 3 (two-sample, unequal variances). For full output — the test statistic, critical values, and both one- and two-tailed p-values — use the **Data Analysis ToolPak**, which offers `t-Test: Paired Two Sample for Means`, `t-Test: Two-Sample Assuming Equal Variances`, and `t-Test: Two-Sample Assuming Unequal Variances`.

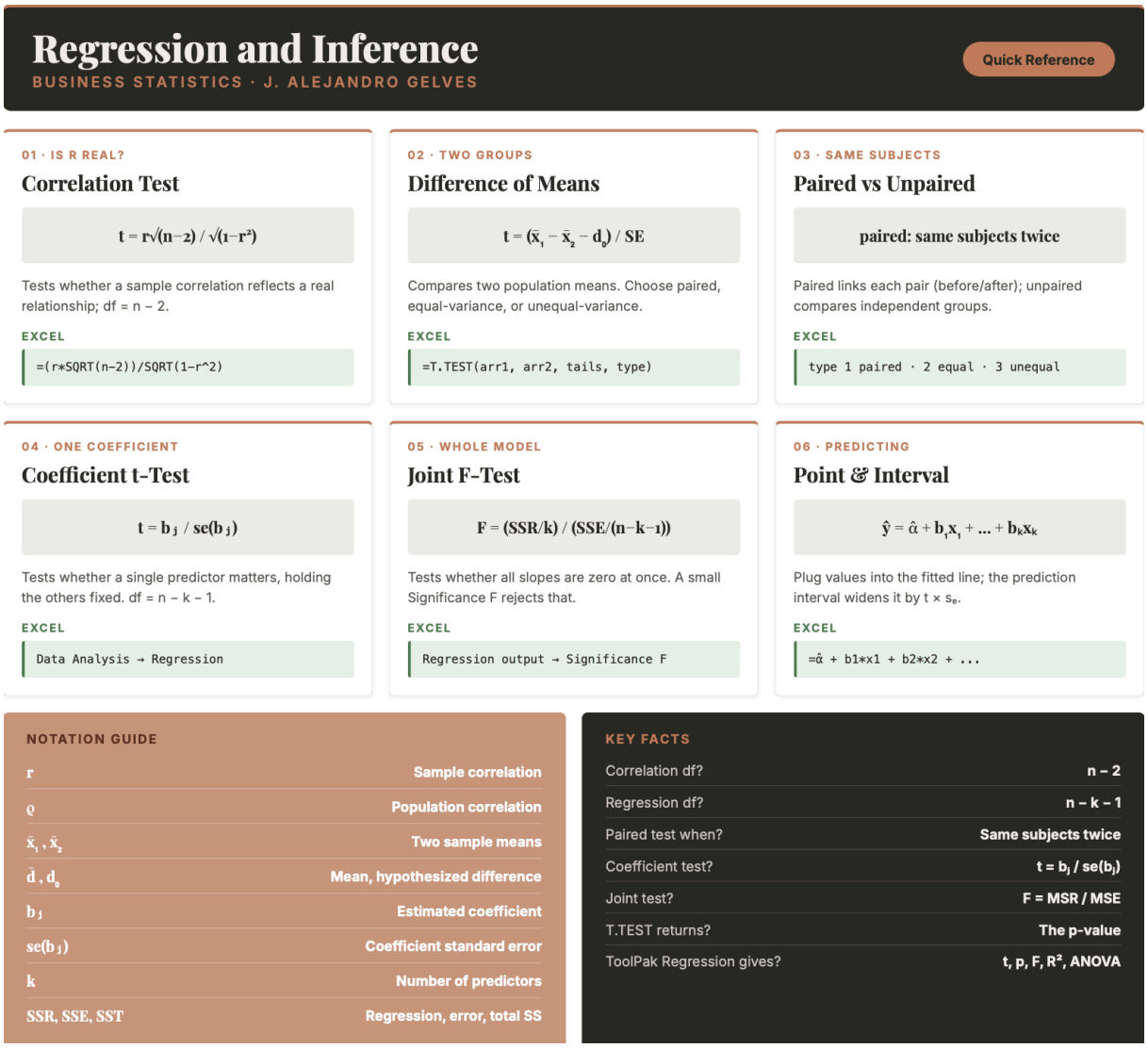
Regression inference. Run `Data → Data Analysis → Regression`. The output reports each coefficient with its standard error, t-statistic, and p-value (the individual coefficient tests), the F-statistic with its **Significance F** (the joint test), and the ANOVA table — everything in this chapter in a single step.

14.5 Excel Function Summary

Below is a list of the Excel functions and tools used in this section:

- `=CORREL(x_range, y_range)` returns the correlation coefficient, the starting point for a correlation test.
- `=T.TEST(array1, array2, tails, type)` returns the p-value of a difference-of-means test. `type` is 1 (paired), 2 (equal variances), or 3 (unequal variances).
- `=T.DIST.2T(x, deg_freedom)`, `=T.DIST(x, df, TRUE)`, and `=T.DIST.RT(x, df)` convert a test statistic to a p-value.
- **Data Analysis ToolPak → t-Test** performs the three difference-of-means tests with full output.
- **Data Analysis ToolPak → Regression** returns coefficients, standard errors, t-statistics, p-values, the F-statistic, R^2, and the ANOVA table.

14.6 Chapter Summary Cheat Sheet



14.7 Exercises

The following exercises will help you test your knowledge of regression and inference. In particular, the exercises work on:

- Determining the significance of a correlation.
- Conducting paired and unpaired tests of means.
- Determining the significance of regression coefficients, individually and jointly.
- Making predictions from a regression model.

Try not to peek at the answers until you have formulated your own answer and double checked your work for any mistakes.

Exercise 1

1. Consider the competing hypotheses $H_0 : \rho = 0$, $H_a : \rho \neq 0$. A sample of 25 observations reveals a correlation coefficient of 0.15 between two variables. At a 5% significance level, can we reject the null hypothesis?

Answer on page 199 →

2. Use the **Hitters** data set to examine the relationship between *Hits* and *Salary*. Calculate the correlation coefficient and test $H_0 : \rho = 0$ against $H_a : \rho \neq 0$ at the 1% significance level.

Download Data

Answer on page 199 →

Exercise 2

1. Use the **Hitters** data set to investigate whether the average number of *Hits* differs between the two leagues (American and National), using the *NewLeague* and *Hits* variables. Test at the 5% significance level. Is there reason to believe the population variances differ?

Download Data

Answer on page 199 →

2. Use the **BrainCancer** data set to test whether males have a higher average survival *time* than females, using the *sex* and *time* variables. Test at the 5% significance level. Is there reason to believe the population variances differ?

Download Data

Answer on page 199 →

Exercise 3

Use the **sleep** data set. At the 1% significance level, is there an effect of the drug on the 10 patients? Treat *group* 1 as the measurement before the drug and *group* 2 as after, so the two groups are the **same** patients measured twice.

Download Data

Answer on page 200 →

Exercise 4

1. Use the **Hitters** data set to investigate the effect of *HmRun*, *RBI*, and *Years* on a player's *Salary*. Which variables are statistically different from zero? Are the variables jointly significant? Does the R^2 suggest a good fit?

   Download Data

Answer on page 200 →

2. José Altuve had 28 home runs, 57 RBIs, and 12 years in the league. What salary does the model predict for him, and what is the 95% prediction interval? (The model predicts a 1987 salary.)

Answer on page 200 →

Part V

Toolkit

15 Excel Basics

This chapter reviews the essential Excel skills used throughout the book. The first sections walk through the Excel interface, how to navigate a worksheet, how references and formulas work, and the built-in functions for working with numbers, dates, text, and lookups. The chapter then closes with a dictionary of the statistical functions used in each chapter, so you can quickly find the right tool when you need it. No prior experience is assumed — if you have used Excel before, treat the early sections as a refresher.

15.1 The Excel Interface

Excel is an **electronic spreadsheet**. A spreadsheet is a grid made up of **cells**, and each cell holds a single piece of data — either text or a number. A cell is identified by its **column** (a letter) and its **row** (a number), so the cell in the first column and first row is `A1`. The **Name Box**, at the left end of the formula bar, always shows the reference of the currently selected cell; you can also type a reference (or a range name) into it and press `Enter` to jump straight to that location.

The **Home tab** collects the formatting tools you will use most often: fonts, number formats, borders, fill colors, and alignment. Formatting changes how a value is displayed without changing the value itself.

What makes a spreadsheet powerful is that cells can be *related to one another* through formulas. Instead of typing a result, you write a formula that references other cells, and Excel keeps the result up to date as those cells change.

A handful of keyboard shortcuts will save you a great deal of time. On Windows use `Ctrl`; on a Mac use `Cmd`.

Action	Shortcut
Copy	`Ctrl/Cmd + C`
Paste	`Ctrl/Cmd + V`
Cut	`Ctrl/Cmd + X`
Undo	`Ctrl/Cmd + Z`
Redo	`Ctrl/Cmd + Y`
Find	`Ctrl/Cmd + F`

To keep your work organized, you can rename a worksheet by double-clicking its tab and typing a new name, and you can color-code a tab by right-clicking it and choosing `Tab Color`. Selecting

a block of data and choosing `Home → Format as Table` (or pressing `Ctrl + T`) turns the range into a structured **Excel Table** with built-in filtering, banded rows, and named columns.

Cells and References

When a formula refers to another cell, it uses a **cell reference**. References come in three forms, and the difference matters as soon as you copy a formula:

- **Relative** (`A1`) adjusts automatically when the formula is copied to another cell.
- **Absolute** (`$A$1`) stays fixed when copied. The dollar signs lock both the column and the row.
- **Mixed** (`$A1` or `A$1`) locks only the column or only the row.

For example, `=A1*2` multiplies the value in `A1` by two. Pressing `F4` (or `Cmd + T` on a Mac) while editing a reference cycles through the relative, absolute, and mixed forms.

Ranges

A **range** is a group of adjacent cells, written as the top-left cell, a colon, and the bottom-right cell. For example, `A1:A10` is a single column of ten cells, and `A1:C10` is a rectangular block. A range is the Excel equivalent of a list of values, and it is what most functions take as an input. You can reference an entire column with `A:A` or an entire row with `1:1`.

15.2 Entering Formulas and Functions

Every calculation in Excel begins with an equals sign (`=`). A **function** relates an input (its arguments) to an output. The workflow is always the same: type `=`, name the function, read the tooltip that Excel shows to remind you of the arguments, fill in the arguments separated by commas, and press `Enter`. If you are unsure how a function works, click the `fx` button next to the formula bar for guided help.

The two functions you will reach for first are `SUM` and `AVERAGE`. To total the values in cells `A2` through `A5`:

```
=SUM(A2:A5)
```

and to average them:

```
=AVERAGE(A2:A5)
```

Once a formula is written, you rarely retype it. Select the cell and drag the small square at its bottom-right corner (the **fill handle**) to copy the formula. Because references are relative by default, copying **down** makes Excel adjust the row numbers, and copying **across** makes it adjust the column letters. This is what lets a single formula compute a whole column of results. Many functions also accept a **criteria** argument that filters the data, such as `=AVERAGEIF(A2:A100, ">=1", B2:B100)`. If a formula cannot be evaluated, Excel returns an error such as `#DIV/0!`, `#VALUE!`, `#N/A`, or `#REF!`.

Order of Operations

Excel evaluates formulas following the standard order of operations: parentheses first, then exponents, then multiplication and division, and finally addition and subtraction. For example, `=2^3*2+2` is evaluated as $(2^3) \times 2 + 2 = 18$, and `=2+4*5` is evaluated as $2 + (4 \times 5) = 22$. When in doubt, add parentheses to make the intended order explicit — `=(2+4)*5` returns 30.

Array Formulas

A few calculations — like a conditional variance — must be entered as **array formulas**, which operate on an entire range at once. In modern Excel these "spill" automatically, but in older versions you must complete them by pressing `Ctrl + Shift + Enter` instead of just `Enter`.

15.3 Navigating and Selecting

Moving quickly around a large worksheet is a skill worth practicing. Turning off the grid with `View → Gridlines` gives a cleaner canvas, especially when building a report or a chart.

The navigation shortcuts below let you move and select without the mouse:

- `Ctrl + Arrow` jumps to the edge of the current block of data in that direction.
- `Ctrl + Shift + Arrow` **selects** everything from the active cell to the edge of the data — the fastest way to grab a whole column of values.
- `F2` (`fn + F2` on some Macs) toggles the active cell into edit mode so you can see and adjust the formula in place.
- `Ctrl + Enter` fills every cell in a selection with the same formula at once.

When you want to keep headers visible while scrolling, use `View → Freeze Panes`. `View → Split` divides the window into independently scrolling panes, and you can `Hide` rows, columns, or entire sheets you do not want to see.

15.4 Range Names

Rather than referring to `$B$2:$B$100`, you can give a range a memorable **name** and use it in formulas and charts. There are three ways to create one: select the range and go to `Formulas → Define Name`; type a name directly into the **Name Box** and press `Enter`; or select a block that includes labels and use `Formulas → Create from Selection` to name each column or row from its header.

Names follow a few rules: they cannot contain spaces, they cannot look like a cell reference (so `Q1` is not allowed as a name), and they must begin with a letter or underscore. Use the **Name Manager** (`Formulas → Name Manager`) to review, edit, or delete existing names. A well-named range such as `Returns` makes `=AVERAGE(Returns)` far easier to read and audit than a raw address.

15.5 Data Types

The main data types in Excel are number, text, logical, date, and error.

- **Number** includes integers and decimals. Numbers are right-aligned in a cell by default.
- **Text** represents labels or strings and is left-aligned by default.
- **Logical** values are `TRUE` or `FALSE`, usually produced by a comparison.
- **Date and time** values are stored as serial numbers (see below), which is why dates can be added and subtracted.
- **Error** values are the result of an invalid operation, such as dividing by zero.

15.6 Working with Dates and Time

Excel stores every date as a **serial number**: day `1` is January 1, 1900, day `2` is January 2, 1900, and so on. The **integer part** of the number counts the days since that starting point, while the **decimal part** represents the time as a fraction of a 24-hour day. So `0` corresponds to `12:00 am`, `0.5` corresponds to `12:00 pm`, and a value like `44000.75` is a specific date at `6:00 pm`.

Because dates are really numbers, the way a cell is *formatted* determines whether you see `1/1/2024` or `45292`. If a calculation returns a date when you wanted a plain number (or vice versa), change the cell's number format under `Home → Number`. To pull just the **time** (the decimal part) out of a date-time value, use `MOD` with a divisor of 1:

```
=MOD(A1, 1)
```

Useful date and time functions include `=TODAY()` and `=NOW()` for the current date and date-time, `=DATE(year, month, day)` to build a date, and `=YEAR()`, `=MONTH()`, `=DAY()`, and `=WEEKDAY()` to extract the parts of a date.

15.7 Working with Text

Text functions let you clean and reshape labels — for example, splitting a "Last, First" name into two columns. The most useful ones are:

- `=TRIM(text)` removes extra spaces, leaving only single spaces between words.
- `=LEN(text)` returns the number of characters in the text.
- `=LEFT(text, n)` and `=RIGHT(text, n)` return the first or last `n` characters.
- `=MID(text, start, n)` returns `n` characters beginning at position `start`.
- `=SEARCH(find, text)` (or `=FIND`) returns the position of a character or substring; `SEARCH` is not case-sensitive, `FIND` is.
- `=UPPER(text)`, `=LOWER(text)`, and `=PROPER(text)` change capitalization.
- `=SUBSTITUTE(text, old, new)` replaces every occurrence of `old` with `new`.
- `=REPLACE(text, start, n, new)` replaces `n` characters starting at a position with new text.

As an example, suppose cell `D6` contains `Doe, John`. The position of the comma is `=SEARCH(",", D6)`, which returns 4. The last name is then `=LEFT(D6, SEARCH(",", D6) - 1)` → `Doe`, and the first name is `=RIGHT(D6, LEN(D6) - SEARCH(",", D6) - 1)` → `John`. To join text back together, use the ampersand operator: `=B6 & " " & A6`, or the `=CONCAT()` function.

15.8 Lookup Functions

Lookup functions find a value in a table based on a key. `=VLOOKUP(lookup_value, table, column_number, [match])` searches down the first column of a table and returns a value from a column to its right; `=HLOOKUP()` does the same across the first row. The modern replacement, `=XLOOKUP(lookup_value, lookup_range, return_range)`, is more flexible because the lookup and return ranges can be anywhere and it can look left as well as right.

The optional match argument controls **exact** versus **approximate** matching. An approximate (binary) search assumes the lookup column is **sorted**: Excel repeatedly compares the target with the middle value and discards half the list each time, which is very fast on large data sets — but it returns wrong answers if the data is not sorted. For most business lookups you want an exact match.

Lookups also support **wildcards** in the search text: `?` matches any single character (so `B?ll` matches `Ball` and `Bell`), and `*` matches any number of characters (so `B*ll` matches `Ball`, `Bell`, and `Barbell`). Use `~` before a `?` or `*` when you need to match the literal character.

15.9 Logical and Conditional Functions

The `=IF(test, value_if_true, value_if_false)` function returns one of two values depending on whether a condition is met, and `IF` statements can be nested inside one another. When you have several conditions to check in order, `=IFS(test1, value1, test2, value2, ...)` is cleaner. The logical helpers `=AND(...)`, `=OR(...)`, and `=NOT(...)` combine or reverse conditions.

A closely related family aggregates values that meet a condition:

- `=COUNTIF(range, criteria)` counts the cells that satisfy a condition, and `=COUNTIFS(...)` handles several conditions at once.
- `=SUMIF(range, criteria, sum_range)` totals the values that meet a condition, and `=SUMIFS(...)` allows several conditions.
- `=AVERAGEIF(range, criteria, average_range)` averages the values that meet a condition.

When a criterion is a comparison, write it as text in quotes, for example `">2"` or `">=1"`. To rank values, `=RANK.AVG(number, range)` returns the position of a value within a list (averaging ties). A common use is to highlight the top items in a chart: rank each value, then use a formula such as `=IF(rank <= 5, value, NA())` so that only the top five carry a value and the rest return `NA()`.

15.10 OFFSET and MATCH

`=MATCH(lookup_value, array, 0)` returns the **position** of a value within a row or column (the `0` requests an exact match). It does not return the value itself — it tells you *where* the value sits, which you can then feed to another function.

`=OFFSET(reference, rows, cols, [height], [width])` returns a range that is a given number of rows and columns away from a starting `reference`. The optional `height` and `width` arguments let it return a whole block rather than a single cell. Combining `MATCH` (to locate a position) with `OFFSET` (to retrieve the value at that position) creates flexible lookups; the same idea is often written with `INDEX` and `MATCH`.

Because `OFFSET` can size a range on the fly, it is the key to **dynamic ranges**. Pairing it with `=COUNTA(range)` — which counts the non-empty cells — lets you define a named range that automatically grows as you add data. A chart built on that name then updates itself whenever new rows arrive, with no manual re-selection.

15.11 Sorting, Filtering, and Dynamic Arrays

The `Data → Sort` and `Data → Filter` tools reorder and subset a table interactively. Excel also provides **dynamic array** functions that return a whole range of results that "spill" into neighboring cells:

- `=SORT(range)` returns the range sorted.
- `=FILTER(range, condition)` returns only the rows that satisfy a condition.
- `=RANK(number, range)` returns the rank of a value.

These combine naturally with the conditional functions. For example, to keep only the years that appear in a second, larger list, you can flag matches with `COUNTIF` and pass the result to `FILTER`:

```
=FILTER(Years, COUNTIF(OtherYears, Years))
```

`COUNTIF` returns a `1` for each year found in `OtherYears` and a `0` otherwise, and `FILTER` keeps only the rows flagged with a `1`.

15.12 Importing and Web Data

To bring outside data into Excel you can open a CSV directly with `File → Open`, or use `Data → Get Data → From Text/CSV` for control over how columns are separated. `Data → Get Data → From Web` pulls tables from a web page — for example, stock history from a finance site. Imported data becomes a **query**; under `Data → Queries & Connections` (and `Data Source Settings`) you can refresh it, change the source, or adjust how it loads. Excel's **Stocks** and **Geography** data types can also enrich a plain list of tickers or country names with live fields. When previewing a large import, freeze the header row with `View → Freeze Panes` and jump to the end with `Ctrl + Down` or `Ctrl + End`.

15.13 PivotTables

A **PivotTable** (`Insert → PivotTable`) automatically summarizes and counts data, making it the primary tool for building frequency and relative frequency distributions. You drag fields into the **Rows**, **Columns**, and **Values** areas, and Excel aggregates the underlying data for you — counting, summing, or averaging — without writing a single formula. PivotTables are used throughout the descriptive statistics chapters; see *Descriptive Statistics II* for a worked example.

15.14 Charts and Visualization

Charts are created from `Insert → Charts`. Choosing the right type matters: use a **bar chart** to compare categories and a **line chart** to show change over time. A good chart maximizes the *data-to-ink ratio* — that is, it removes anything that does not convey information. In practice that means removing gridlines, adding subtle borders to bars, cleaning up the axes (sensible tick marks and units), placing the title inside the plot, and adding data labels by right-clicking a series and choosing `Add Data Labels`.

To compare two series, add a second set of values to the chart. When the two series are on very different scales — a count and a percentage, say — plot one of them on a **secondary axis** using a **combo chart** (`Chart Design → Change Chart Type → Combo`). You can **highlight** a particular bar, such as the maximum or the top few, by adding a helper column that returns the value only for the rows to highlight and `NA()` for the rest; because Excel does not plot `NA()`, only the highlighted bars appear, and setting the series overlap to `100%` layers them on top of the originals. A **dummy variable** that switches between `0` and `1` can be plotted as a shaded band to mark periods such as recessions. Once you have styled a chart the way you like, right-click it and choose `Save as Template` to reuse the formatting later.

15.15 The Data Analysis ToolPak

Some analyses — such as regression — require the **Analysis ToolPak**, an add-in that ships with Excel but is turned off by default. To enable it on Windows, go to `File → Options → Add-ins → Manage: Excel Add-ins → Go`, check `Analysis ToolPak`, and click `OK`. On a Mac, use `Tools → Excel Add-ins`. Once enabled, the ToolPak appears under `Data → Data Analysis` and provides one-click tools for descriptive statistics, regression, random number generation, and more.

15.16 Statistical Function Dictionary

The remainder of the chapter lists the statistical Excel functions used in the book, grouped by the chapter in which they first appear. Each is explained in detail, with worked examples, in its home chapter.

General Purpose

- `=SUM(range)` adds all of the numbers in a range.
- `=COUNT(range)` counts the cells that contain numbers, while `=COUNTA(range)` counts all non-empty cells.

- `=MIN(range)` and `=MAX(range)` return the smallest and largest values in a range.
- `=ABS(number)` returns the absolute value of a number.
- `=SQRT(number)` returns the square root of a number.
- `=IF(test, value_if_true, value_if_false)` returns one of two values depending on whether a condition is met.
- `=AND(...)`, `=OR(...)`, and `=NOT(...)` combine or reverse logical conditions.

Tables and Charts (Descriptive Stats II)

- **PivotTable** (`Insert → PivotTable`) automatically summarizes and counts data, making it the primary tool for building frequency and relative frequency distributions.
- `=IFS(test1, value1, test2, value2, ...)` checks multiple conditions in order and returns the value for the first one that is true. It is used to assign each numerical observation to its class (bin) when building a frequency distribution.

Measures of Central Location (Descriptive Stats III)

- `=AVERAGE(range)` calculates the mean of a range of numbers.
- `=MEDIAN(range)` calculates the median.
- `=MODE(range)` returns the single most frequent value, while `=MODE.MULT(range)` returns all modes when the data is multimodal.
- `=SUMPRODUCT(array1, array2)` multiplies corresponding values in two ranges and sums the results. Dividing by `=SUM(weights)` gives the weighted mean.
- `=GEOMEAN(range)` calculates the geometric mean. For rates of change, apply the function to the growth factors (`1 + r`) and subtract one at the end.
- `=AVERAGEIF(range, criteria, average_range)` calculates the mean of the values that meet a given condition.

Measures of Dispersion (Descriptive Stats IV)

- `=COUNTIF(range, criteria)` counts the number of cells that meet a condition.
- `=MINIFS(min_range, criteria_range, criteria)` and `=MAXIFS(max_range, criteria_range, criteria)` return the minimum and maximum values filtered by one or more conditions.

- `=VAR.S(range)` calculates the sample variance; `=VAR.P(range)` calculates the population variance.
- `=STDEV.S(range)` calculates the sample standard deviation (the square root of the sample variance); `=STDEV.P(range)` calculates the population standard deviation.
- `=VAR(IF(criteria_range="condition", value_range))` and `=STDEV(IF(criteria_range="con value_range))` calculate a conditional variance and standard deviation. Excel has no native `VARIF` or `STDEVIF`, so these are entered as array formulas.
- **Coefficient of Variation** has no dedicated function; compute it as `=STDEV.S(range) / AVERAGE(range)`.

Measures of Shape and Position (Descriptive Stats V)

- `=PERCENTILE.INC(array, k)` returns the kth percentile of a range using the formula $L_p = 1 + k(n-1)$, where k is expressed as a decimal between 0 and 1. `=PERCENTILE.EXC(array, k)` uses the alternative formula $L_p = k(n+1)$.
- `=QUARTILE.INC(array, quart)` returns the five point summary. The argument `quart` takes values 0 (minimum), 1 (Q_1), 2 (median), 3 (Q_3), and 4 (maximum).
- `=STANDARDIZE(x, mean, std_dev)` calculates the z-score for a value. It can also be computed manually as `=(x - AVERAGE(range)) / STDEV.S(range)`.
- `=SKEW(range)` calculates the skewness of a range.
- `=KURT(range)` calculates the excess kurtosis of a range.

Measures of Association (Regression I)

- `=COVARIANCE.S(array1, array2)` calculates the sample covariance between two variables; `=COVARIANCE.P()` calculates the population covariance.
- `=CORREL(array1, array2)` calculates the correlation coefficient, which is always between -1 and 1.
- `=CORREL(array1, array2)^2` calculates the coefficient of determination R^2 by squaring the correlation coefficient.

Regression (Regression II)

- **Data Analysis ToolPak → Regression** returns the coefficients, R^2, adjusted R^2, standard error, and the ANOVA table in a single output.

- `=SLOPE(y_range, x_range)` and `=INTERCEPT(y_range, x_range)` return the slope and intercept of a simple regression line.
- `=FORECAST.LINEAR(x, y_range, x_range)` returns the predicted value of y for a given x.
- `=RSQ(y_range, x_range)` returns the R^2, and `=STEYX(y_range, x_range)` returns the residual standard error.

Probability Distributions (Probability I–III)

- `=FACT(number)`, `=COMBIN(n, x)`, and `=PERMUT(n, x)` return the factorial, number of combinations, and number of permutations. Use `=COMBINA()` and `=PERMUTATIONA()` when repetition is allowed.
- `=BINOM.DIST(x, n, p, cumulative)`, `=POISSON.DIST(x, mean, cumulative)`, and `=HYPGEOM.DIST(sample_s, number_sample, population_s, number_pop, cumulative)` return discrete probabilities (`FALSE` for the PMF, `TRUE` for the cumulative distribution). `=BINOM.INV(n, p, alpha)` returns a binomial quantile.
- `=NORM.DIST(x, mean, std_dev, cumulative)` and `=NORM.S.DIST(z, cumulative)` return normal probabilities, while `=NORM.INV(probability, mean, std_dev)` and `=NORM.S.INV(probability)` return normal quantiles. `=EXPON.DIST(x, lambda, cumulative)` returns exponential probabilities.

Answers to Exercises

1 Descriptive Stats I

EXERCISE 1, QUESTION 1 back to page 11

- *The "Title" variable represents the names of books. Therefore, this is a categorical variable.*
- *"Price" represents the cost of each book in a numeric format, making it a numerical variable.*
- *"Year Published" indicates the publication year of each book. It is numerical.*
- *If "Rating" represents a numerical score based on a continuous scale (e.g., average user ratings on a platform like Goodreads), it is numerical because arithmetic operations like averaging or comparing differences are meaningful. If "Rating" represents predefined categories (e.g., "Excellent," "Good," "Fair," "Poor") or is interpreted as ranks without meaningful differences between values, it would be categorical.*

EXERCISE 1, QUESTION 2 back to page 11

- *The measurement scale is nominal for "Title" since these are labels used to identify each book and do not have a numerical meaning or order.*
- *If "Rating" represents a score (e.g., 4.2, 4.0) given to each book, it is numerical and could be considered interval data because the scale represents a meaningful difference, but it may not have an absolute zero or meaningful ratios (e.g., a book rated 4.0 is not "twice as good" as one rated 2.0).*
- *"Price" is a measurable quantity with a meaningful zero (e.g., a book priced at $0 means it is free), making it ratio data.*
- *"Year" is interval data because the zero point is arbitrary (year 0 does not represent the absence of time) and differences between years are meaningful (e.g., 1897 - 1818 = 79 years).*

EXERCISE 2, QUESTION 1 back to page 11

The variable Year is measured on the interval scale because the observations can be ranked, categorized and measured when using this kind of scale. However, there is no true zero point so we cannot calculate meaningful ratios between years.

EXERCISE 2, QUESTION 2 back to page 12

The variable Quarter is measured on the ordinal scale, even though it contains numbers.

EXERCISE 2, QUESTION 3 back to page 12

The variable Deliveries is measured on the ratio scale. It is the strongest level of measurement because it allows us to categorize and rank the data as well as find meaningful differences between observations. Also, with a true zero point, we can interpret the ratios between observations.

EXERCISE 3 back to page 12

- *Customer ID: Categorical, Nominal. These are arbitrary labels used only for identification. There is no order or numerical meaning; you can only determine if two IDs are the same or different.*
- *Satisfaction Level: Categorical, Ordinal. The ratings have a clear order (higher numbers indicate greater satisfaction), but the intervals between levels are not necessarily equal (the difference between 4 and 5 may not be perceived the same as between 1 and 2). Ratios are not meaningful.*
- *Visit Duration: Numerical, Ratio. Time in minutes has a true zero (0 minutes means no time spent), equal intervals, and meaningful ratios (e.g., 60 minutes is twice as long as 30 minutes). All arithmetic operations are valid.*
- *Beverage Type: Categorical, Nominal. These are unordered categories/labels for different drinks. You can group or count them, but there is no natural order or numerical relationship between categories like "Latte" and "Tea".*

2 Descriptive Stats II

EXERCISE 1, QUESTION 1 back to page 20

The most common diagnosis is Meningioma, a slow-growing tumor that forms from the membranous layers surrounding the brain and spinal cord. The diagnosis represents about 48.28% of the sample.

Diagnosis	Frequency	Relative
Meningioma	42	48.28%
LG glioma	9	10.34%
HG glioma	22	25.29%
Other	14	16.09%

In Excel, build a PivotTable: drag the `diagnosis` *field to the Rows area and again to the Values area (Excel counts each category). To add the relative frequency, drag* `diagnosis` *to the Values area a second time and set* `Value Field Settings → Show Values As → % of Column Total`*. Alternatively, use* `=COUNTIF(range, "Meningioma")` *for each category and divide by the total* `=COUNTA(range)`*.*

EXERCISE 1, QUESTION 2 back to page 20

The majority of diagnoses are Meningioma. Low grade glioma is the least common. High grade glioma and Other have about the same frequency.

With the PivotTable selected, go to `Insert → Charts → Clustered Column` *to create the bar chart.*

EXERCISE 1, QUESTION 3 back to page 20

33 people did not survive Meningioma — the highest number of non-survivals. The survival rate of Meningioma is only 21.43%.

Status	Meningioma	LG glioma	HG glioma	Other
0	33	5	5	9
1	9	4	17	5

In Excel, build a PivotTable with `diagnosis` *in the Columns area,* `status` *in the Rows area, and* `status` *in the Values area as a count. To obtain the survival rates, set* `Show Values As → % of Column Total`*; the survival rate for Meningioma is* $9/42 = 21.43\%$.

EXERCISE 2, QUESTION 1 back to page 20

The highest frequency is in the $80 < x \leq 90$ bin. 8 temperatures were between $50 < x \leq 60$ degrees.

Interval	Frequency
$(50, 60]$	8
$(60, 70]$	25
$(70, 80]$	52
$(80, 90]$	54
$(90, 100]$	14

In Excel, add a helper column with an `=IFS()` *formula to assign each temperature to its class, then build a PivotTable to count the observations in each bin, exactly as shown in the chapter.*

EXERCISE 2, QUESTION 2 back to page 20

The temperature was 5.23% of the time between 50 and 60; the temperature was 70 or less 33 times; the temperature was above 70, 78.43% of the time.

Interval	Frequency	Relative	Cumulative	Rel. Cumulative
$(50, 60]$	8	5.23%	8	5.23%
$(60, 70]$	25	16.34%	33	21.57%
$(70, 80]$	52	33.99%	85	55.56%
$(80, 90]$	54	35.29%	139	90.85%
$(90, 100]$	14	9.15%	153	100.00%

In the PivotTable, drag the bin field to the Values area a second time and use `Show Values As → % of Column Total` *for the relative frequency, and* `Show Values As → Running Total In` *for the cumulative and relative cumulative distributions.*

EXERCISE 2, QUESTION 3 back to page 20

The distribution is not perfectly symmetric. It is skewed slightly to the left.
With the PivotTable selected, insert a Clustered Column chart and set the `Gap Width` *to* 0% to turn it into a histogram.

EXERCISE 3, QUESTION 1 back to page 21

From 1901 through 2000, both portfolios have behaved very similarly. Returns are between −3% and 3%, there is no trend, and positive (negative) returns for X seem to match with positive (negative) returns of Y.
In Excel, add a Year column running from 1901 to 2000. Select the Year, X, and Y columns and go to `Insert → Charts → Line`. Excel plots one line per portfolio and adds a legend automatically. Use `Chart Design → Change Colors` (or right-click a series → `Format Data Series`) to give X and Y distinct colors.

EXERCISE 4, QUESTION 1 back to page 21

Department	Frequency
Customer Service	6
Finance	8
HR	3
IT	2
Marketing	7
Operations	6
R&D	3
Sales	5

In Excel, build a PivotTable with `dept` *in the Rows area and again in the Values area as a count, or use* `=COUNTIF(range, "Finance")` *for each department.*

EXERCISE 4, QUESTION 2 back to page 21

Department	Relative
Customer Service	15.0%
Finance	20.0%
HR	7.5%
IT	5.0%
Marketing	17.5%
Operations	15.0%
R&D	7.5%
Sales	12.5%

In the PivotTable, drag `dept` *to the Values area a second time and set* `Show Values As → % of Column Total`. *Alternatively, divide each* `=COUNTIF()` *result by the total* `=COUNTA(range)`.

EXERCISE 4, QUESTION 3 back to page 21

The highest frequency is in the Finance department with 8 employees. The relative frequency is 20%. With the PivotTable selected, go to `Insert → Charts → Clustered Column` *to create the bar chart.*

EXERCISE 5, QUESTION 1 back to page 21

The highest frequency is in the $[50,60)$ *bin with 13 students.*

Bin	Frequency
$[50, 60)$	13
$[60, 70)$	11
$[70, 80)$	9
$[80, 90)$	9
$[90, 100)$	7

In Excel, add a helper column with an `=IFS()` *formula to assign each score to its class, build a PivotTable to count the scores in each bin, then insert a Clustered Column chart and set the* `Gap Width` *to* 0% to turn it into a histogram.

3 Descriptive Statistics III

EXERCISE 1, QUESTION 1 back to page 29

To find the mean we will use the following formula ($\frac{1}{n}\sum_{i=i}^{n} x_i$). The summation of the values is 51 and the number of observations is 5. The mean is $51/5 = 10.2$.
The median is found by locating the middle value when data is sorted in ascending order. $x_{sorted} = [8, 9, 10, 12, 12]$. The median in this example is 10.
The mode is the value with the highest frequency. In this example the mode is 12 since it is repeated twice and all other numbers appear only once.
To verify in Excel, enter the data in cells `A1:A5` *and use:*

- `=AVERAGE(8,9,10,12,12)` → 10.2
- `=MEDIAN(8,9,10,12,12)` → 10
- `=MODE(8,9,10,12,12)` → 12

EXERCISE 1, QUESTION 2 back to page 29

The mean is −2.67, the median is −3.5, the mode is −4. *To verify in Excel, enter the data in cells* `A1:A6` *and use:*

- `=AVERAGE(-4,0,-6,1,-3,-4)` → −2.67
- `=MEDIAN(-4,0,-6,1,-3,-4)` → −3.5
- `=MODE(-4,0,-6,1,-3,-4)` → −4

EXERCISE 1, QUESTION 3 back to page 29

The mean is 18.33, the median is 20, the data is bimodal with both 15 and 20 being modes.
To verify in Excel, enter the data in cells `A1:A9` *and use:*

- `=AVERAGE(20,15,25,20,10,15,25,20,15)` → 18.33
- `=MEDIAN(20,15,25,20,10,15,25,20,15)` → 20
- `=MODE.MULT(20,15,25,20,10,15,25,20,15)` → 15 and 20 (note: use `MODE.MULT` instead of `MODE` to capture all modes)

EXERCISE 2, QUESTION 1 back to page 29

The mean price for Country Hill is 1.87. The mean price for Minute Maid is 2.09.
To verify in Excel, assuming `PriceCH` *is in column D and* `PriceMM` *is in column E, use:*

- `=AVERAGE(D2:D1071)` → 1.87
- `=AVERAGE(E2:E1071)` → 2.09

EXERCISE 2, QUESTION 2 back to page 29

The mean price at store 1 for Country Hill is 1.80. *The juice is cheaper at store 1.The mean price at each store is:*

```
# A tibble: 5 x 2
```

```
  StoreID MeanCH
    <dbl>  <dbl>
1       1   1.80
2       2   1.84
3       3   1.93
4       4   1.95
5       7   1.84
```

To verify in Excel, use `AVERAGEIF` *to filter by store. Assuming* `StoreID` *is in column C and* `PriceCH` *is in column D:*

- `=AVERAGEIF(C2:C1071, 1, D2:D1071)` → store 1 mean
- `=AVERAGEIF(C2:C1071, 2, D2:D1071)` → store 2 mean
- `=AVERAGEIF(C2:C1071, 3, D2:D1071)` → store 3 mean

EXERCISE 2, QUESTION 3 back to page 29

Purchasers of Country Hill at store 1 paid a median price of 1.76 for Country Hill juice. This once again was the lowest price.The median price for Country Hill purchasers at each store is given by:

```
# A tibble: 5 x 2
  StoreID MedianCH
    <dbl>    <dbl>
1       1     1.76
2       2     1.86
3       3     1.99
4       4     1.99
5       7     1.86
```

To verify in Excel, use `MEDIAN` *with* `IF` *as an array formula. Assuming* `Purchase` *is in column A,* `StoreID` *in column C, and* `PriceCH` *in column D. Enter with* `Ctrl+Shift+Enter`*:*

- `=MEDIAN(IF((C2:C1071=1)*(A2:A1071="CH"), D2:D1071))` → store 1 median

Repeat changing the store number for each store.

EXERCISE 3, QUESTION 1 back to page 30

The average price of sale is found by using the weighted average formula. $\frac{\sum w_i x_i}{\sum w_i}$ The weights (w_i) are given by the number of shares bought and the values (x_i) are the prices. The weighted average is 246.802.
To verify in Excel use:

- `=SUMPRODUCT({250.34,234.59,270.45},{80,120,50}) / SUM(80,120,50)` → 246.802

EXERCISE 3, QUESTION 2 back to page 30

The average if equal shares were bought would be 251.7933.

- `=AVERAGE(250.34,234.59,270.45)` → 251.793

EXERCISE 4, QUESTION 1 back to page 30

The inflation rate is the percentage change in the CPI. The inflation rate for each period is shown in the table below:

1-2	2-3	3-4	4-5
30%	23.08%	12.5%	16.67%

EXERCISE 4, QUESTION 2 back to page 30

*The average growth rate is 20.38. Use the geometric mean formula with compounding.

$$\bar{x}_g = (1.3 \times 1.2308 \times 1.125 \times 1.1667)^{1/4} - 1$$

$$\bar{x}_g = 0.2038$$

You can use the command `GEOMEAN(1.3,1.2308,1.125,1.1667)-1` in Excel to verify the result.

EXERCISE 5 back to page 31

At the end of 4 years it is predicted that you would have 1621.17 dollars. Each year you would have gained 12.84% on average.
To verify in Excel use:

- `=AVERAGE(0.173, 0.196, 0.068, 0.082)` → 0.12975 arithmetic mean
- `=GEOMEAN(1.173, 1.196, 1.068, 1.082)-1` → 0.1173 geometric mean
- `=1000*(1+0.1173)^4` → 1621.17

EXERCISE 6, QUESTION 1 back to page 31

The mean and the median salary of the hitters is 535 and 425 respectively.Due to right-skew from superstar salaries, the median better represents the typical player.
To verify in Excel, assuming salaries are in column S, use:

- `=AVERAGE(S2:S323)` → 535
- `=MEDIAN(S2:S323)` → 425

EXERCISE 6, QUESTION 2 back to page 31

For players with ≥1 home run: mean higher (around 600+), median still lower than mean — skewness persists, median again more representative.

To verify in Excel, assuming salaries are in column S and home runs in column C:

- `=AVERAGEIF(C2:C323, ">=1", S2:S323)` → mean salary
- `=MEDIAN(IF(C2:C323>=1, S2:S323))` → median salary

4 Descriptive Stats IV

EXERCISE 1, QUESTION 1 back to page 43

The mean is 60, the Range is 94, the MAD is 28, the variance is 1186 and the variance is 34.44.
Here is the table of deviations:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\lvert x_i - \bar{x}\rvert$
70	10	100	10
68	8	64	8
4	-56	3136	56
98	38	1444	38

The variance averages out the squared deviations $(x_i - \bar{x})^2$, the MAD averages out the absolute deviations $\lvert x_i - \bar{x}\rvert$, and the standard deviation is the square root of the variance.

EXERCISE 1, QUESTION 2 back to page 43

The mean is -2, *Range is* 7, *the MAD is* 2.33, *the variance is* 7.6 *and the standard deviation is* 2.76.
Here is the table of deviations from the mean:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\lvert x_i - \bar{x}\rvert$
-4	-2	4	2
0	2	4	2
-6	-4	16	4
1	3	9	3
-3	-1	1	1
0	2	4	2

EXERCISE 1, QUESTION 3 back to page 43

The mean is 6, *Range is* 8, *the MAD is* 2, *the variance is* 8 *and the standard deviation is* 2.83.
Here is the table of deviations from the mean:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\lvert x_i - \bar{x} \rvert$
2	-4	16	4
8	2	4	2
4	-2	4	2
10	4	16	4
6	0	0	0
6	0	0	0

The range is $Range = 10 - 2 = 8$.
The MAD is $mad = (4 + 2 + 2 + 4 + 0 + 0)/6 = 12/6 = 2$.
The sample variance is $s^2 = (16 + 4 + 4 + 16 + 0 + 0)/(6 - 1) = 40/5 = 8$.
The standard deviation is $s = \sqrt{8} = 2.83$.

EXERCISE 2, QUESTION 1 back to page 44

For the sample taken, GBTC has the less variation. The standard deviation of GBTC is 9.43*, which is less than* 16.57 *for VTI or* 50.38 *for TSLA.*

EXERCISE 2, QUESTION 2 back to page 44

The answer is the same, since the MAD for GBTC is 8.46 *which is lower than* 14.27 *for VTI or* 41.67 *for TSLA.*

EXERCISE 2, QUESTION 3 back to page 44

By considering the magnitudes of the stock prices, it seems like VTI is the less volatile stock. VTI has a CV of 0.08 *which is lower than* 0.44 *for GBTC or* 0.18 *for TSLA. In fact, by CV Bitcoin seems to be the most risky asset.*

EXERCISE 3, QUESTION 1 back to page 44

The best performing stock on average is stock X. It has an average return of -0.078% *vs.* 0.097% *for stock Y. The safest stock is stock X as well, since the standard deviation is* 1.062 *percentage points vs.* 1.14 *percentage points for stock Y.*

EXERCISE 3, QUESTION 2 back to page 44

*The Sharpe Ratio measures the excess return per unit of risk taken. Stock X has the better Sharpe Ratio -0.106 vs. -0.115 for stock Y. Stock X is recommended since it provides a higher excess return per unit of risk taken.

EXERCISE 4 back to page 44

Start by computing the sample mean:

$$\bar{x} = \frac{2.1 + (-0.5) + 3.0 + 1.2 + (-1.8) + 0.7 + 2.5 + (-0.3)}{8} = \frac{6.9}{8} = 0.8625$$

Here is the table of deviations from the mean:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\lvert x_i - \bar{x} \rvert$
2.1	1.2375	1.5314	1.2375
-0.5	-1.3625	1.8564	1.3625
3.0	2.1375	4.5689	2.1375
1.2	0.3375	0.1139	0.3375
-1.8	-2.6625	7.0889	2.6625
0.7	-0.1625	0.0264	0.1625
2.5	1.6375	2.6814	1.6375
-0.3	-1.1625	1.3514	1.1625

The sample variance is:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1} = \frac{19.2188}{7} = 2.7455$$

The sample standard deviation is:

$$s = \sqrt{2.7455} = 1.657$$

The Mean Absolute Deviation is:

$$mad = \frac{\sum \lvert x_i - \bar{x} \rvert}{n} = \frac{10.7}{8} = 1.3375$$

The Coefficient of Variation is:

$$CV = \frac{s}{\bar{x}} \times 100 = \frac{1.657}{0.8625} \times 100 = 192.1\%$$

The CV is about 192%*, which is large: the standard deviation is almost twice the mean. This indicates the returns are highly variable relative to their average — the mean is small compared with typical month-to-month fluctuations. In practice, a CV this large signals that the project's returns are unreliable; an investor cannot count on earning close to the average return in any given month.*

EXERCISE 5, QUESTION 1 back to page 45

The sample mean is:

$$\bar{x} = \frac{120 + (-85) + 200 + (-150) + 95 + (-60) + 310 + (-220) + 175 + (-45)}{10} = \frac{340}{10} = 34$$

The bettor is profitable on average, earning \$34 *per week.*
Here is the table of deviations from the mean:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$\lvert x_i - \bar{x} \rvert$
120	86	7,396	86
-85	-119	14,161	119
200	166	27,556	166
-150	-184	33,856	184
95	61	3,721	61
-60	-94	8,836	94
310	276	76,176	276
-220	-254	64,516	254
175	141	19,881	141
-45	-79	6,241	79

The sample variance is:

$$s^2 = \frac{262{,}340}{9} = 29{,}149$$

The sample standard deviation is:

$$s = \sqrt{29{,}149} = \$170.73$$

The MAD is:

$$mad = \frac{86 + 119 + 166 + 184 + 61 + 94 + 276 + 254 + 141 + 79}{10} = \frac{1{,}460}{10} = \$146$$

The bettor is profitable on average but faces large week-to-week swings — a typical week deviates from the mean by about \$146 *to* \$171 *depending on the measure used.*

EXERCISE 5, QUESTION 2 back to page 45

$$CV = \frac{170.73}{34} \times 100 = 502\%$$

A CV of 502% *is extremely large (i.e., the standard deviation is five) times the mean. This indicates that the average weekly profit of* \$34 *is tiny relative to the typical week-to-week swings of* \$171*. The bettor cannot reliably count on earning near the mean in any given week. A high CV signals that the strategy is highly risky relative to its average reward.*

EXERCISE 5, QUESTION 3 back to page 45

With a risk-free rate of \$0*:*

$$SR = \frac{\bar{x} - R_f}{s} = \frac{34 - 0}{170.73} = 0.199$$

A Sharpe ratio of 0.199 *is low. As a point of reference, a Sharpe ratio above* 1 *is generally considered acceptable in finance. The bettor earns only* \$0.20 *of profit per dollar of risk taken — suggesting that while the strategy is marginally profitable, the risk taken to achieve that profit is not well compensated. A more disciplined betting strategy or better odds selection would be needed to improve the risk-adjusted return.*

5 Descriptive Stats V

EXERCISE 1, QUESTION 1 back to page 58

The minimum is 0, the first quartile is 2, second quartile is 4, third quartile is 8, and maximum is 30. 30 is an outlier since it is beyond $Q_3 + 1.5 \times IQR$.
Quartiles are calculated with the location formula $L_p = 1 + k(n-1)$. The sorted data is $\{0, 1, 3, 4, 6, 10, 30\}$ with $n = 7$:

- $L_{25} = 1 + 0.25(6) = 2.5 \Rightarrow Q_1 = 1 + 0.5(3 - 1) = 2$
- $L_{50} = 1 + 0.50(6) = 4 \Rightarrow Median = 4$
- $L_{75} = 1 + 0.75(6) = 5.5 \Rightarrow Q_3 = 6 + 0.5(10 - 6) = 8$

In Excel we get the five point summary with `=QUARTILE.INC()`:

```
=QUARTILE.INC(A:A, 0)   /* Minimum = 0  */
=QUARTILE.INC(A:A, 1)   /* Q1      = 2  */
=QUARTILE.INC(A:A, 2)   /* Median  = 4  */
=QUARTILE.INC(A:A, 3)   /* Q3      = 8  */
=QUARTILE.INC(A:A, 4)   /* Maximum = 30 */
```

The interquartile range is needed to determine if there are any outliers. The $IQR = Q_3 - Q_1 = 6$. Since $Q_3 + 1.5 \times IQR = 8 + 1.5(6) = 17$, everything beyond 17 is an outlier — so 30 is an outlier.

```
=QUARTILE.INC(A:A, 3) - QUARTILE.INC(A:A, 1)   /* IQR = 6  */
=QUARTILE.INC(A:A, 3) + 1.5 * 6                 /* UL  = 17 */
```

EXERCISE 1, QUESTION 2 back to page 58

If we use the z-score instead we find that 30 is **not** an outlier since the z-score is $Z_{30} = 2.15$. This observation is only 2.15 standard deviations away from the mean.
In Excel we make a quick calculation of the z-score to confirm our result. The z-score is $Z_i = \frac{x_i - \bar{x}}{s}$:

```
=STANDARDIZE(30, AVERAGE(A:A), STDEV.S(A:A))   /* = 2.15 */
```

Since $2.15 < 3$, the value 30 is not flagged as an outlier by the z-score method, even though the IQR method flagged it. The two methods can disagree.

EXERCISE 1, QUESTION 3 back to page 58

Chebyshev's theorem states that $1 - \frac{1}{z^2}$ of the data lies between z standard deviations from the mean. *Substituting the* z-score found in 2. we get that at least 78.34% of the data lies within that many standard deviations of the mean. In Excel:

```
=1 - 1/(2.15^2)    /*  0.7834, i.e. 78.34% */
```

EXERCISE 2, QUESTION 1 back to page 59

$$\bar{x} = \frac{2+2+2+6+13}{5} = \frac{25}{5} = 5$$

$$s = \sqrt{\frac{9+9+9+1+64}{4}} = \sqrt{23} = 4.796$$

In Excel:

```
=AVERAGE(A2:A6)    /* mean = 5      */
=STDEV.S(A2:A6)    /* s    = 4.796  */
```

EXERCISE 2, QUESTION 2 back to page 59

x_i	$x_i - \bar{x}$	$\frac{x_i-\bar{x}}{s}$	$\left(\frac{x_i-\bar{x}}{s}\right)^3$	$\left(\frac{x_i-\bar{x}}{s}\right)^4$
2	−3	−0.625	−0.244	0.153
2	−3	−0.625	−0.244	0.153
2	−3	−0.625	−0.244	0.153
6	1	0.208	0.009	0.002
13	8	1.667	4.630	7.716

In Excel, put the standardized value in a helper column and copy it down, then square and cube it:

```
=(A2 - AVERAGE($A$2:$A$6)) / STDEV.S($A$2:$A$6)   /* z    */
=B2^3                                             /* z^3  */
=B2^4                                             /* z^4  */
```

EXERCISE 2, QUESTION 3 back to page 59

$$Sk = \frac{5}{(4)(3)} \sum \left(\frac{x_i - \bar{x}}{s}\right)^3 = \frac{5}{12}(-0.244 - 0.244 - 0.244 + 0.009 + 4.630)$$

$$= \frac{5}{12}(3.907) = 1.628$$

6 Regression I

EXERCISE 1, QUESTION 1 back to page 66

The covariance is 14.9 and the correlation is 0.96. The results indicate that there is a strong direct relationship between the two variables.
Start by finding the mean of each variable: $\bar{x} = 19.8$ and $\bar{y} = 16.6$. Then build the deviations table:

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	Product
20	17	0.2	0.4	0.08
21	19	1.2	2.4	2.88
15	12	−4.8	−4.6	22.08
18	13	−1.8	−3.6	6.48
25	22	5.2	5.4	28.08

$$s_{xy} = \frac{0.08 + 2.88 + 22.08 + 6.48 + 28.08}{4} = \frac{59.6}{4} = 14.9$$

$$r = \frac{14.9}{sd(x) \times sd(y)} = \frac{14.9}{3.49 \times 4.34} = 0.96$$

To verify in Excel, enter x in column A and y in column B and use:

- `=COVARIANCE.S(A2:A6, B2:B6)` → 14.9
- `=CORREL(A2:A6, B2:B6)` → 0.96

EXERCISE 1, QUESTION 2 back to page 66

The covariance is 0.85 and the correlation is 0.148. The results indicate that there is a very weak direct relationship between the two variables. They might be unrelated.
To verify in Excel, enter w in column A and z in column B and use:

- `=COVARIANCE.S(A2:A6, B2:B6)` → 0.85
- `=CORREL(A2:A6, B2:B6)` → 0.148

EXERCISE 2, QUESTION 1 back to page 66

The correlation coefficient is −0.78. This is indicative of a moderately strong inverse relationship between mpg and hp. As horsepower increases, fuel efficiency tends to decrease.
In Excel, assuming `mpg` *is in column A and* `hp` *is in column B:*

- `=CORREL(A2:A33, B2:B33)` → −0.78

EXERCISE 2, QUESTION 2 back to page 67

The scatter diagram is downward sloping and most points are close to the trend line. It is what was expected from a correlation coefficient of −0.78.

In Excel, select the `hp` *and* `mpg` *columns and go to* `Insert → Charts → Scatter`*. To add the trend line, right-click any point and choose* `Add Trendline → Linear`*.*

EXERCISE 2, QUESTION 3 back to page 67

The coefficient of determination is 0.60. This value is not very close to one. This is expected since miles per gallon can also vary because of the car's weight and fuel efficiency. It makes sense that `hp` only explains 60% of the total variation.
In Excel, square the correlation coefficient:

- `=CORREL(A2:A33, B2:B33)^2` → 0.60

EXERCISE 3, QUESTION 1 back to page 67

It seems like there is a direct relationship between both variables. The more debt a graduate takes on, the higher the earnings tend to be.
In Excel, select the `MD_EARN_WNE_P10` *and* `GRAD_DEBT_MDN` *columns and go to* `Insert → Charts → Scatter`*. Add a linear trend line by right-clicking a point and choosing* `Add Trendline → Linear`*.*

EXERCISE 3, QUESTION 2 back to page 67

The correlation coefficient shows a moderate direct relationship between earnings and debt, $r = 0.46$. The coefficient of determination indicates that only 21% of the variation in earnings can be explained by debt.
In Excel, assuming `GRAD_DEBT_MDN` *is in column A and* `MD_EARN_WNE_P10` *is in column B:*

- `=CORREL(A2:A1000, B2:B1000)` → 0.46
- `=CORREL(A2:A1000, B2:B1000)^2` → 0.21

Note: use `=CORREL()` *rather than* `=COVARIANCE.S()` *when there are missing values, as CORREL handles them more gracefully in Excel.*

EXERCISE 4, QUESTION 1 back to page 67

There is a direct relationship between bedrooms and price. As the number of bedrooms increases, sale prices tend to rise. This is intuitive since larger homes generally command higher market values.
In Excel, select the `bedrooms` *and* `Price` *columns and go to* `Insert → Charts → Scatter`*. Add a linear trend line by right-clicking a point and choosing* `Add Trendline → Linear`*.*

EXERCISE 4, QUESTION 2 back to page 67

The correlation coefficient is $r = 0.92$, indicating a strong direct relationship between bedrooms and sale price. Larger homes tend to sell for higher prices, though many other factors (location, condition, amenities) also influence price, which prevents a perfect correlation.
In Excel, assuming `bedrooms` *is in column A and* `Price` *is in column B:*

- `=CORREL(A2:A348, B2:B348)` → 0.92

EXERCISE 4, QUESTION 3 back to page 67

The coefficient of determination is $R^2 = 0.85$, so about 85% of the variation in home prices is explained by the number of bedrooms alone. The remaining variation is likely driven by factors such as location, number of bathrooms, lot size, school district quality, and the overall condition of the property.
In Excel, square the correlation coefficient:

- `=CORREL(A2:A348, B2:B348)^2` → 0.85

7 Regression II

EXERCISE 1, QUESTION 1 back to page 81

The regression line is $\hat{y} = -4.93 + 1.09x$. For each unit increase in x, y increases on average by 1.09.
Start with the deviations from the mean ($\bar{x} = 19.8$, $\bar{y} = 16.6$):

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	Product	$(x_i - \bar{x})^2$
20	17	0.2	0.4	0.08	0.04
21	19	1.2	2.4	2.88	1.44
15	12	−4.8	−4.6	22.08	23.04
18	13	−1.8	−3.6	6.48	3.24
25	22	5.2	5.4	28.08	27.04

$$\hat{\beta} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{59.6}{54.8} = 1.09$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 16.6 - 1.09(19.8) = -4.93$$

To verify in Excel, enter x in column A and y in column B. A single `LINEST` *returns both coefficients, with the intercept last:*

```
=LINEST(B2:B6, A2:A6, TRUE, TRUE)
```

- `=INDEX(LINEST(B2:B6,A2:A6,TRUE,TRUE),1,1)` → $\hat{\beta} = 1.09$
- `=INDEX(LINEST(B2:B6,A2:A6,TRUE,TRUE),1,2)` → $\hat{\alpha} = -4.93$

EXERCISE 1, QUESTION 2 back to page 81

SST is 69.2, SSR is 64.82 and SSE is 4.38 (note that $SSR + SSE = SST$). The R^2 is $\frac{SSR}{SST} = 0.94$ and the Standard Error estimate is 1.21. Both indicate a great fit of the regression line to the data.
All four numbers come out of the same `LINEST` *block, so spill it once with* `=LINEST(B2:B6, A2:A6, TRUE, TRUE)` *and read off:*

- `=INDEX(LINEST(B2:B6,A2:A6,TRUE,TRUE),5,1)` → $SSR = 64.82$
- `=INDEX(LINEST(B2:B6,A2:A6,TRUE,TRUE),5,2)` → $SSE = 4.38$
- `=INDEX(LINEST(B2:B6,A2:A6,TRUE,TRUE),3,1)` → $R^2 = 0.94$
- `=INDEX(LINEST(B2:B6,A2:A6,TRUE,TRUE),3,2)` → $s_e = 1.21$

Then $SST = SSR + SSE = 69.2$, which `=DEVSQ(B2:B6)` confirms.

EXERCISE 1, QUESTION 3 back to page 81

If $x = 32$ then $\hat{y} = -4.93 + 1.09(32) = 29.87$. The regression is a good fit, so we can feel good about our prediction. However, we would be concerned about the small sample size of the data.
In Excel:

- `=TREND(B2:B6, A2:A6, 32)` → 29.87

EXERCISE 2, QUESTION 1 back to page 82

An extra year of education increases the annual salary by about $\$5,300$ (the slope). A person with no education would be expected to earn about $\$17,258$ (the intercept).
In Excel, assuming `Salary` *is in column B and* `Education` *is in column C:*

```
=LINEST(B2:B101, C2:C101, TRUE, TRUE)
```

- `=INDEX(LINEST(B2:B101,C2:C101,TRUE,TRUE),1,1)` → 5.30
- `=INDEX(LINEST(B2:B101,C2:C101,TRUE,TRUE),1,2)` → 17.26

EXERCISE 2, QUESTION 2 back to page 82

The R^2 is 0.67 and the standard error is about 21. The line is a moderately good fit. A person with 16 years of education is predicted to earn about $\$102,000$.
In Excel:

- `=INDEX(LINEST(B2:B101,C2:C101,TRUE,TRUE),3,1)` → $R^2 = 0.67$
- `=INDEX(LINEST(B2:B101,C2:C101,TRUE,TRUE),3,2)` → $s_e = 21$
- `=TREND(B2:B101, C2:C101, 16)` → 102.08

EXERCISE 3, QUESTION 1 back to page 82

Approximately 36% of the sample owns a home (mean = 0.3625).
In Excel, assuming `OwnHome` *is in column A and* `Food` *is in column B, first delete any rows with missing values. Then add a helper column C with the dummy variable:*

```
=IF(A2="Yes", 1, 0)
```

Copy it down for all rows and take the average:

- `=AVERAGE(C2:C81)` → 0.3625

You can also skip the helper column entirely — the same proportion is `=AVERAGE(IF(A2:A81="Yes",1,0))`.

EXERCISE 3, QUESTION 2 back to page 82

The intercept (≈ 6473) is the average food expenditure of individuals without homes. The slope (≈ -3418) is the difference in food expenditure between individuals who own a home and those who do not — home owners spend about $\$3,418$ less on food than non-owners.
In Excel, with `Food` *in column B and the dummy in column C:*

```
=LINEST(B2:B81, C2:C81, TRUE, TRUE)
```

- `=INDEX(LINEST(B2:B81,C2:C81,TRUE,TRUE),1,2)` → intercept = 6473
- `=INDEX(LINEST(B2:B81,C2:C81,TRUE,TRUE),1,1)` → slope = −3418

Because a single dummy just splits the sample in two, the two predictions are the two group means:

- `=TREND(B2:B81, C2:C81, 0)` → 6473, the same as `=AVERAGEIF(A2:A81,"No",B2:B81)`
- `=TREND(B2:B81, C2:C81, 1)` → 3055, the same as `=AVERAGEIF(A2:A81,"Yes",B2:B81)`

EXERCISE 3, QUESTION 3 back to page 82

The regression line is $\widehat{dummy} = 1.432 - 0.000204\,Food$. This is a linear probability model: it predicts the likelihood of owning a home from food expenditure. The slope tells us how that likelihood changes as food expenditure increases by one dollar — in general, the likelihood of owning a home decreases as food expenditure increases. The scatter plot shows that most home owners have food expenditures below $\$6,000$, while non-owners are mostly above it.
In Excel, with the dummy in column C and `Food` *in column B:*

- `=INDEX(LINEST(C2:C81,B2:B81,TRUE,TRUE),1,2)` → intercept = 1.432
- `=INDEX(LINEST(C2:C81,B2:B81,TRUE,TRUE),1,1)` → slope = −0.000204

For the scatter plot, select `Food` *and the dummy variable, go to* `Insert → Charts → Scatter`*, and add a linear trend line by right-clicking a point and choosing* `Add Trendline → Linear`.

EXERCISE 4, QUESTION 1 back to page 82

With an $R^2 = 0.81$, the model fits the data very well.
In Excel, first isolate Japan: select the data, go to `Data → Filter`*, and filter* `Country.Name` *to* `Japan`*. Copy the* 62 Japan rows to a new sheet. Assuming `Year` is in column C and `Population` is in column D:

- `=INDEX(LINEST(D2:D63,C2:C63,TRUE,TRUE),3,1)` → $R^2 = 0.81$

EXERCISE 4, QUESTION 2 back to page 83

The prediction for 2030 is about 140 million people (140,268,585).
In Excel:

- `=TREND(D2:D63, C2:C63, 2030)` → 140,268,585

EXERCISE 4, QUESTION 3 back to page 83

After looking at the scatter plot, it seems unlikely that the population in Japan will hit 140 million. In the most recent years the population has been decreasing, so the straight-line trend overstates the future population — the prediction should be treated with caution.
In Excel, select the `Year` *and* `Population` *columns for Japan, go to* `Insert → Charts → Scatter`*, and add a linear trend line by right-clicking a point and choosing* `Add Trendline → Linear`.

8 Probability I

EXERCISE 1, QUESTION 1 back to page 100

- $P(D) = 1/5 = 0.2$ since all events are equally likely.
- $P(B^c) = 4/5 = 0.8$
- $P(A \cup C \cup E) = P(A + C + E) = 3/5 = 0.6$.

EXERCISE 1, QUESTION 2 back to page 100

- *Events* A and B are not mutually exclusive since they share some of the same elements.
- *The events are not exhaustive since the union of both doesn't create the sample space.*

EXERCISE 1, QUESTION 3 back to page 100

- *The probability is* 68.8%.
- *The events are mutually exclusive. If someone is classified as obese, the person is not classified again as overweight.*
- *The events are not exhaustive since there are people in the U.S. that have a normal weight.*
- *The probability that the person drawn has normal weight is* 31.2%.

EXERCISE 2, QUESTION 1 back to page 100

From the multiplication rule,

$$P(A|B) * P(B) = P(A \cap B)$$

substituting values yields:

$$P(A \cap B) = 0.45 * 0.3 = 0.135$$

From the addition rule,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

substituting yields:

$$P(A \cup B) = 0.65 + 0.3 - 0.135 = 0.815$$

From the multiplication rule once again,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

substituting values yields,

$$P(B|A) = 0.135/0.65 = 0.2076923$$

EXERCISE 2, QUESTION 2 back to page 100

From the complement rule we have that $P(A^c) = 0.6$ and $P(B^c) = 0.5$.
Using the multiplication rule,

$$P(A^c|B^c) = \frac{P(A^c \cap B^c)}{P(B^c)}$$

substituting yields:

$$P(A^c|B^c) = 0.24/0.5 = 0.48$$

From the addition rule,

$$P(A^c \cup B^c) = P(A^c) + P(B^c) - P(A^c \cap B^c)$$

substituting yields:

$$P(A^c \cup B^c) = 0.6 + 0.5 - 0.24 = 0.86$$

The event that has no elements of A or B is given by $P(A^c \cap B^c)$. Therefore $P(A \cup B) = 1 - 0.24 = 0.76$ has all the elements of A and B.

EXERCISE 2, QUESTION 3 back to page 100

In short, the problem states $P(A) = 0.4$, $P(B) = 0.6$, and $P(A|B) = 0.5$. Where A and B are events of stocks rising in price. The question asks for:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Using the multiplication rule,

$$P(A \cap B) = 0.5 * 0.6 = 0.3$$

Hence,

$$P(A \cup B) = 0.4 + 0.6 - 0.3 = 0.7$$

The events are not mutually exclusive since $P(A \cap B) = 0.3 \neq 0$. The events are also not independent since $P(A|B) = 0.5 \neq 0.4 = P(A)$.

EXERCISE 3, QUESTION 1 back to page 101

Below is the joint probability table. The $P(A) = 0.26 + 0.34 = 0.6$, $P(A \cap B) = 0.26$, $P(A|B) = 0.26/0.4 = 0.65$, and $P(B|A^c) = 0.14/0.4 = 0.35$. Events A and B are not independent since $P(A) \neq P(A|B)$. The events are not mutually exclusive since $P(A \cap B) = 0.26 \neq 0$.

	B	B^c	Total
A	0.26	0.34	0.6
A^c	0.14	0.26	0.4
Total	0.4	0.6	1

EXERCISE 4, QUESTION 1 back to page 101

The contingency table below shows the counts for each combination of Crash Type and Light Condition:

	Daylight	Not Daylight	Total
Head-on	166	108	274
Not Head-On	3258	1326	4584
Total	3424	1434	4858

In Excel, build a PivotTable: drag `Crash Type` *to the Rows area,* `Light Condition` *to the Columns area, and either field to the Values area as a Count. To turn the counts into a joint probability table, use* `Value Field Settings → Show Values As → % of Grand Total`*. You can also count any single cell directly. Assuming* `Crash Type` *is in column B and* `Light Condition` *in column C:*

- `=COUNTIFS(B:B, "Head-on", C:C, "Daylight")` → 166

EXERCISE 4, QUESTION 2 back to page 101

a) The probability of a Head-On crash is:

$$P(\text{Head-on}) = \frac{166 + 108}{4858} = \frac{274}{4858} = 0.056$$

b) The probability that the crash is in daylight is:

$$P(\text{Daylight}) = \frac{166 + 3258}{4858} = \frac{3424}{4858} = 0.70$$

c) The probability that a crash is Head-On given daylight is:

$$P(\text{Head-on} \mid \text{Daylight}) = \frac{P(\text{Head-on} \cap \text{Daylight})}{P(\text{Daylight})} = \frac{166}{3424} = 0.048$$

In Excel these follow from the PivotTable counts. For example, part a) is `=COUNTIF(B:B,"Head-on") / COUNTA(B2:B4859)`.

EXERCISE 4, QUESTION 3 back to page 101

The two variables are dependent since $P(\text{Head-on} \mid \text{Daylight}) \neq P(\text{Head-on})$, that is $0.048 \neq 0.056$.

EXERCISE 5, QUESTION 1 back to page 101

$P(A^c) = 1 - P(A) = 1 - 0.7 = 0.3$, $P(A \cap B) = P(B|A)P(A) = 0.55(0.70) = 0.385$, $P(A^c \cap B) = P(B|A^c)P(A^c) = 0.10(0.30) = 0.03$, $P(B) = P(A \cap B) + P(A^c \cap B) = 0.385 + 0.03 = 0.415$, and $P(A|B) = \frac{P(A \cap B)}{P(B)} = 0.385/0.415 = 0.9277$.

EXERCISE 5, QUESTION 2 back to page 101

Let the event of failing be F, the event of not failing be NF, the event of having a tutor be T, and the event of not having a tutor be NT. The probability of failing the course is 0.25. $(F) = P(F \cap T) + P(F \cap T^c) = P(F|T)P(T) + P(F|T^c)P(T^c) = 0.10(0.50) + 0.40(0.50) = 0.05 + 0.20 = 0.25$ The probability of not having a tutor, given that she failed the course is 0.2. $P(T|F) = \frac{P(F \cap T)}{P(F \cap T) + P(F \cap T^c)} = 0.05/0.25 = 0.20$.

EXERCISE 6, QUESTION 1 back to page 102

$3! = 3 \times 2 \times 1 = 6$, $4! = 4 \times 3 \times 2 \times 1 = 24$, $C_6^8 = 28$, and $P_6^8 = 20{,}160$.
In Excel:

- `=FACT(3)` → 6
- `=FACT(4)` → 24
- `=COMBIN(8, 6)` → 28
- `=PERMUT(8, 6)` → 20,160

EXERCISE 6, QUESTION 2 back to page 102

If order doesn't matter, there are 252 ways. If order matters, then there are 30,240 ways.
In Excel:

- `=COMBIN(10, 5)` → 252
- `=PERMUT(10, 5)` → 30,240

9 Probability II

EXERCISE 1, QUESTION 1 back to page 111

The expected value is 10.75 and the standard deviation is 5.31. The probability of $x < 15$ is 0.65.
In Excel, enter the values x in column A and the probabilities $P(X{=}x)$ *in column B:*

- `=SUMPRODUCT(A2:A5, B2:B5)` → 10.75 (expected value)
- `=SQRT(SUMPRODUCT((A2:A5-10.75)^2, B2:B5))` → 5.31 (standard deviation)

- $P(x < 15) = 0.35 + 0.3 = 0.65$

EXERCISE 1, QUESTION 2 back to page 111

The expected value is -17.4 and the standard deviation is 6.86. The probability of $x \geq -9$ is 0.25.
In Excel, with y in column A and P(Y=y) in column B:

- `=SUMPRODUCT(A2:A5, B2:B5)` → -17.4
- `=SQRT(SUMPRODUCT((A2:A5-(-17.4))^2, B2:B5))` → 6.86
- $P(y \geq -9) = 0.15 + 0.1 = 0.25$

EXERCISE 1, QUESTION 3 back to page 111

Both funds have the same expected return of 6. The safest return comes from fund 1 since the standard deviation is only 11.14 vs. 31.05 for fund 2.
In Excel, enter the probabilities in column A, Fund 1 in column B, and Fund 2 in column C:

- `=SUMPRODUCT(A2:A4, B2:B4)` → 6 (Fund 1 expected return)
- `=SUMPRODUCT(A2:A4, C2:C4)` → 6 (Fund 2 expected return)
- `=SQRT(SUMPRODUCT((B2:B4-6)^2, A2:A4))` → 11.14 (Fund 1 std. dev.)
- `=SQRT(SUMPRODUCT((C2:C4-6)^2, A2:A4))` → 31.05 (Fund 2 std. dev.)

Since both funds have the same expected return, choose Fund 1 — it achieves that return with much less risk.

EXERCISE 2, QUESTION 1 back to page 112

The expected return of the portfolio is 10.4 and the standard deviation is 14.60.
The portfolio weights are $\alpha = 2/5 = 0.4$ for X and $\beta = 3/5 = 0.6$ for Y. In Excel:

- Expected return $\alpha R_1 + \beta R_2$: `=0.4*8 + 0.6*12` → 10.4
- Standard deviation $\sqrt{\alpha^2\sigma_1^2 + \beta^2\sigma_2^2 + 2\alpha\beta\rho\sigma_1\sigma_2}$: `=SQRT(0.4^2*12^2 + 0.6^2*20^2 + 2*0.4*0.6*0.4*12*20)` → 14.60

EXERCISE 3, QUESTION 1 back to page 112

$P(Z = 1) = 0.31$, and $P(Z \geq 2) = 0.57$. The expected value is $np = 1.75$ and the standard deviation is $\sqrt{np(1-p)} = 1.067$.
In Excel:

- `=BINOM.DIST(1, 5, 0.35, FALSE)` → 0.31
- `=1 - BINOM.DIST(1, 5, 0.35, TRUE)` → 0.57

EXERCISE 3, QUESTION 2 back to page 112

$P(W > 160) = 0.14$, and $P(155 \leq W \leq 165) = 0.45$. The expected value is $np = 154$ and the standard deviation is $\sqrt{np(1-p)} = 5.95$.
In Excel:

- `=1 - BINOM.DIST(160, 200, 0.77, TRUE)` → 0.14
- `=BINOM.DIST(165, 200, 0.77, TRUE) - BINOM.DIST(154, 200, 0.77, TRUE)` → 0.45

EXERCISE 3, QUESTION 3 back to page 112

The probabilities are the same. Each event has a probability of 0.3456. If the probability changes to 0.5 now the event of two women and two men is more likely.
In Excel:

- `=BINOM.DIST(3, 4, 0.6, FALSE)` → 0.3456 (three men)
- `=BINOM.DIST(2, 4, 0.6, FALSE)` → 0.3456 (two men)
- `=BINOM.DIST(3, 4, 0.5, FALSE)` → 0.25
- `=BINOM.DIST(2, 4, 0.5, FALSE)` → 0.375

At $p = 0.5$, having two of each is the most likely outcome.

EXERCISE 4, QUESTION 1 back to page 112

The $P(S = 2) = 0.25$ and $P(S \geq 2) = 0.44$. The expected value and the variance is 1.5.
In Excel:

- `=POISSON.DIST(2, 1.5, FALSE)` → 0.25
- `=1 - POISSON.DIST(1, 1.5, TRUE)` → 0.44

EXERCISE 4, QUESTION 2 back to page 112

The $P(T = 14) = 0.039$ and $P(18 \leq T \leq 23) = 0.49$.
In Excel:

- `=POISSON.DIST(14, 20, FALSE)` → 0.039
- `=POISSON.DIST(23, 20, TRUE) - POISSON.DIST(17, 20, TRUE)` → 0.49

EXERCISE 4, QUESTION 3 back to page 112

The probability of administering more than 8 but less than 12 shots is 0.3.
If 84 shots are administered on average weekly, then 12 are administered daily. Using this daily mean in Excel:

- `=POISSON.DIST(11, 12, TRUE) - POISSON.DIST(8, 12, TRUE)` → 0.30

EXERCISE 5, QUESTION 1 back to page 113

$P(X = 0) = 0.58$, $P(X = 1) = 0.37$, and $P(X \leq 1) = 0.94$.
In Excel (*successes in population* $r = 3$, population $N = 25$, sample $n = 4$):

- `=HYPGEOM.DIST(0, 4, 3, 25, FALSE)` → 0.58
- `=HYPGEOM.DIST(1, 4, 3, 25, FALSE)` → 0.37

- `=HYPGEOM.DIST(1, 4, 3, 25, TRUE)` → 0.94

EXERCISE 5, QUESTION 2 back to page 113

The probability is 0.078.
In Excel (*successes in population* $r = 25$, population $N = 100$, sample $n = 20$):

- `=1 - HYPGEOM.DIST(7, 20, 25, 100, TRUE)` → 0.078

EXERCISE 5, QUESTION 3 back to page 113

The probability of matching two white balls is 5%. Winning the jackpot is extremely unlikely! A probability of 0.00000000512. It is more likely to be struck by lightning according to the CDC.
In Excel, treat the five drawn white balls as the successes in a population of 59, with the player's five picks as the sample:

- `=HYPGEOM.DIST(2, 5, 5, 59, FALSE)` → 0.05 (match two white balls)

For the jackpot, multiply the probability of matching all five white balls by the probability of matching the single Powerball (*1 of 39*):

- `=HYPGEOM.DIST(5, 5, 5, 59, FALSE) * HYPGEOM.DIST(1, 1, 1, 39, FALSE)` → 0.00000000512

10 Probability III

EXERCISE 1, QUESTION 1 back to page 119

The height of the density function $f(x) = 0.1667$, the mean is 1, standard deviation is 1.73, and $P(X \leq -1) = 0.1667$.
The uniform distribution has no Excel function, so each quantity is computed directly from its formula:

- Height $f(x) = \frac{1}{b-a}$: `=1/(4-(-2))` → 0.1667
- Mean $\mu = \frac{a+b}{2}$: `=(-2+4)/2` → 1
- Standard deviation $\sigma = \sqrt{\frac{(b-a)^2}{12}}$: `=SQRT((4-(-2))^2/12)` → 1.73
- $P(X \leq -1) = \frac{x-a}{b-a}$: `=(-1-(-2))/(4-(-2))` → 0.1667

EXERCISE 1, QUESTION 2 back to page 119

The probability that you will arrive on time is 0.875. If the time of the errand is 30 minutes, then the probability goes down to 0.75.
There is a 120 minute interval in which the provider can arrive, so $P(X > t) = 1 - \frac{t}{120}$. In Excel:

- `=1 - 15/120` → 0.875
- `=1 - 30/120` → 0.75

EXERCISE 2, QUESTION 1 back to page 119

$P(-0.67 \leq Z \leq -0.23) = 0.158$, $P(0 \leq Z \leq 1.96) = 0.475$, $P(-1.28 \leq Z \leq 0) = 0.4$ and $P(Z > 4.2) \approx 0$.
Use the `=NORM.S.DIST()` *function for the standard normal distribution:*

- `=NORM.S.DIST(-0.23, TRUE) - NORM.S.DIST(-0.67, TRUE)` → 0.158
- `=NORM.S.DIST(1.96, TRUE) - NORM.S.DIST(0, TRUE)` → 0.475
- `=NORM.S.DIST(0, TRUE) - NORM.S.DIST(-1.28, TRUE)` → 0.4
- `=1 - NORM.S.DIST(4.2, TRUE)` → ≈ 0

EXERCISE 2, QUESTION 2 back to page 119

$P(Y > 7.6) = 0.005386$, $P(7.4 \leq Y \leq 10.6) = 0.0071$, a y such that $P(Y > y) = 0.025$ is 6.42, and a y such that $P(y \leq Y \leq 2.5)$ is -2.56.
Use `=NORM.DIST()` *for probabilities and* `=NORM.INV()` *for quantiles:*

- `=1 - NORM.DIST(7.6, 2.5, 2, TRUE)` → 0.005386
- `=NORM.DIST(10.6, 2.5, 2, TRUE) - NORM.DIST(7.4, 2.5, 2, TRUE)` → 0.0071
- $P(Y > y) = 0.025$ means y is the 97.5th percentile: `=NORM.INV(0.975, 2.5, 2)` → 6.42
- Since 2.5 is the mean, $P(y \leq Y \leq 2.5) = 0.4943$ leaves $0.5 - 0.4943 = 0.0057$ in the left tail: `=NORM.INV(0.0057, 2.5, 2)` → -2.56

EXERCISE 2, QUESTION 3 back to page 119

The probability is 10.56%. A game lasting no more than 2.069 hours would be in the bottom 1%.
In Excel:

- `=NORM.DIST(2.5, 3, 0.4, TRUE)` → 0.1056
- `=NORM.INV(0.01, 3, 0.4)` → 2.069

EXERCISE 3, QUESTION 1 back to page 120

The standard deviation is equal to the mean 0.1. $P(0.10 \leq S \leq 0.2) = 0.2325$
The rate is $\lambda = 1/\mu = 1/0.1 = 10$. In Excel:

- `=EXPON.DIST(0.2, 10, TRUE) - EXPON.DIST(0.1, 10, TRUE)` → 0.2325

EXERCISE 3, QUESTION 2 back to page 120

The mean time between car arrivals is $1/360 = 0.002778$ hours. The probability that the next car will arrive within the next 10 seconds is 0.6321.
Ten seconds is $10/3600 = 1/360$ of an hour. Using a rate of $\lambda = 360$ cars per hour in Excel:

- `=EXPON.DIST(1/360, 360, TRUE)` → 0.6321

11 Inference I

EXERCISE 1, QUESTION 1 back to page 129

In a modern version of Excel, type the following in the top-left cell of an empty block; it will spill into a 10×1000 grid where each column is a sample of size 10:

- `=RANDARRAY(10, 1000, 100, 200, FALSE)`

In an older version, enter `=100 + 100*RAND()` *in the top-left cell and copy it across* 1000 columns and down 10 rows.
Because the values are drawn from a uniform distribution on $[100, 200]$, the population mean is $\mu = \frac{100+200}{2} = 150$ and the population standard deviation is $\sigma = \frac{200-100}{\sqrt{12}} \approx 28.87$. (The values recalculate on every edit; to freeze a sample, copy the block and use `Paste Special → Values`.)

EXERCISE 1, QUESTION 2 back to page 129

Below the block, compute the mean of each column and copy it across all 1000 columns:

- `=AVERAGE(A1:A10)` → the mean of the first sample

This produces 1000 sample means. Summarize them with:

- `=AVERAGE(...)` → about 150, matching the population mean (unbiasedness)
- `=STDEV.S(...)` → about 9.1

The standard deviation of the sample means is far smaller than the population standard deviation (≈ 28.87), and it matches the standard error $\sigma/\sqrt{n} = 28.87/\sqrt{10} \approx 9.13$.

EXERCISE 1, QUESTION 3 back to page 129

Select the 1000 sample means and go to `Insert → Charts → Histogram` (or use the bins-and-PivotTable method from Descriptive Statistics II). Although the population is uniform, the distribution of the sample means is bell-shaped — this is the Central Limit Theorem in action.
Using a normal distribution with mean 150 and standard error 9.13:

- `=NORM.DIST(160, 150, 9.13, TRUE) - NORM.DIST(140, 150, 9.13, TRUE)` → about 0.73

There is roughly a 73% probability that the sample mean falls between 140 and 160. Because the numbers are random, your exact value will vary slightly each time the sheet recalculates.

EXERCISE 2, QUESTION 1 back to page 129

The expected value is 80 (equal to the population mean). The standard error is 1.4, and the probability is 98.38%.
In Excel:

- `=14/SQRT(100)` → 1.4 (standard error)
- `=NORM.DIST(85, 80, 1.4, TRUE) - NORM.DIST(77, 80, 1.4, TRUE)` → 0.9838

EXERCISE 2, QUESTION 2 back to page 129

The probabilities are 24.66% and 1.8%.
For the mean of four cars, use the standard error $3.5/\sqrt{4}$:

- `=1 - NORM.DIST(35, 33.8, 3.5/SQRT(4), TRUE)` → 0.2466

For all four cars, first find the probability that one car exceeds 35, then raise it to the fourth power (the draws are independent):

- `=(1 - NORM.DIST(35, 33.8, 3.5, TRUE))^4` → 0.018

EXERCISE 3, QUESTION 1 back to page 130

The expected value is 0.75 (the same as the population). The standard error is $\sqrt{p(1-p)/n} = 0.03$, and the probability is 0.8975.
In Excel:

- `=SQRT(0.75*0.25/200)` → 0.0306
- `=NORM.DIST(0.8, 0.75, SQRT(0.75*0.25/200), TRUE) - NORM.DIST(0.7, 0.75, SQRT(0.75*0.25/200), TRUE)` → 0.8975

EXERCISE 3, QUESTION 2 back to page 130

The probability with a sample of 50 is 69.29%. With a sample of 200 it rises to 84.33%. As the sample size increases, the standard error shrinks, so the sampling distribution of $\hat{p}$ becomes tighter around 0.23 and more of its area lies above 0.20.
In Excel:

- `=1 - NORM.DIST(0.2, 0.23, SQRT(0.23*0.77/50), TRUE)` → 0.6929
- `=1 - NORM.DIST(0.2, 0.23, SQRT(0.23*0.77/200), TRUE)` → 0.8433

EXERCISE 4, QUESTION 1 back to page 130

The process appears to be out of control. In the early samples the machine is not filling the cans with enough drink; it recovers in later periods, but the early behavior is a warning sign.
Build a control chart. The center line is 350, and the limits are three standard errors away, where the standard error is $10/\sqrt{10}$:

- `=350 + 3*(10/SQRT(10))` → 359.49 (upper limit)
- `=350 - 3*(10/SQRT(10))` → 340.51 (lower limit)

Plot the eight sample means together with the two limits: put the sample numbers, the sample means, and two columns holding the constant limits side by side, then insert a Line or Scatter chart. Several early samples fall below the lower limit of 340.51, confirming the process is out of control.

EXERCISE 4, QUESTION 2 back to page 130

The process looks in control. All five proportions fall within three standard errors of the mean.

The center line is 0.01, and the limits use the standard error $\sqrt{0.01 \times 0.99/1000}$:

- `=0.01 + 3*SQRT(0.01*0.99/1000)` → 0.0194 (upper limit)
- `=0.01 - 3*SQRT(0.01*0.99/1000)` → 0.0006 (lower limit)

Plotting the five proportions against these limits shows every point inside the band, so the process is under control.

12 Inference II

EXERCISE 1, QUESTION 1 back to page 137

For an exponential distribution with rate $\lambda = 0.02$, the mean and standard deviation are both $1/\lambda = 50$. To draw an exponential value in Excel, transform a uniform random number:

- `=-LN(RAND())/0.02`

Fill a block of 10,000 rows and 50 columns with this formula; each row is a sample of size 50. (The values recalculate on every edit — copy the block and use `Paste Special → Values` to freeze them.)

EXERCISE 1, QUESTION 2 back to page 137

In the column just after the data, average each row and copy the formula down all 10,000 rows:

- `=AVERAGE(A2:AX2)` → the mean of the first sample

Summarizing the 10,000 sample means:

- `=AVERAGE(...)` → about 50, matching the population mean
- `=STDEV.S(...)` → about 7.07, which matches the standard error $\sigma/\sqrt{n} = 50/\sqrt{50} \approx 7.07$

EXERCISE 1, QUESTION 3 back to page 137

With the standard error stored in a cell (*call it* `SE` ≈ 7.07), the limits are the sample mean plus or minus $z_{0.05} \times SE$:

- `=FirstMean - NORM.S.INV(0.95)*SE` → lower limit
- `=FirstMean + NORM.S.INV(0.95)*SE` → upper limit

Because a typical sample mean sits near 50, the interval will usually contain the population mean. About 90% of such intervals do.

EXERCISE 1, QUESTION 4 back to page 137

Rather than build 10,000 pairs of limits, note that an interval includes 50 exactly when the sample mean is within $z_{0.05} \times SE$ of 50. Add a helper column:

- `=IF(ABS(SampleMean - 50) <= NORM.S.INV(0.95)*SE, 1, 0)`

Copy it down and sum it with `=SUM(...)`. *About* 9,000 of the 10,000 intervals contain the population mean — that is exactly what a 90% confidence level means.

EXERCISE 2, QUESTION 1 back to page 137

The 90% interval is [94.52, 114.68] and the 95% interval is [92.44, 116.76]. The higher the confidence level, the wider the interval.
Since σ is unknown, use the t-distribution. `CONFIDENCE.T` returns the margin of error directly:

- `=CONFIDENCE.T(0.10, 28.8, 24)` → 10.08, so the 90% interval is 104.6 ± 10.08
- `=CONFIDENCE.T(0.05, 28.8, 24)` → 12.16, so the 95% interval is 104.6 ± 12.16

EXERCISE 2, QUESTION 2 back to page 138

The interval for $n = 16$ is [30.75, 66.61] and for $n = 25$ is [34.79, 62.57]. As the sample size grows, the interval gets narrower and more precise.
In Excel:

- `=CONFIDENCE.T(0.05, 33.64, 16)` → 17.93, so the interval is 48.68 ± 17.93
- `=CONFIDENCE.T(0.05, 33.64, 25)` → 13.89, so the interval is 48.68 ± 13.89

EXERCISE 3 back to page 138

The 95% interval for Group 1 is $[-0.53, 2.03]$ and for Group 2 is [0.90, 3.76].
The data is sorted by group, so with `extra` *in column A the ten Group 1 values are in* `A2:A11` *and the ten Group 2 values in* `A12:A21`. *For each group, find the mean and the margin of error:*

- `=AVERAGE(A2:A11)` → 0.75 for Group 1, and `=AVERAGE(A12:A21)` → 2.33 for Group 2
- `=CONFIDENCE.T(0.05, STDEV.S(A2:A11), 10)` → the margin of error, giving `AVERAGE ± margin`

(*You can also get each group's mean with* `=AVERAGEIF(B:B, "1", A:A)`.) *Drug 2 appears more effective. Its interval lies entirely above zero, so it is unlikely to have no effect, and its mean increase in sleep is* 2.33 *hours, well above Drug 1's* 0.75 *hours.*

EXERCISE 4, QUESTION 1 back to page 138

The 90% interval is [0.319, 0.481] and the 95% interval is [0.304, 0.496]. Neither includes 0.5, so at both levels we can conclude the population proportion differs from 0.5.
With $\hat{p} = 0.4$ and $n = 100$, the standard error is `=SQRT(0.4*0.6/100)` $= 0.049$. The margins of error are:

- `=NORM.S.INV(0.95) * SQRT(0.4*0.6/100)` → 0.081, giving 0.4 ± 0.081
- `=NORM.S.INV(0.975) * SQRT(0.4*0.6/100)` → 0.096, giving 0.4 ± 0.096

EXERCISE 4, QUESTION 2 back to page 138

The 95% confidence interval is $[0.128,\ 0.186]$.
The data is in tidy form with columns `Hair`, `Eye`, `Sex`, *and* `n` (*the count*). *Total the Hazel rows and divide by the overall total to get the sample proportion:*

- `=SUMIF(Eye_range, "Hazel", n_range)` → 93 Hazel-eyed students
- `=SUM(n_range)` → 592 students, so $\hat{p} = 93/592 = 0.157$

Then the margin of error and interval are:

- `=NORM.S.INV(0.975) * SQRT(0.157*(1-0.157)/592)` → 0.029, giving $0.157 \pm 0.029 = [0.128,\ 0.186]$

13 Inference III

EXERCISE 1, QUESTION 1 back to page 144

The test statistic is -1.6. The two-tailed p-value is 13.04%, which is greater than 5%, so we cannot reject the null. The population mean is not statistically different from 50.
In Excel:

- `=(46 - 50) / (10 / SQRT(16))` → -1.6 (test statistic)
- `=T.DIST.2T(ABS(-1.6), 15)` → 0.1304 (two-tailed p-value)

The critical value `=T.INV.2T(0.05, 15)` *is* 2.131. Since $|-1.6| < 2.131$, we fail to reject the null.

EXERCISE 1, QUESTION 2 back to page 144

The null hypothesis cannot be rejected: the left-tailed p-value of 1.9% is greater than the 1% significance level.
Enter the data in a range (*say* `A2:A7`). *In Excel:*

- `=AVERAGE(A2:A7)` → 92 and `=STDEV.S(A2:A7)` → 6.96
- `=(92 - 100) / (6.96 / SQRT(6))` → -2.82 (test statistic)
- `=T.DIST(-2.82, 5, TRUE)` → 0.019 (left-tailed p-value)

EXERCISE 1, QUESTION 3 back to page 144

The null hypothesis can be rejected: the right-tailed p-value of 0.2% is less than the 10% significance level.
Enter the data in a range (*say* `A2:A11`). *In Excel:*

- `=AVERAGE(A2:A11)` → 256.4 and `=STDEV.S(A2:A11)` → 38.28
- `=(256.4 - 210) / (38.28 / SQRT(10))` → 3.83 (test statistic)
- `=T.DIST.RT(3.83, 9)` → 0.002 (right-tailed p-value)

EXERCISE 2 back to page 144

The claim that the average waiting time is 92 minutes can be rejected at the 10%, 5%, and 1% significance levels.
With the `waiting` *column in column B (272 observations), in Excel:*

- `=AVERAGE(B2:B273)` → 70.9 and `=STDEV.S(B2:B273)` → 13.6
- `=(70.9 - 92) / (13.6 / SQRT(272))` → −25.6 (test statistic)
- `=T.DIST.2T(ABS(-25.6), 271)` → approximately 0 (two-tailed p-value)

The sample mean is so far below 92 that the p-value is essentially zero, so we reject the null at any conventional level.

EXERCISE 3, QUESTION 1 back to page 144

The hypotheses are $H_0 : p = 0.4$ and $H_a : p \neq 0.4$. The two-tailed p-value is 0.102, greater than 0.05, so we cannot reject the null. The population proportion is not significantly different from 0.4.
In Excel:

- `=(0.48 - 0.4) / SQRT(0.4*(1-0.4)/100)` → 1.633 (test statistic)
- `=2*(1 - NORM.S.DIST(1.633, TRUE))` → 0.102 (two-tailed p-value)

EXERCISE 3, QUESTION 2 back to page 144

The sample proportion is $128/320 = 0.4$. The left-tailed p-value of 3.6% is less than 5%, so we reject the null and conclude the population proportion is less than 0.45.
In Excel:

- `=(0.4 - 0.45) / SQRT(0.45*(1-0.45)/320)` → −1.80 (test statistic)
- `=NORM.S.DIST(-1.80, TRUE)` → 0.036 (left-tailed p-value)

EXERCISE 3, QUESTION 3 back to page 145

The hypotheses are $H_0 : p \leq 0.5$ and $H_a : p > 0.5$. The right-tailed p-value of 0.26 is greater than the 1% significance level, so we cannot reject the null. We therefore **cannot** conclude that more than 50% of the population is below 10.
Enter the data in a range (say `A2:A11`*). In Excel:*

- `=COUNTIF(A2:A11, "<10") / COUNT(A2:A11)` → 0.6 (sample proportion)
- `=(0.6 - 0.5) / SQRT(0.5*(1-0.5)/10)` → 0.632 (test statistic)
- `=1 - NORM.S.DIST(0.632, TRUE)` → 0.264 (right-tailed p-value)

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We reject the null hypothesis: the proportion of hazel-eyed students is significantly different from 5%.
The data is tidy, with columns `Hair`, `Eye`, `Sex`, *and* `n`. *In Excel:*

- `=SUMIF(Eye_range, "Hazel", n_range)` → 93 and `=SUM(n_range)` → 592, so $\hat{p} = 93/592 = 0.157$

- `=(0.157 - 0.05) / SQRT(0.05*(1-0.05)/592)` → 11.96 (test statistic)
- `=2*(1 - NORM.S.DIST(11.96, TRUE))` → approximately 0 (two-tailed p-value)

The sample proportion of 15.7% is nearly 12 standard errors above the claimed 5%, so we reject the null decisively.

14 Regression and Inference

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We cannot reject the null: the p-value of 0.47 is greater than 0.05.
In Excel:

- `=(0.15*SQRT(25-2)) / SQRT(1 - 0.15^2)` → 0.73 (test statistic)
- `=T.DIST.2T(0.73, 23)` → 0.47 (two-tailed p-value)

EXERCISE 1, QUESTION 2 back to page 152

The correlation is 0.44 and the test statistic is 7.89. The p-value is approximately 0, so we reject the null hypothesis — the correlation between Hits and Salary is highly significant.
In Excel (*there are* 263 players with a recorded salary):

- `=CORREL(Hits_range, Salary_range)` → 0.44
- `=(0.44*SQRT(263-2)) / SQRT(1 - 0.44^2)` → 7.89 (test statistic)
- `=T.DIST.2T(7.89, 261)` → approximately 0

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There is no strong reason to believe the variances differ — players are recruited from a common pool. At the 5% level the difference in mean Hits is not significant, so we cannot reject the null.
Copy the Hits for `NewLeague = "A"` *into one column and the Hits for* `NewLeague = "N"` *into another* (*or filter the data*). *This is an unpaired, two-tailed test assuming equal variances:*

- `=T.TEST(A_range, N_range, 2, 2)` → 0.28 (p-value)

Since $0.28 > 0.05$, we fail to reject the null.

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There may be reason to believe the variances differ, so we use an unequal-variance test. At the 5% level we cannot reject $H_0 : \mu_{male} - \mu_{female} \leq 0$ — there is no evidence that males survive longer.
Run `Data → Data Analysis → t-Test: Two-Sample Assuming Unequal Variances` *on the male and female survival times. The male mean* (26.78) is actually **below** the female mean (28.10), giving a test statistic of -0.31. Because the difference points opposite to the alternative, the one-tailed p-value is about 0.62 — far above 0.05 — so we fail to reject the null.

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The drug has an effect: we reject $H_0 : \bar{d} = 0$. The mean difference in extra sleep is statistically different from zero.
Because the same patients are measured twice, use a paired, two-tailed test. With the Group 1 values in `A2:A11` *and the Group 2 values in* `A12:A21`*:*

- `=T.TEST(A2:A11, A12:A21, 2, 1)` → 0.0028 (p-value)

Since $0.0028 < 0.01$, we reject the null at the 1% level.

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RBI and Years are statistically significant (p-values near 0*)*: salary rises with experience and with RBIs. Home runs are **not** significant (p-value about 0.14). The F-statistic is highly significant, so the predictors are jointly significant. However, the R^2 and adjusted R^2 are only about 0.32, so the model explains just 32% of the variation in salary — more variables would be needed for a better fit.
Copy `HmRun`, `RBI`, *and* `Years` *into three adjacent columns (the ToolPak needs a contiguous X range), then run* `Data → Data Analysis → Regression` *with* `Salary` *as the Input Y Range and those three columns as the Input X Range. The output gives:*

- each coefficient's **p-value** (the individual tests): RBI and Years significant, HmRun not
- the **Significance F** (the joint test): approximately 0
- **R Square** and **Adjusted R Square**: about 0.33 and 0.32

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The predicted salary is 619.93 (thousand dollars), with a 95% prediction interval of $[-129.89, 1369.7]$.
Using the coefficients from the regression output ($\hat{\alpha} = -90.09$, $b_{HmRun} = -7.35$, $b_{RBI} = 9.16$, $b_{Years} = 32.82$), the point prediction is:

- `=-90.09 + (-7.35)*28 + 9.16*57 + 32.82*12` → 619.93

The prediction interval widens the estimate by $t \times s_e$, where $t =$ `T.INV.2T(0.05, 259)` ≈ 1.97 and s_e is the regression's standard error ($\approx$ 381 for a new observation). This gives $619.93 \pm 749.8 = [-129.89, 1369.7]$. The interval is very wide — and even includes negative salaries — because the model explains only about a third of the variation in salary.